

## SBS Commands

### 12.1 0x00 ManufacturerAccess() and 0x44 ManufacturerBlockAccess()

*ManufacturerBlockAccess()* provides a method of reading and writing data in the Manufacturer Access System (MAC). This block MAC access method is a new standard for the bq40zxy family. The MAC command is sent via *ManufacturerBlockAccess()* by the SMBus block protocol. The result is returned on *ManufacturerBlockAccess()* via an SMBus block read.

Example: Send a MAC *Gauging()* to enable IT via *ManufacturerBlockAccess()*.

1. With Impedance Track disabled, send *Gauging()* (0x0021) to *ManufacturerBlockAccess()*
  - (a) SMBus block write. Command = 0x44. Data = 21 00 (data must be sent in Little Endian)
2. IT is enabled, *ManufacturingStatus()[GAUGEN\_EN]* = 1.

Example: Read *Chemical ID()* (0x0006) via *ManufacturerBlockAccess()*

1. Send *Chemical ID()* to *ManufacturerBlockAccess()*.
  - (a) SMBus block write. Command = 0x44. Data sent = 06 00 (data must be sent in Little Endian)
2. Read the result from *ManufacturerBlockAccess()*.
  - (a) SMBus block read. Command = 0x44. Data read = 06 00 00 01 (each data entity is returned in Little Endian).
  - (b) The first 2 bytes, "06 00", is the MAC command.
  - (c) The second 2 bytes, "00 01", is the chem ID returning in Little Endian. That is 0x0100, chem ID 100.

For backwards compatibility with the bq30zxy families, sending MAC commands via *ManufacturerAccess()* (0x00) as well as the returning data on *ManufacturerData()* are supported in bq40z50. Note that MAC commands are sent through *ManufacturerAccess()* (0x00) by an SMBus write word protocol. The result reading from *ManufacturerData()* does not include the MAC command.

Example: Send a MAC *Gauging()* to enable IT via *ManufacturerAccess()*.

1. With Impedance Track disabled, send *Gauging()* (0x0021) to *ManufacturerAccess()*.
  - (a) SMBus word write. Command = 0x00. Data = 00 21
2. IT is enabled, *ManufacturingStatus()[GAUGEN\_EN]* = 1.

Example: Read *Chemical ID()* (0x0006) via *ManufacturerAccess()*

1. Send *Chemical ID()* to *ManufacturerAccess()*.
  - (a) SMBus word write. Command = 0x00. Data sent = 00 06
2. Read the result from *ManufacturerData()*.
  - (a) SMBus block read. Command = 0x23. Data read = 00 01
  - (b) That is 0x0100, chem ID 100.

The *ManufacturerAccess()* and *ManufacturerBlockAccess()* are interchangeable. The result can be read from *ManufacturerData()* or *ManufacturerBlockAccess()* regardless of how the MAC command is sent.

**Table 12-1. ManufacturerAccess() Command List**

| Command | Function                   | Access | Format | Data Read on 0x44 or 0x23 | Data Read on 0x2F | Available in SEALED Mode | Type  | Unit  |
|---------|----------------------------|--------|--------|---------------------------|-------------------|--------------------------|-------|-------|
| 0x0001  | DeviceType                 | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0002  | FirmwareVersion            | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0003  | HardwareVersion            | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0004  | IFChecksum                 | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0005  | StaticDFSSignature         | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0006  | ChemID                     | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0008  | StaticChemDFSSignature     | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0009  | AllDFSSignature            | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0010  | ShutdownMode               | W      | —      | —                         | —                 | Yes                      | Hex   | —     |
| 0x0011  | SleepMode                  | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0013  | AutoCCOset                 | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x001D  | FuseToggle                 | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x001E  | PrechargeFET               | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x001F  | ChargeFET                  | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0020  | DischargeFET               | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0021  | Gauging                    | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0022  | FETControl                 | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0023  | LifetimeDataCollection     | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0024  | PermanentFailure           | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0025  | BlackBoxRecorder           | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0026  | Fuse                       | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0027  | LEDDisplayEnable           | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0028  | LifetimeDataReset          | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0029  | PermanentFailureData Reset | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x002E  | LifetimeDataFlush          | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x002F  | LifetimeDataSpeedUp Mode   | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x002A  | BlackBoxRecorderReset      | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x002B  | LEDToggle                  | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x002C  | LEDDisplayPress            | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x002D  | CalibrationMode            | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0030  | SealDevice                 | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0035  | SecurityKeys               | R/W    | Block  | Yes                       | —                 | —                        | Hex   | —     |
| 0x0037  | AuthenticationKey          | R/W    | Block  | —                         | Yes               | —                        | Hex   | —     |
| 0x0041  | DeviceReset                | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0x0050  | SafetyAlert                | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0051  | SafetyStatus               | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0052  | PFAAlert                   | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0053  | PFStatus                   | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0054  | OperationStatus            | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0055  | ChargingStatus             | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0056  | GaugingStatus              | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0057  | ManufacturingStatus        | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0058  | AFERRegister               | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0060  | LifetimeDataBlock1         | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0061  | LifetimeDataBlock2         | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0062  | LifetimeDataBlock3         | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0070  | ManufacturerInfo           | R      | Block  | Yes                       | —                 | Yes                      | Hex   | —     |
| 0x0071  | DASatus1                   | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0072  | DASatus2                   | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0073  | GaugeStatus1               | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |

**Table 12-1. ManufacturerAccess() Command List (continued)**

| Command | Function                             | Access | Format | Data Read on 0x44 or 0x23 | Data Read on 0x2F | Available in SEALED Mode | Type  | Unit  |
|---------|--------------------------------------|--------|--------|---------------------------|-------------------|--------------------------|-------|-------|
| 0x0074  | GaugeStatus2                         | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0075  | GaugeStatus3                         | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0076  | CBStatus                             | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0077  | StateofHealth                        | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0078  | FilteredCapacity                     | R      | Block  | Yes                       | —                 | Yes                      | Mixed | Mixed |
| 0x0F00  | ROMMode                              | W      | —      | —                         | —                 | —                        | Hex   | —     |
| 0xF080  | ExitCalibrationOutput                | R/W    | Block  | Yes                       | —                 | —                        | Hex   | —     |
| 0xF081  | OutputCCandADCfor Calibration        | R/W    | Block  | Yes                       | —                 | —                        | Hex   | —     |
| 0xF082  | OutputShortedCCand ADCforCalibration | R/W    | Block  | Yes                       | —                 | —                        | Hex   | —     |

### 12.1.1 ManufacturerAccess() 0x0000

A read word on this command returns the lowest 16 bits of the *OperationStatus()* data.

### 12.1.2 ManufacturerAccess() 0x0001 Device Type

The device can be checked for the IC part number. The IC part number returns on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAA, where:

| Value | Description |
|-------|-------------|
| AAaa  | Device Type |

### 12.1.3 ManufacturerAccess() 0x0002 Firmware Version

The device can be checked for the firmware version of the IC. The firmware revision returns on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: ddDDvvVVbbBBTTzzZZRREE, where:

| Value | Description             |
|-------|-------------------------|
| DDdd  | Device Number           |
| VVvv  | Version                 |
| BBbb  | Build Number            |
| TT    | Firmware Type           |
| ZZzz  | Impedance Track Version |
| RR    | Reserved                |
| EE    | Reserved                |

### 12.1.4 ManufacturerAccess() 0x0003 Hardware Version

The device can be checked for the hardware version of the IC. The hardware revision returns on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*.

### 12.1.5 ManufacturerAccess() 0x0004 Instruction Flash Signature

The device can return the instruction flash signature. The IF signature returns on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*.

### 12.1.6 ManufacturerAccess() 0x0005 Static DF Signature

The device can return the data flash checksum. The signature of all static DF returns on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*. MSB is set to 1 if the calculated signature does not match the signature stored in DF.

### 12.1.7 ManufacturerAccess() 0x0006 Chemical ID

This command returns the chemical ID of the OCV tables used in the gauging algorithm. The chemical ID returns on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*.

### 12.1.8 ManufacturerAccess() 0x0008 Static Chem DF Signature

The device can return the data flash checksum. The signature of all static chemistry DF returns on subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*. MSB is set to 1 if the calculated signature does not match the signature stored in DF.

### 12.1.9 ManufacturerAccess() 0x0009 All DF Signature

The device can return the data flash checksum. The signature of all DF parameters returns on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*. MSB is set to 1 if the calculated signature does not match the signature stored in DF. It is normally expected that this signature will change due to updates of lifetime, gauging, and other information.

### 12.1.10 ManufacturerAccess() 0x0010 SHUTDOWN Mode

To reduce power consumption, the device can be sent to SHUTDOWN mode before shipping. After sending this command, the *OperationStatus() [SDM] = 1*, an internal counter will start, the CHG and DSG FETs will be turned off when the counter reaches **Ship FET Off Time**. When the counter reaches Ship Delay time, the device will enter SHUTDOWN mode if no charger present is detected.

If the device is SEALED, this feature requires the command to be sent twice in a row within 4 seconds (for safety purposes). If the device is in UNSEALED or FULL ACCESS mode, sending the command the second time will cancel the delay and enter shutdown immediately.

To wake up the device, a voltage > **Charger Present Threshold** must apply to the PACK pin. The device will power up and a full reset is applied.

### 12.1.11 ManufacturerAccess() 0x0011 SLEEP Mode

If the sleep conditions are met, the device can be sent to sleep with *ManufacturerAccess()*.

| Status   | Condition                                                                                                                     | Action                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enable   | 0x0011 to <i>ManufacturerAccess()</i>                                                                                         | <i>OperationStatus() [SLEEP] = 1</i>                                                                                                                                                                                                                                                                       |
| Activate | <b>DA Configuration[NR] = 0 AND</b><br><i>OperationStatus() [PRES] = 0 AND</i><br><i> Current()  &lt; Power:Sleep Current</i> | Turn off CHG FET, DSG FET, PCHG FET<br>Device goes to sleep.<br>Device wakes up every Power: <b>Sleep Voltage Time</b> period to measure voltage and temperature.<br>Device wakes up every Power: <b>Sleep Current Time</b> period to measure current.                                                     |
| Activate | <b>DA Configuration[NR] = 1 AND</b><br><i> Current()  &lt; Power:Sleep Current</i>                                            | Turn off DSG FET, PCHG FET<br>Turn off CHG FET if <b>DA Configuration[SLEEPCHG] = 0</b><br>Device goes to sleep.<br>Device wakes up every Power: <b>Sleep Voltage Time</b> period to measure voltage and temperature.<br>Device wakes up every Power: <b>Sleep Current Time</b> period to measure current. |
| Exit     | <b>DA Configuration[NR] = 0 AND</b><br><i>OperationStatus() [PRES] = 1</i>                                                    | <i>OperationStatus() [SLEEP] = 0</i><br>Return to NORMAL mode                                                                                                                                                                                                                                              |
| Exit     | <i> Current()  &gt; Configuration:Sleep Current</i>                                                                           | <i>OperationStatus() [SLEEP] = 0</i><br>Return to NORMAL mode                                                                                                                                                                                                                                              |
| Exit     | Wake Comparator trips                                                                                                         | <i>OperationStatus() [SLEEP] = 0</i><br>Return to NORMAL mode                                                                                                                                                                                                                                              |
| Exit     | <i>SafetyAlert()</i> flag or <i>PFAAlert()</i> flag set                                                                       | <i>OperationStatus() [SLEEP] = 0</i><br>Return to NORMAL mode                                                                                                                                                                                                                                              |

### 12.1.12 ManufacturerAccess() 0x0013 AutoCCOffset

This command manually starts an Auto CC Offset calibration. The calibration takes about 16 s.

This value is updated to CC Auto Offset, and is used for cell current measurement when the device is in CHARGING or DISCHARGING state. This offset is not used during RELAX mode. The cell current measurement is a current measurement taken simultaneously as the cell voltage measurement.

### 12.1.13 ManufacturerAccess() 0x001D Fuse Toggle

This command manually activates/deactivates the FUSE output to ease testing during manufacturing. If the *OperationStatus()[FUSE]* = 0 indicates the FUSE output is low. Sending this command toggles the FUSE output to be high and the *OperationStatus()[FUSE]* = 1.

### 12.1.14 ManufacturerAccess() 0x001E PCHG FET Toggle

This command turns on/off the PCHG FET drive function to ease testing during manufacturing. If the *ManufacturingStatus()[PCHG\_TEST]* = 0, sending this command will turn on the PCHG FET and the *ManufacturingStatus()[PCHG\_TEST]* = 1 and vice versa. This toggling command is only enabled if *ManufacturingStatus()[FET\_EN]* = 0, indicating an FW FET control is not active and manual control is allowed. A reset clears the [PCHG\_TEST] flag and turns off the PCHG FET.

### 12.1.15 ManufacturerAccess() 0x001F CHG FET Toggle

This command turns on/off the CHG FET drive function to ease testing during manufacturing. If the *ManufacturingStatus()[CHG\_TEST]* = 0, sending this command turns on the CHG FET and the *ManufacturingStatus()[CHG\_TEST]* = 1 and vice versa. This toggling command is only enabled if *ManufacturingStatus()[FET\_EN]* = 0, indicating an FW FET control is not active and manual control is allowed. A reset clears the [CHG\_TEST] flag and turns off the CHG FET.

### 12.1.16 ManufacturerAccess() 0x0020 DSG FET Toggle

This command turns on/off DSG FET drive function to ease testing during manufacturing. If the *ManufacturingStatus()[DSG\_TEST]* = 0, sending this command turns on the DSG FET and the *ManufacturingStatus()[DSG\_TEST]* = 1 and vice versa. This toggling command is only enabled if *ManufacturingStatus()[FET\_EN]* = 0, indicating an FW FET control is not active and manual control is allowed. A reset clears the [DSG\_TEST] flag and turns off the DSG FET.

### 12.1.17 ManufacturerAccess() 0x0021 Gauging

This command enables or disables the gauging function to ease testing during manufacturing. The initial setting is loaded from **Mfg Status Init[GAUGE\_EN]**. If the *ManufacturingStatus()[GAUGE\_EN]* = 0, sending this command will enable gauging and the *ManufacturingStatus()[GAUGE\_EN]* = 1 and vice versa. In UNSEALED mode, the *ManufacturingStatus()[GAUGE\_EN]* status is copied to **Mfg Status Init[GAUGE\_EN]** when the command is received by the gauge. The device remains on its latest gauging status prior to a reset.

### 12.1.18 ManufacturerAccess() 0x0022 FET Control

This command disables/enables control of the CHG, DSG, and PCHG FET by the firmware. The initial setting is loaded from **Mfg Status Init[FET\_EN]**. If the *ManufacturingStatus()[FET\_EN]* = 0, sending this command allows the FW to control the PCHG, CHG, and DSG FETs and the *ManufacturingStatus()[FET\_EN]* = 1 and vice versa.

In UNSEALED mode, the *ManufacturingStatus()[FET\_EN]* status is copied to **Mfg Status Init[FET\_EN]** when the command is received by the gauge. The device remains on its latest FET control status prior to a reset.

### 12.1.19 ManufacturerAccess() 0x0023 Lifetime Data Collection

This command disables/enables Lifetime Data Collection to help streamline production testing. The initial setting is loaded from **Mfg Status Init[LF\_EN]**. If the *ManufacturingStatus()[LF\_EN]* = 0, sending this command starts the Lifetime Data Collection and the *ManufacturingStatus()[LF\_EN]* = 1 and vice versa.

In UNSEALED mode, the *ManufacturingStatus()[LF\_EN]* status is copied to **Mfg Status Init[LF\_EN]** when the command is received by the gauge. The device remains on its latest Lifetime Data Collection setting prior to a reset.

### 12.1.20 ManufacturerAccess() 0x0024 Permanent Failure

This command disables/enables Permanent Failure to help streamline production testing.

The initial setting is loaded from **Mfg Status Init[PF\_EN]**. If the *ManufacturingStatus()[PF\_EN]* = 0, sending this command enables Permanent Failure protections and the *ManufacturingStatus()[PF\_EN]* = 1 and vice versa.

In UNSEALED mode, *ManufacturingStatus()[PF\_EN]* status is copied to **Mfg Status Init[PF\_EN]** when the command is received by the gauge. The device remains on its PF enable/disable setting prior to a reset.

### 12.1.21 ManufacturerAccess() 0x0025 Black Box Recorder

This command enables/disables Black Box Recorder function to help streamline production testing. The initial setting is loaded from **Mfg Status Init[BBR\_EN]**. If the *ManufacturingStatus()[BBR\_EN]* = 0, sending this command enables the Black Box Recorder and the *ManufacturingStatus()[BBR\_EN]* = 1 and vice versa.

In UNSEALED mode, the *ManufacturingStatus()[BBR\_EN]* status is copied to **Mfg Status Init[BBR\_EN]** when the command is received by the gauge. The device remains on its latest Black Box Recorder enable/disable setting prior to a reset.

### 12.1.22 ManufacturerAccess() 0x0026 Fuse

This command disables/enables firmware-based fuse activation to ease testing during manufacturing. The initial setting is loaded from **Mfg Status Init[FUSE\_EN]**. If the *ManufacturingStatus()[FUSE\_EN]* = 0, sending this command allows the FW to control the FUSE output and the *ManufacturingStatus()[FUSE\_EN]* = 1 and vice versa.

In UNSEALED mode, the *ManufacturingStatus()[FUSE\_EN]* status is copied to **Mfg Status Init[FUSE\_EN]** when the command is received by the gauge. The device remains on its latest Fuse Control setting prior to a reset.

### 12.1.23 ManufacturerAccess() 0x0027 LED DISPLAY Enable

This command enables or disables the LED display function to ease testing during manufacturing. The initial setting is loaded from **Mfg Status Init[LED\_EN]**. If the *ManufacturingStatus()[LED\_EN]* = 0, sending this command will enable the LED display and the *ManufacturingStatus()[LED\_EN]* = 1 and vice versa. In UNSEALED mode, the *ManufacturingStatus()[LED\_EN]* status is copied to **Mfg Status Init[LED\_EN]** when the command is received by the gauge. The device remains on its latest setting prior to a reset.

### 12.1.24 ManufacturerAccess() 0x0028 Lifetime Data Reset

Sending this command resets Lifetime Data in data flash to help streamline production testing.

### 12.1.25 ManufacturerAccess() 0x0029 Permanent Fail Data Reset

Sending this command resets PF data in data flash to help streamline production testing.

### 12.1.26 ManufacturerAccess() 0x002A Black Box Recorder Reset

Sending this command resets the Black Box Recorder data in data flash to help streamline production testing.



### 12.1.27 ManufacturerAccess() 0x002B LED TOGGLE

This command toggles the LED display on or off to help streamline testing during manufacturing. When the LED display is off, the *OperationStatus()[LED]* = 0. Sending this command turns on all LED displays with *OperationStatus()[LED]* set to 1, and vice versa.

### 12.1.28 ManufacturerAccess() 0x002C LED DISPLAY PRESS

This command simulates a low-high-low detection of the  $\overline{\text{DISP}}$  pin, activating the LED display according to the LED Support data flash setting. This command forces RSOC to 100% in order to demonstrate all LEDs in use, the full speed, and the brightness.

### 12.1.29 ManufacturerAccess() 0x002D CALIBRATION Mode

This command disables/enables entry into CALIBRATION mode. Status is indicated by the *ManufacturingStatus()[CAL\_EN]* flag. CALIBRATION mode is disabled upon a reset.

| Status  | Condition                                                                          | Action                                                                                                                                                                              |
|---------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disable | <i>ManufacturingStatus()[CAL_EN]</i> = 1 AND 0x002D to <i>ManufacturerAccess()</i> | <i>ManufacturingStatus()[CAL_EN]</i> = 0<br>Disable output of ADC and CC raw data on <i>ManufacturingData()</i>                                                                     |
| Enable  | <i>ManufacturingStatus()[CAL_EN]</i> = 0 AND 0x002D to <i>ManufacturerAccess()</i> | <i>ManufacturingStatus()[CAL_EN]</i> = 1<br>Enable output of ADC and CC raw data on <i>ManufacturingData()</i> , controllable with 0xF081 and 0xF082 on <i>ManufacturerAccess()</i> |

### 12.1.30 ManufacturerAccess() 0x002E Lifetime Data Flush

This command flushes the RAM Lifetime Data to data flash to help streamline evaluation testing.

### 12.1.31 ManufacturerAccess() 0x002F Lifetime Data SPEED UP Mode

For ease of evaluation testing, this command enables a lifetime SPEED UP mode where every 1 s in real time counts as 2 hours in FW time. When the lifetime SPEED UP mode is enabled, the *ManufacturingStatus()[LT\_TEST]* = 1.

The SPEED UP mode will be disabled if this command is sent again when *[LT\_TEST]* = 1, the MAC *LifetimeDataReset()* command is sent, the MAC *SealDevice()* command is sent, or the device is reset.

### 12.1.32 ManufacturerAccess() 0x0030 Seal Device

This command seals the device for the field, disabling certain SBS commands and access to data flash. See [Table 12-1](#) and [Chapter 12](#) for details.

When the device is sealed, the *OperationStatus()[SEC1, SEC0]* = 1,1. All the test features in *ManufacturingStatus()* will also be disabled.

### 12.1.33 ManufacturerAccess() 0x0035 Security Keys

This is a read/write command for 2-word UNSEAL and FULL ACCESS keys.

When reading the keys, data can be read from *ManufacturerData()* or *ManufacturerBlockAccess()*. The keys are returned in the following format: aaAabbBBccCCddDD, where:

| Value | Description                        |
|-------|------------------------------------|
| AAaa  | First word of the UNSEAL key       |
| BBbb  | Second word of the UNSEAL key      |
| CCcc  | First word of the FULL ACCESS key  |
| DDdd  | Second word of the FULL ACCESS key |

The default UNSEAL key is 0x0414 and 0x3672. The default FULL ACCESS key is 0xFFFF and 0xFFFF. It is highly recommended to change the UNSEAL and FULL ACCESS keys from default.

The keys can only be changed through the *ManufacturerBlockAccess()*.

Example: Change UNSEAL key to 0x1234, 0x5678, and leave the FULL ACCESS as default.

Send an SMBus block write with Command = 0x44.

Data = MAC command + New UNSEAL key + New FULL ACCESS KEY  
= 35 00 34 12 78 56 FF FF FF FF

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**NOTE:** The first word of the keys cannot be the same. That means an UNSEAL key with 0xABCD 0x1234 and FULL ACCESS key with 0xABCD 0x5678 are not valid because the first word is the same.

This is because the first word is used as a “detection” for the right command. This also means the first word cannot be the same as any existing MAC command.

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### 12.1.34 ManufacturerAccess() 0x0037 Authentication Key

This command enables the update of the authentication key into the device. The device must be in FULL ACCESS mode for the authentication key to update.

To update a new authentication key:

- Send the *AuthenticationKey()* + the new 128-bit authentication key to *ManufacturerBlockAccess()*, OR
- Send the *AuthenticationKey()* to *ManufacturerAccess()*, then send the 128-bit authentication key to *Authentication()*.

There is no direct read access to the authentication key. After writing the new authentication to the gauge, the gauge will generate an all-zero challenge and provide the corresponding response for verification.

To verify the new authentication key:

- Read the response from *ManufacturerBlockAccess()* after updating the new authentication key, OR
- Read the response from *Authentication()* after updating the new authentication key.

### 12.1.35 ManufacturerAccess() 0x0041 Device Reset

This command resets the device.

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**NOTE:** Command 0x0012 also resets the device (for backwards compatibility with the bq30zxy device).

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### 12.1.36 ManufacturerAccess() 0x0050 SafetyAlert

This command returns the *SafetyAlert()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|      |      |      |      |           |      |           |      |           |      |      |      |      |      |      |     |
|------|------|------|------|-----------|------|-----------|------|-----------|------|------|------|------|------|------|-----|
| 31   | 30   | 29   | 28   | 27        | 26   | 25        | 24   | 23        | 22   | 21   | 20   | 19   | 18   | 17   | 16  |
| RSVD | RSVD | RSVD | RSVD | UTD       | UTC  | PCHG<br>C | CHGV | CHGC      | OC   | CTOS | CTO  | PTOS | PTO  | RSVD | OTF |
| 15   | 14   | 13   | 12   | 11        | 10   | 9         | 8    | 7         | 6    | 5    | 4    | 3    | 2    | 1    | 0   |
| RSVD | CUVC | OTD  | OTC  | ASCD<br>L | RSVD | ASCC<br>L | RSVD | AOLD<br>L | RSVD | OCD2 | OCD1 | OCC2 | OCC1 | COV  | CUV |



**RSVD (Bits 31–28):** Reserved. Do not use.

**UTD (Bit 27):** Undertemperature During Discharge

1 = Detected

0 = Not Detected

**UTC (Bit 26):** Undertemperature During Charge

1 = Detected

0 = Not Detected

**PCHGC (Bit 25):** Over-Precharge Current

1 = Detected

0 = Not Detected

**CHGV (Bit 24):** Overcharging Voltage

1 = Detected

0 = Not Detected

**CHGC (Bit 23):** Overcharging Current

1 = Detected

0 = Not Detected

**OC (Bit 22):** Overcharge

1 = Detected

0 = Not Detected

**CTOS (Bit 21):** Charge Timeout Suspend

1 = Detected

0 = Not Detected

**CTO (Bit 20):** Charge Timeout

1 = Detected

0 = Not Detected

**PTOS (Bit 19):** Precharge Timeout Suspend

1 = Detected

0 = Not Detected

**PTO (Bit 18):** Precharge Timeout

1 = Detected

0 = Not Detected

**RSVD (Bit 17):** Reserved. Do not use.

**OTF (Bit 16):** Overtemperature FET

1 = Detected

0 = Not Detected

**RSVD (Bit 15):** Reserved. Do not use.

**CUVC (Bit 14):** Cell Undervoltage Compensated

1 = Detected

0 = Not Detected

**OTD (Bit 13):** Overtemperature During Discharge

1 = Detected

0 = Not Detected

**OTC (Bit 12):** Overtemperature During Charge

1 = Detected

0 = Not Detected

**ASCDL (Bit 11):** Short-Circuit During Discharge Latch

1 = Detected

0 = Not Detected

**RSVD (Bit 10):** Reserved. Do not use.

**ASCCL (Bit 9):** Short-Circuit During Charge Latch

1 = Detected

0 = Not Detected

**RSVD (Bit 8):** Reserved. Do not use.

**AOLDL (Bit 7):** Overload During Discharge Latch

1 = Detected

0 = Not Detected

**RSVD (Bit 6):** Reserved. Do not use.

**OCD2 (Bit 5):** Overcurrent During Discharge 2

1 = Detected

0 = Not Detected

**OCD1 (Bit 4):** Overcurrent During Discharge 1

1 = Detected

0 = Not Detected

**OCC2 (Bit 4):** Overcurrent During Charge 2

1 = Detected

0 = Not Detected

**OCC1 (Bit 2):** Overcurrent During Charge 1

1 = Detected

0 = Not Detected

**COV (Bit 1):** Cell Overvoltage

1 = Detected

0 = Not Detected

**CUV (Bit 0):** Cell Undervoltage

1 = Detected

0 = Not Detected

### 12.1.37 ManufacturerAccess() 0x0051 SafetyStatus

This command returns the *SafetyStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|      |      |      |      |           |      |           |      |           |      |      |      |      |      |      |     |
|------|------|------|------|-----------|------|-----------|------|-----------|------|------|------|------|------|------|-----|
| 31   | 30   | 29   | 28   | 27        | 26   | 25        | 24   | 23        | 22   | 21   | 20   | 19   | 18   | 17   | 16  |
| RSVD | RSVD | RSVD | RSVD | UTD       | UTC  | PCHG<br>C | CHGV | CHGC      | OC   | RSVD | CTO  | RSVD | PTO  | RSVD | OTF |
| 15   | 14   | 13   | 12   | 11        | 10   | 9         | 8    | 7         | 6    | 5    | 4    | 3    | 2    | 1    | 0   |
| RSVD | CUVC | OTD  | OTC  | ASCD<br>L | ASCD | ASCC<br>L | ASCC | AOLD<br>L | AOLD | OCD2 | OCD1 | OCC2 | OCC1 | COV  | CUV |

**RSVD (Bits 31–28):** Reserved. Do not use.

**UTD (Bit 27):** Undertemperature During Discharge

1 = Detected

0 = Not Detected

**UTC (Bit 26):** Undertemperature During Charge

1 = Detected

0 = Not Detected

**PCHGC (Bit 25):** Over-Precharge Current

1 = Detected

0 = Not Detected

**CHGV (Bit 24):** Overcharging Voltage

1 = Detected

0 = Not Detected

**CHGC (Bit 23):** Overcharging Current

1 = Detected

0 = Not Detected

**OC (Bit 22):** Overcharge

1 = Detected

0 = Not Detected

**RSVD (Bit 21):** Reserved. Do not use.

**CTO (Bit 20):** Charge Timeout

1 = Detected

0 = Not Detected

**RSVD (Bit 19):** Reserved. Do not use.

**PTO (Bit 18):** Precharge Timeout

1 = Detected

0 = Not Detected

**RSVD (Bit 17):** Reserved. Do not use.

**OTF (Bit 16):** Overtemperature FET

1 = Detected

0 = Not Detected

**RSVD (Bit 15):** Reserved. Do not use.

**CUVC (Bit 14):** Cell Undervoltage Compensated

1 = Detected

0 = Not Detected

**OTD (Bit 13):** Overtemperature During Discharge

1 = Detected

0 = Not Detected

**OTC (Bit 12):** Overtemperature During Charge

1 = Detected

0 = Not Detected

**ASCDL (Bit 11):** Short-circuit During Discharge Latch

1 = Detected

0 = Not Detected

**ASCD (Bit 10):** Short-circuit During Discharge

1 = Detected

0 = Not Detected

**ASCCL (Bit 9):** Short-circuit During Charge Latch

1 = Detected

0 = Not Detected

**ASCC (Bit 8):** Short-circuit During Charge

1 = Detected

0 = Not Detected

**AOLDL (Bit 7):** Overload During Discharge Latch

1 = Detected

0 = Not Detected

**AOLD (Bit 6):** Overload During Discharge

1 = Detected

0 = Not Detected

**OCD2 (Bit 5):** Overcurrent During Discharge 2

1 = Detected

0 = Not Detected

**OCD1 (Bit 4):** Overcurrent During Discharge 1

1 = Detected

0 = Not Detected

**OCC2 (Bit 3):** Overcurrent During Charge 2

1 = Detected

0 = Not Detected

**OCC1 (Bit 2):** Overcurrent During Charge 1

1 = Detected

0 = Not Detected

**COV (Bit 1):** Cell Overvoltage

1 = Detected

0 = Not Detected

**CUV (Bit 0):** Cell Undervoltage

1 = Detected

0 = Not Detected

### 12.1.38 ManufacturerAccess() 0x0052 PFAAlert

This command returns the *PFAAlert()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|      |      |      |      |      |      |      |      |      |      |      |      |       |       |           |           |
|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-----------|-----------|
| 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19    | 18    | 17        | 16        |
| TS4  | TS3  | TS2  | TS1  | RSVD | RSVD | OPNC | RSVD | RSVD | 2LVL | AFEC | AFER | FUSE  | RSVD  | DFET<br>F | CFET<br>F |
| 15   | 14   | 13   | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3     | 2     | 1         | 0         |
| RSVD | RSVD | RSVD | VIMA | VIMR | CD   | IMP  | CB   | QIM  | SOTF | RSVD | SOT  | S OCD | S OCC | S OV      | S UV      |

**TS4 (Bit 31):** Open Thermistor–TS4 Failure

1 = Detected

0 = Not Detected

**TS3 (Bit 30):** Open Thermistor–TS3 Failure

1 = Detected

0 = Not Detected

**TS2 (Bit 29):** Open Thermistor–TS2 Failure

1 = Detected

0 = Not Detected

**TS1 (Bit 28):** Open Thermistor–TS1 Failure

1 = Detected

0 = Not Detected

**RSVD (Bits 27–26):** Reserved. Do not use.

**OPNC (Bit 25):** Open Cell Tab Connection Failure

1 = Detected

0 = Not Detected

**RSVD (Bits 24–23):** Reserved. Do not use.

**2LVL (Bit 22):** Second Level Protector Failure

1 = Detected

0 = Not Detected

**AFEC (Bit 21):** AFE Communication Failure

1 = Detected

0 = Not Detected

**AFER (Bit 20):** AFE Register Failure

1 = Detected

0 = Not Detected

**FUSE (Bit 19):** Chemical Fuse Failure

1 = Detected

0 = Not Detected

**DFETF (Bit 17):** Discharge FET Failure

1 = Detected

0 = Not Detected

**CFETF (Bit 16):** Charge FET Failure

1 = Detected

0 = Not Detected

**RSVD (Bits 15–13):** Reserved. Do not use.

**VIMA (Bit 12):** Voltage Imbalance While Pack Is Active Failure

1 = Detected

0 = Not Detected

**VIMR (Bit 11):** Voltage Imbalance While Pack Is At Rest Failure

1 = Detected

0 = Not Detected

**CD (Bit 10):** Capacity Degradation Failure

1 = Detected

0 = Not Detected

**IMP (Bit 9):** Impedance Failure

1 = Detected

0 = Not Detected

**CB (Bit 8):** Cell Balancing Failure

1 = Detected

0 = Not Detected

**QIM (Bit 7):** QMax Imbalance Failure

1 = Detected

0 = Not Detected

**SOTF (Bit 6):** Safety Overtemperature FET Failure

1 = Detected

0 = Not Detected

**RSVD (Bit 5):** Reserved. Do not use.

**SOT (Bit 4):** Safety Overtemperature Cell Failure

1 = Detected

0 = Not Detected

**SOCD (Bit 3):** Safety Overcurrent in Discharge

1 = Detected

0 = Not Detected

**SOCC (Bit 2):** Safety Overcurrent in Charge

1 = Detected

0 = Not Detected

**SOV (Bit 1):** Safety Cell Overvoltage Failure

1 = Detected

0 = Not Detected

**SUV (Bit 0):** Safety Cell Undervoltage Failure

1 = Detected

0 = Not Detected

### 12.1.39 ManufacturerAccess() 0x0053 PFStatus

This command returns the *PFStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|      |      |      |      |      |     |             |     |     |      |      |      |      |      |           |           |
|------|------|------|------|------|-----|-------------|-----|-----|------|------|------|------|------|-----------|-----------|
| 31   | 30   | 29   | 28   | 27   | 26  | 25          | 24  | 23  | 22   | 21   | 20   | 19   | 18   | 17        | 16        |
| TS4  | TS3  | TS2  | TS1  | RSVD | DFW | OPN<br>CELL | IFC | PTC | 2LVL | AFEC | AFER | FUSE | RSVD | DFET<br>F | CFET<br>F |
| 15   | 14   | 13   | 12   | 11   | 10  | 9           | 8   | 7   | 6    | 5    | 4    | 3    | 2    | 1         | 0         |
| RSVD | RSVD | RSVD | VIMA | VIMR | CD  | IMP         | CB  | QIM | SOTF | RSVD | SOT  | SOCD | SOCC | SOV       | SUV       |

**TS4 (Bit 31):** Open Thermistor–TS4 Failure

1 = Detected

0 = Not Detected

**TS3 (Bit 30):** Open Thermistor–TS3 Failure

1 = Detected



0 = Not Detected

**TS2 (Bit 29):** Open Thermistor–TS2 Failure

1 = Detected

0 = Not Detected

**TS1 (Bit 28):** Open Thermistor–TS1 Failure

1 = Detected

0 = Not Detected

**RSVD (Bit 27):** Reserved. Do not use.

**DFW (Bit 26):** Data Flash Wearout Failure

1 = Detected

0 = Not Detected

**OPNCELL (Bit 25):** Open Cell Tab Connection Failure

1 = Detected

0 = Not Detected

**IFC (Bit 24):** Instruction Flash Checksum Failure

1 = Detected

0 = Not Detected

**PTC (Bit 23):** PTC Failure

1 = Detected

0 = Not Detected

**2LVL (Bit 22):** Second Level Protector Failure

1 = Detected

0 = Not Detected

**AFEC (Bit 21):** AFE Communication Failure

1 = Detected

0 = Not Detected

**AFER (Bit 20):** AFE Register Failure

1 = Detected

0 = Not Detected

**FUSE (Bit 19):** Chemical Fuse Failure

1 = Detected

0 = Not Detected

**RSVD (Bit 18):** Reserved. Do not use.

**DFETF (Bit 17):** Discharge FET Failure

1 = Detected

0 = Not Detected

**CFETF (Bit 16):** Charge FET Failure

1 = Detected

0 = Not Detected

**RSVD (Bits 15–13):** Reserved. Do not use.

**VIMA (Bit 12):** Voltage Imbalance while Pack is Active Failure

1 = Detected

0 = Not Detected

**VIMR (Bit 11):** Voltage Imbalance while Pack At Rest Failure

1 = Detected

0 = Not Detected

**CD (Bit 10):** Capacity Degradation Failure

1 = Detected

0 = Not Detected

**IMP (Bit 9):** Impedance Failure

1 = Detected

0 = Not Detected

**CB (Bit 8):** Cell Balancing Failure

1 = Detected

0 = Not Detected

**QIM (Bit 7):** QMax Imbalance Failure

1 = Detected

0 = Not Detected

**SOTF (Bit 6):** Safety Overtemperature FET Failure

1 = Detected

0 = Not Detected

**RSVD (Bit 5):** Reserved. Do not use.

**SOT (Bit 4):** Safety Overtemperature Cell Failure

1 = Detected

0 = Not Detected

**SOCD (Bits 3):** Safety Overcurrent in Discharge

1 = Detected

0 = Not Detected

**SOCC (Bits 2):** Safety Overcurrent in Charge

1 Detected

0 Not Detected

**SOV (Bit 1):** Safety Cell Overvoltage Failure

1 = Detected

0 = Not Detected

**SUV (Bit 0):** Safety Cell Undervoltage Failure

1 = Detected

0 = Not Detected

#### 12.1.40 ManufacturerAccess() 0x0054 OperationStatus

This command returns the *OperationStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|           |      |            |    |           |           |             |      |            |      |                   |      |              |      |     |      |
|-----------|------|------------|----|-----------|-----------|-------------|------|------------|------|-------------------|------|--------------|------|-----|------|
| 31        | 30   | 29         | 28 | 27        | 26        | 25          | 24   | 23         | 22   | 21                | 20   | 19           | 18   | 17  | 16   |
| RSVD      | RSVD | EMSH<br>UT | CB | SLPC<br>C | SLPA<br>D | SMBL<br>CAL | INIT | SLEE<br>PM | XL   | CAL<br>OFFS<br>ET | CAL  | AUTO<br>CALM | AUTH | LED | SDM  |
| 15        | 14   | 13         | 12 | 11        | 10        | 9           | 8    | 7          | 6    | 5                 | 4    | 3            | 2    | 1   | 0    |
| SLEE<br>P | XCHG | XDSG       | PF | SS        | SDV       | SEC1        | SEC0 | BTP<br>INT | RSVD | FUSE              | RSVD | PCHG         | CHG  | DSG | PRES |

**RSVD (Bits 31–30):** Reserved. Do not use.

**EMSHUT (Bit 29):** Emergency Shutdown

- 1 = Active
- 0 = Inactive

**CB (Bit 28):** Cell balancing status

- 1 = Active
- 0 = Inactive

**SLPCC (Bit 27):** CC Measurement in SLEEP mode

- 1 = Active
- 0 = Inactive

**SLPAD (Bit 26):** ADC Measurement in SLEEP mode

- 1 = Active
- 0 = Inactive

**SMBL CAL (Bit 25):** Auto CC calibration when the bus is low. This bit may not be read by the host because the FW will clear it when a communication is detected.

- 1 = Auto CC calibration starts
- 0 = When the bus is high or communication is detected for the case of *[IN\_SYSTEM\_SLEEP] = 1*.

**INIT (Bit 24):** Initialization after full reset

- 1 = Active
- 0 = Inactive

**SLEEP M (Bit 23):** SLEEP mode triggered via command

- 1 = Active
- 0 = Inactive

**XL (Bit 22):** 400-kHz SMBus mode

- 1 = Active
- 0 = Inactive

**CAL\_OFFSET (Bit 21):** Calibration Output (raw CC offset data).

- 1 = Active when MAC *OutputShortedCCADCCal()* is sent and the raw shorted CC data for calibration is available.
- 0 = When the raw shorted CC data for calibration is not available.

**CAL (Bit 20):** Calibration Output (raw ADC and CC data)

- 1 = Active when either the MAC *OutputCCADCCal()* or *OutputShortedCCADCCal()* is sent and the raw CC and ADC data for calibration is available.
- 0 = When the raw CC and ADC data for calibration is not available.

**AUTOCAL M (Bit 19):** Auto CC Offset Calibration by MAC *AutoCCOffset()*

- 1 = The gauge receives the MAC *AutoCCOffset()* and starts the auto CC offset calibration.
- 0 = Clear when the calibration is completed.

**AUTH (Bit 18):** Authentication in progress

- 1 = Active
- 0 = Inactive

**LED (Bit 17):** LED Display

- 1 = LED display is on.
- 0 = LED display is off.

**SDM (Bit 16):** Shutdown triggered via command

- 1 = Active
- 0 = Inactive

**SLEEP (Bit 15):** SLEEP mode conditions met

- 1 = Active
- 0 = Inactive

**XCHG (Bit 14):** Charging disabled

- 1 = Active
- 0 = Inactive

**XDSG (Bit 13):** Discharging disabled

- 1 = Active
- 0 = Inactive

**PF (Bit 12):** PERMANENT FAILURE mode status

- 1 = Active
- 0 = Inactive

**SS (Bit 11):** SAFETY mode status

- 1 = Active
- 0 = Inactive

**SDV (Bit 10):** Shutdown triggered via low pack voltage

- 1 = Active
- 0 = Inactive

**SEC1, SEC0 (Bits 9–8):** SECURITY mode

- 0, 0 = Reserved
- 0, 1 = Full Access
- 1, 0 = Unsealed
- 1, 1 = Sealed

**BTP\_INT (Bit 7):** Battery Trip Point Interrupt. Setting and clearing this bit depends on various conditions.

See [Section 6.9](#) for details.

**RSVD (Bit 6):** Reserved. Do not use.

**FUSE (Bit 5):** Fuse status

- 1 = Active
- 0 = Inactive

**RSVD (Bit 4):** Reserved. Do not use.

**PCHG (Bit 3):** Precharge FET status

- 1 = Active
- 0 = Inactive

**CHG (Bit 2):** CHG FET status

- 1 = Active
- 0 = Inactive

**DSG (Bit 1):** DSG FET status

- 1 = Active
- 0 = Inactive

**PRES (Bit 0):** System present low

1 = Active

0 = Inactive

### 12.1.41 ManufacturerAccess() 0x0055 ChargingStatus

This command returns the *ChargingStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|     |     |      |    |    |    |    |    |      |      |      |      |      |      |     |     |
|-----|-----|------|----|----|----|----|----|------|------|------|------|------|------|-----|-----|
|     |     |      |    |    |    |    |    | 23   | 22   | 21   | 20   | 19   | 18   | 17  | 16  |
|     |     |      |    |    |    |    |    | RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | CCC | CVR |
| 15  | 14  | 13   | 12 | 11 | 10 | 9  | 8  | 7    | 6    | 5    | 4    | 3    | 2    | 1   | 0   |
| CCR | VCT | MCHG | IN | HV | MV | LV | PV | RSVD | OT   | HT   | STH  | RT   | STL  | LT  | UT  |

**RSVD (Bits 18–23):** Reserved. Do not use.

**CCC (Bit 17):** Charging Loss Compensation

1 = Active

0 = Inactive

**CVR (Bit 16):** Charging Voltage Rate of Change

1 = Active

0 = Inactive

**CCR (Bit 15):** Charging Current Rate of Change

1 = Active

0 = Inactive

**VCT (Bit 14):** Charge Termination

1 = Active

0 = Inactive

**MCHG (Bit 13):** Maintenance Charge

1 = Active

0 = Inactive

**IN (Bit 12):** Charge Inhibit

1 = Active

0 = Inactive

**HV (Bit 11):** High Voltage Region

1 = Active

0 = Inactive

**MV (Bit 10):** Mid Voltage Region

1 = Active

0 = Inactive

**LV (Bit 9):** Low Voltage Region

1 = Active

0 = Inactive

**PV (Bit 8):** Precharge Voltage Region

1 = Active

0 = Inactive

**RSVD (Bits 7):** Reserved. Do not use.

**OT (Bit 6):** Overtemperature Region

1 = Active

0 = Inactive

**HT (Bit 5):** High Temperature Region

1 = Active

0 = Inactive

**STH (Bit 4):** Standard Temperature High Region

1 = Active

0 = Inactive

**RT (Bit 3):** Recommended Temperature Region

1 = Active

0 = Inactive

**STL (Bit 2):** Standard Temperature Low Region

1 = Active

0 = Inactive

**LT (Bit 1):** Low Temperature Region

1 = Active

0 = Inactive

**UT (Bit 0):** Undertemperature Region

1 = Active

0 = Inactive

#### 12.1.42 ManufacturerAccess() 0x0056 GaugingStatus

This command returns the *GaugingStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|      |      |             |     |     |       |      |      | 23   | 22   | 21   | 20         | 19   | 18 | 17   | 16  |
|------|------|-------------|-----|-----|-------|------|------|------|------|------|------------|------|----|------|-----|
|      |      |             |     |     |       |      |      | RSVD | RSVD | RSVD | OCV<br>FR  | LDMD | RX | QMax | VDQ |
| 15   | 14   | 13          | 12  | 11  | 10    | 9    | 8    | 7    | 6    | 5    | 4          | 3    | 2  | 1    | 0   |
| NSFM | RSVD | SLP<br>QMax | QEN | VOK | R_DIS | RSVD | REST | CF   | DSG  | EDV  | BAL_<br>EN | TC   | TD | FC   | FD  |

**RSVD (Bits 21 –23):** Reserved. Do not use.

**OCVFR (Bit 20):** Open Circuit Voltage in Flat Region (during RELAX)

1 = Detected

0 = Not Detected

**LDMD (Bit 19):** LOAD mode

1 = Constant Power

0 = Constant Current

**RX (Bit 18):** Resistance Update (toggles after every resistance update)

**QMax (Bit 17):** QMax Update (toggles after every QMax update)

**VDQ (Bit 16):** Discharge Qualified for Learning (opposite of the R\_DIS flag)

1 = Detected

0 = Not Detected

**NSFM (Bit 15):** Negative Scale Factor Mode

- 1 = Negative Ra Scaling Factor Detected
- 0 = Negative Ra Scaling Factor Not Detected

**RSVD (Bit 14):** Reserved. Do not use.

**SLPQMax (Bit 13):** OCV update in SLEEP mode

- 1 = Active. OCV reading in process
- 0 = Inactive. Completed OCV reading

**QEN (Bit 12):** Impedance Track Gauging (Ra and QMax updates are enabled.)

- 1 = Enabled
- 0 = Disabled

**VOK (Bit 11):** Voltages are OK for QMax update. This flag is updated at exit of the RELAX mode.

- 1 = A DOD is saved for next QMax update.
- 0 = No DOD saved and QMax update is not possible.

**R\_DIS (Bit 10):** Resistance Updates

- 1 = Disabled
- 0 = Enabled

**RSVD (Bit 9):** Reserved. Do not use.

**REST (Bit 8):** Rest

- 1 = OCV Reading Taken
- 0 = OCV Reading Not Taken or Not in RELAX

**CF (Bit 7):** Condition Flag

- 1 = *MaxError()* > Max Error Limit (Condition Cycle Needed)
- 0 = *MaxError()* < Max Error Limit (Condition Cycle Not Needed)

**DSG (Bit 6):** Discharge/Relax

- 1 = Charging Not Detected
- 0 = Charging Detected

**EDV (Bit 5):** End-of-Discharge Termination Voltage

- 1 = Termination voltage reached during discharge
- 0 = Termination voltage not reached, or not in DISCHARGE mode

**BAL\_EN (Bit 4):** Cell Balancing

- 1 = Cell balancing is possible if enabled.
- 0 = Cell balancing is not allowed.

**TC (Bit 3):** Terminate Charge

- 1 = Detected
- 0 = Not Detected

**TD (Bit 2):** Terminate Discharge

- 1 = Detected
- 0 = Not Detected

**FC (Bits 1):** Fully Charged

- 1 = Detected
- 0 = Not Detected

**FD (Bit 0):** Fully Discharged

- 1 = Detected
- 0 = Not Detected

### 12.1.43 ManufacturerAccess() 0x0057 ManufacturingStatus

This command returns the *ManufacturingStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

|          |         |       |        |          |        |        |         |
|----------|---------|-------|--------|----------|--------|--------|---------|
| 15       | 14      | 13    | 12     | 11       | 10     | 9      | 8       |
| CAL_TEST | LT_TEST | RSVD  | RSVD   | RSVD     | RSVD   | LED_EN | FUSE_EN |
| 7        | 6       | 5     | 4      | 3        | 2      | 1      | 0       |
| BBR_EN   | PF_EN   | LF_EN | FET_EN | GAUGE_EN | DSG_EN | CHG_EN | PCHG_EN |

**CAL\_TEST (Bit 15):** CALIBRATION mode

1 = Enabled

0 = Disabled

**LT\_TEST (Bit 14):** LIFETIME SPEED UP mode

1 = Enabled

0 = Disabled

**RSVD (Bits 13–10):** Reserved. Do not use.

**LED\_EN (Bit 9):** LED Display

1 = LED display is on.

0 = LED display is off.

**FUSE\_EN (Bit 8):** Fuse Action

1 = Enabled

0 = Disabled

**BBR\_EN (Bit 7):** Black Box Recorder

1 = Enabled

0 = Disabled

**PF\_EN (Bit 6):** Permanent Failure

1 = Enabled

0 = Disabled

**LF\_EN (Bit 5):** Lifetime Data Collection

1 = Enabled

0 = Disabled

**FET\_EN (Bit 4):** All FET Action

1 = Enabled

0 = Disabled

**GAUGE\_EN (Bit 3):** Gas Gauging

1 = Enabled

0 = Disabled

**DSG\_EN (Bit 2):** Discharge FET Test

1 = Discharge FET test activated

0 = Disabled

**CHG\_EN (Bit 1):** Charge FET Test

1 = Charge FET test activated

0 = Disabled

**PCHG\_EN (Bit 0):** Precharge FET Test



1 = Precharge FET test activated  
0 = Disabled

#### 12.1.44 ManufacturerAccess() 0x0058 AFE Register

This command returns the *AFERegister()* values on *ManufacturerBlockAccess()* or *ManufacturerData()*. These are the AFE hardware registers and are intended for internal debug use only.

| Status   | Condition                             |
|----------|---------------------------------------|
| Activate | 0x0058 to <i>ManufacturerAccess()</i> |

**Action:** Output AFE Register values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: AABBCDDDEEFFGGHHIIJJKKLLMMNNOOPPQQRRSSTTUU where:

| Value | Description                                                                                          |
|-------|------------------------------------------------------------------------------------------------------|
| AA    | AFE Interrupt Status. AFE Hardware interrupt status (for example, wake time, push-button, and so on) |
| BB    | AFE FET Status. AFE FET status (for example, CHG FET, DSG FET, PCHG FET, FUSE input, and so on)      |
| CC    | AFE RXIN. AFE I/O port input status                                                                  |
| DD    | AFE Latch Status. AFE protection latch status                                                        |
| EE    | AFE Interrupt Enable. AFE interrupt control settings                                                 |
| FF    | AFE Control. AFE FET control enable setting                                                          |
| GG    | AFE RXIEN. AFE I/O input enable settings                                                             |
| HH    | AFE RLOUT. AFE I/O pins output status                                                                |
| II    | AFE RHOUT. AFE I/O pins output status                                                                |
| JJ    | AFE RHINT. AFE I/O pins interrupt status                                                             |
| KK    | AFE Cell Balance. AFE cell balancing enable settings and status                                      |
| LL    | AFE ADC/CC Control. AFE ADC/CC Control settings                                                      |
| MM    | AFE ADC Mux Control. AFE ADC channel selections                                                      |
| NN    | AFE LED Control                                                                                      |
| OO    | AFE Control. AFE control on various HW based features                                                |
| PP    | AFE Timer Control. AFE comparator and timer control                                                  |
| QQ    | AFE Protection. AFE protection delay time control                                                    |
| RR    | AFE OCD. AFE OCD settings                                                                            |
| SS    | AFE SCC. AFE SCC settings                                                                            |
| TT    | AFE SCD1. AFE SCD1 settings                                                                          |
| UU    | AFE SCD2. AFE SCD2 settings                                                                          |

#### 12.1.45 ManufacturerAccess() 0x0060 Lifetime Data Block 1

This command returns the Lifetime Data with the following format:

aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIjjJkkKKllLLmmMMNNOOPPQQRRSS.

| Value | Description            |
|-------|------------------------|
| AAaa  | Cell 1 Max Voltage     |
| BBbb  | Cell 2 Max Voltage     |
| CCcc  | Cell 3 Max Voltage     |
| DDdd  | Cell 4 Max Voltage     |
| EEee  | Cell 1 Min Voltage     |
| FFff  | Cell 2 Min Voltage     |
| GGgg  | Cell 3 Min Voltage     |
| HHhh  | Cell 4 Min Voltage     |
| Iiii  | Max Delta Cell Voltage |
| JJjj  | Max Charge Current     |
| KKkk  | Max Discharge Current  |

| Value | Description         |
|-------|---------------------|
| LLll  | Max Avg Dsg Current |
| MMmm  | Max Avg Dsg Power   |
| NN    | Max Temp Cell       |
| OO    | Min Temp Cell       |
| PP    | Max Delta Cell temp |
| QQ    | Max Temp Int Sensor |
| RR    | Min Temp Int Sensor |
| SS    | Max Temp Fet        |

#### 12.1.46 ManufacturerAccess() 0x0061 Lifetime Data Block 2

This command returns the Lifetime Data with the following format:

AABBCCDDEEFFGGHH.

| Value | Description           |
|-------|-----------------------|
| AA    | No. of Shutdowns      |
| BB    | No. of Partial Resets |
| CC    | No. of Full Resets    |
| DD    | No. of WDT resets     |
| EE    | CB Time Cell 1        |
| FF    | CB Time Cell 2        |
| GG    | CB Time Cell 3        |
| HH    | CB Time Cell 4        |

#### 12.1.47 ManufacturerAccess() 0x0062 Lifetime Data Block 3

This command returns the Lifetime Data with the following format:

aaAAbbBBccCCddDDeeEEffFFggGGhhHH.

| Value | Description       |
|-------|-------------------|
| AAaa  | Total FW Runtime  |
| BBbb  | Time Spent in UT  |
| CCcc  | Time Spent in LT  |
| DDdd  | Time Spent in STL |
| EEee  | Time Spent in RT  |
| FFff  | Time Spent in STH |
| GGgg  | Time Spent in HT  |
| HHhh  | Time Spent in OT  |

#### 12.1.48 ManufacturerAccess() 0x0063 Lifetime Data Block 4

This command returns the Lifetime Data with the following format:

aaAAbbBBccCCddDDeeEEffFFggGGhhHHILLmmMMnnNNooOOppPP.

| Value | Description        |
|-------|--------------------|
| AAaa  | No. of COV Events  |
| BBbb  | Last COV Event     |
| CCcc  | No. of CUV Events  |
| DDdd  | Last CUV Event     |
| EEee  | No. of OCD1 Events |
| FFff  | Last OCD1 Event    |
| GGgg  | No. of OCD2 Events |

| Value | Description        |
|-------|--------------------|
| HHhh  | Last OCD2 Event    |
| IIii  | No. of OCC1 Events |
| JJjj  | Last OCC1 Event    |
| KKkk  | No. of OCC2 Events |
| LLll  | Last OCC2 Event    |
| MMmm  | No. of AOLD Events |
| NNnn  | Last AOLD Event    |
| OOoo  | No. of ASCD Events |
| PPpp  | Last ASCD Event    |

### 12.1.49 ManufacturerAccess() 0x0064 Lifetime Data Block 5

This command returns the Lifetime Data with the following format:

aaAAbbBBccCCddDDeeEEffFFggGGhhHHiILLmmMMnnNNooOOppPP.

| Value | Description            |
|-------|------------------------|
| AAaa  | No. of ASCC Events     |
| BBbb  | Last ASCC Event        |
| CCcc  | No. of OTC Events      |
| DDdd  | Last OTC Event         |
| EEee  | No. of OTD Events      |
| FFff  | Last OTD Event         |
| GGgg  | No. of OTF Events      |
| HHhh  | Last OTF Event         |
| IIii  | No. Valid Charge Term  |
| JJjj  | Last Valid Charge Term |
| KKkk  | No. of Qmax Updates    |
| LLll  | Last Qmax Update       |
| MMmm  | No. of Ra Updates      |
| NNnn  | Last Ra Update         |
| OOoo  | No. of Ra Disable      |
| PPpp  | Last Ra Disable        |

### 12.1.50 ManufacturerAccess() 0x0070 ManufacturerInfo

This command returns ManufacturerInfo on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                                                                                                            |
|----------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0070 to <i>ManufacturerAccess()</i> | Output 32 bytes of ManufacturerInfo on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> in the following format: AABCCDDEEFFGGHHIIJJKLLMMNN<br>OOPPQQRRSSTTUUVVWWXXVVZZ112233 445566 |

### 12.1.51 ManufacturerAccess() 0x0071 DAStatus1

This command returns the Cell Voltages, Pack Voltage, Bat Voltage, Cell Currents, Cell Powers, Power, and Average Power on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0071 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 32 bytes of data on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGGhhHHiIlJjJkkKKiILLmmMMnnNNooOOppPP where:

| Value | Description                                                                                                               | Unit |
|-------|---------------------------------------------------------------------------------------------------------------------------|------|
| AAaa  | Cell Voltage 1                                                                                                            | mV   |
| BBbb  | Cell Voltage 2                                                                                                            | mV   |
| CCcc  | Cell Voltage 3                                                                                                            | mV   |
| DDdd  | Cell Voltage 4                                                                                                            | mV   |
| EEee  | BAT Voltage. Voltage at the BAT pin. This is different than <i>Voltage()</i> , which is the sum of all the cell voltages. | mV   |
| FFff  | PACK Voltage                                                                                                              | mV   |
| GGgg  | Cell Current 1. Simultaneous current measured during Cell Voltage1 measurement                                            | mA   |
| HHhh  | Cell Current 2. Simultaneous current measured during Cell Voltage2 measurement                                            | mA   |
| IIii  | Cell Current 3. Simultaneous current measured during Cell Voltage3 measurement                                            | mA   |
| JJjj  | Cell Current 4. Simultaneous current measured during Cell Voltage 4 measurement                                           | mA   |
| KKkk  | Cell Power 1. Calculated using Cell Voltage1 and Cell Current 1 data                                                      | mA   |
| LLll  | Cell Power 2. Calculated using Cell Voltage2 and Cell Current 2 data                                                      | cW   |
| MMmm  | Cell Power 3. Calculated using Cell Voltage3 and Cell Current 3 data                                                      | cW   |
| NNnn  | Cell Power 4. Calculated using Cell Voltage4 and Cell Current 4 data                                                      | cW   |
| OOoo  | Power calculated by <i>Voltage()</i> × <i>Current()</i>                                                                   | cW   |
| PPpp  | Average Power                                                                                                             | cW   |

### 12.1.52 ManufacturerAccess() 0x0072 DStatus2

This command returns the internal temp sensor, TS1, TS2, TS3, TS4, Cell Temp, and FETTemp on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0072 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 14 bytes of temperature data values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGG where:

| Value | Description      | Unit  |
|-------|------------------|-------|
| AAaa  | Int Temperature  | 0.1°K |
| BBbb  | TS1 Temperature  | 0.1°K |
| CCcc  | TS2 Temperature  | 0.1°K |
| DDdd  | TS3 Temperature  | 0.1°K |
| EEee  | TS4 Temperature  | 0.1°K |
| FFff  | Cell Temperature | 0.1°K |
| GGgg  | FET Temperature  | 0.1°K |

### 12.1.53 ManufacturerAccess() 0x0073 GaugeStatus1

This command instructs the device to return Impedance Track related gauging information on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0073 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 32 bytes of IT data values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGGhhHHIIllJJkkKKIILLmmMMnnNNooOOppPPqqQQ where:

| Value | Description                                                                                                                                              | Unit |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| AAaa  | True Rem Q. True remaining capacity in mAh from IT simulation before any filtering or smoothing function. This value can be negative or higher than FCC. | mAh  |
| BBbb  | True Rem E. True remaining energy in cWh from IT simulation before any filtering or smoothing function. This value can be negative or higher than FCC.   | cWh  |

| Value | Description                                                                                              | Unit               |
|-------|----------------------------------------------------------------------------------------------------------|--------------------|
| CCcc  | Initial Q. Initial capacity calculated from IT simulation                                                | mAh                |
| DDdd  | Initial E. Initial energy calculated from IT simulation                                                  | cWh                |
| EEee  | True FCC Q. True full charge capacity from IT simulation without the effects of any smoothing function   | mAh                |
| FFff  | True FCC E. True full charge energy from IT simulation without the effects of any smoothing function     | cWh                |
| GGgg  | T <sub>sim</sub> . Temperature during the last simulation run.                                           | 0.1°K              |
| HHhh  | T <sub>ambient</sub> . Current assumed ambient temperature used by the IT algorithm for thermal modeling | 0.1°K              |
| IIii  | RaScale 0. Ra table scaling factor of Cell 1                                                             | —                  |
| JJjj  | RaScale 1. Ra table scaling factor of Cell 2                                                             | —                  |
| KKkk  | RaScale 2. Ra table scaling factor of Cell 3                                                             | —                  |
| LLll  | RaScale 3. Ra table scaling factor of Cell 4                                                             | —                  |
| MMmm  | CompRes 0. Last temperature compensated Resistance of Cell 1                                             | 2 <sup>-10</sup> Ω |
| NNnn  | CompRes 1. Last temperature compensated Resistance of Cell 2                                             | 2 <sup>-10</sup> Ω |
| OOoo  | CompRes 2. Last temperature compensated Resistance of Cell 3                                             | 2 <sup>-10</sup> Ω |
| PPpp  | CompRes 3. Last temperature compensated Resistance of Cell 4                                             | 2 <sup>-10</sup> Ω |

### 12.1.54 ManufacturerAccess() 0x0074 GaugeStatus2

This command instructs the device to return Impedance Track related gauging information on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0074 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 32 bytes of IT data values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: AABBCDDDEEFFggGGhhHHIIjjJJkkKKIILLmmMMnnNNooOOppPPqqQQrrRRssSS where:

| Value    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 | Unit |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| AA       | Pack Grid. Active pack grid point (minimum of CellGrid0 to Cell Grid3). This data is only valid during DISCHARGE mode when [R_DIS] = 0. If [R_DIS] = 1 or not discharging, this value is not updated.                                                                                                                                                                                                                                       | —    |
| BB       | BB: LStatus—Learned status of resistance table<br>Bit 3   Bit 2   Bit 1   Bit 0<br>QMax   ITEN   CF1   CF0<br>CF1, CF0: QMax Status<br>0,0 = Battery OK<br>0,1 = QMax is first updated in learning cycle.<br>1,0 = QMax and resistance table updated in learning cycle<br>ITEN: IT enable<br>0 = IT disabled<br>1 = IT enabled<br>QMax: QMax update in field<br>0 = QMax has not been updated in the field<br>1 = QMax updated in the field | —    |
| CC       | Cell Grid 0. Active grid point of Cell 1. This data is only valid during DISCHARGE mode when [R_DIS] = 0. If [R_DIS] = 1 or not discharging, this value is not updated.                                                                                                                                                                                                                                                                     | —    |
| DD       | Cell Grid 1. Active grid point of Cell 2. This data is only valid during DISCHARGE mode when [R_DIS] = 0. If [R_DIS] = 1 or not discharging, this value is not updated.                                                                                                                                                                                                                                                                     | —    |
| EE       | Cell Grid 2. Active grid point of Cell 3. This data is only valid during DISCHARGE mode when [R_DIS] = 0. If [R_DIS] = 1 or not discharging, this value is not updated.                                                                                                                                                                                                                                                                     | —    |
| FF       | Cell Grid 3. Active grid point of Cell 4. This data is only valid during DISCHARGE mode when [R_DIS] = 0. If [R_DIS] = 1 or not discharging, this value is not updated.                                                                                                                                                                                                                                                                     | —    |
| GGggHHhh | State Time. Time past since last state change (DISCHARGE, CHARGE, REST)                                                                                                                                                                                                                                                                                                                                                                     | s    |
| IIii     | DOD0_0. Depth of discharge for Cell 1                                                                                                                                                                                                                                                                                                                                                                                                       | —    |
| JJjj     | DOD0_1. Depth of discharge for Cell 2                                                                                                                                                                                                                                                                                                                                                                                                       | —    |
| KKkk     | DOD0_2. Depth of discharge for Cell 3                                                                                                                                                                                                                                                                                                                                                                                                       | —    |
| LLll     | DOD0_3. Depth of discharge for Cell 4                                                                                                                                                                                                                                                                                                                                                                                                       | —    |
| MMmm     | DOD0 Passed Q. Passed capacity since the last DOD0 update                                                                                                                                                                                                                                                                                                                                                                                   | mAh  |

| Value | Description                                             | Unit  |
|-------|---------------------------------------------------------|-------|
| NNnn  | DOD0 Passed E. Passed energy since last DOD0 update     | cWh   |
| OOoo  | DOD0 Time. Time passed since the last DOD0 update       | hr/16 |
| PPpp  | DODEOC 0. Depth of discharge at end of charge of Cell 1 | —     |
| QQqq  | DODEOC 1. Depth of discharge at end of charge of Cell 2 | —     |
| RRrr  | DODEOC 2. Depth of discharge at end of charge of Cell 3 | —     |
| SSss  | DODEOC 3. Depth of discharge at end of charge of Cell 4 | —     |

### 12.1.55 ManufacturerAccess() 0x0075 GaugeStatus3

This command instructs the device to return Impedance Track related gauging information on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0075 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 24 bytes of IT data values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAAbbBBccCCddDDeeEEffFFggGGhhHHIiIlJjJkKkKILL where:

| Value | Description                                                                                                | Unit  |
|-------|------------------------------------------------------------------------------------------------------------|-------|
| AAaa  | QMax 0. QMax of Cell 1                                                                                     | mAh   |
| BBbb  | QMax 1. QMax of Cell 2                                                                                     | mAh   |
| CCcc  | QMax 2. QMax of Cell 3                                                                                     | mAh   |
| DDdd  | QMax 3. QMax of Cell 4                                                                                     | mAh   |
| EEee  | QMax DOD0_0. DOD0 saved to be used for next QMax update of Cell 1. The value is only valid when [VOK] = 1. | —     |
| FFff  | QMax DOD0_1. DOD0 saved to be used for next QMax update of Cell 2. The value is only valid when [VOK] = 1. | —     |
| GGgg  | QMax DOD0_2. DOD0 saved to be used for next QMax update of Cell 3. The value is only valid when [VOK] = 1. | —     |
| HHhh  | QMax DOD0_3. DOD0 saved to be used for next QMax update of Cell 4. The value is only valid when [VOK] = 1. | —     |
| Illi  | QMax Passed Q. Pass capacity since last QMax DOD value is saved.                                           | mAh   |
| JJjj  | QMax Time. Time passed since last QMax DOD value is saved.                                                 | hr/16 |
| KKkk  | Temp k. Thermal Model temperature factor                                                                   | —     |
| LLll  | Temp a. Thermal Model temperature                                                                          | —     |

### 12.1.56 ManufacturerAccess() 0x0076 CBStatus

This command instructs the device to return cell balance time information on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0076 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 8 bytes of IT data values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAAbbBBccCCddDD where:

| Value | Description                                                   | Unit |
|-------|---------------------------------------------------------------|------|
| AAaa  | Cell Balance Time 0. Calculated cell balancing time of Cell 1 | s    |
| BBbb  | Cell Balance Time 1. Calculated cell balancing time of Cell 2 | s    |
| CCcc  | Cell Balance Time 2. Calculated cell balancing time of Cell 3 | s    |
| DDdd  | Cell Balance Time 3. Calculated cell balancing time of Cell 4 | s    |

### 12.1.57 ManufacturerAccess() 0x0077 State-of-Health

This command returns the state-of-health FCC in mAh and energy in cWh with the following format:

aaAAbbBB.

| Value | Description            | Unit |
|-------|------------------------|------|
| AAaa  | State-of-Health FCC    | mAh  |
| BBbb  | State-of-Health energy | cWh  |

### 12.1.58 ManufacturerAccess() 0x0078 FilterCapacity

This command instructs the device to return the filtered remaining capacity and full charge capacity even if **[SMOOTH]** = 0 on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                                                                 |
|----------|---------------------------------------------------------------------------|
| Activate | 0x0078 to <i>ManufacturerBlockAccess()</i> or <i>ManufacturerAccess()</i> |

**Action:** Output 8 bytes of IT data values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAAbbBBccCCddDD where:

| Value | Description                   | Unit |
|-------|-------------------------------|------|
| AAaa  | Filtered remaining capacity   | mAh  |
| BBbb  | Filtered remaining energy     | mWh  |
| CCcc  | Filtered full charge capacity | mAh  |
| DDdd  | Filtered full charge energy   | mWh  |

### 12.1.59 ManufacturerAccess() 0x0F00 ROM Mode

This command sends the device into ROM mode in preparation for firmware re-programming. To enter ROM mode, the device must be in FULL ACCESS mode. To return from ROM mode to FW mode, issue the SMBus command 0x08.

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**NOTE:** Command 0x0033 also puts the device in ROM mode (for backwards compatibility with the bq30zxy device).

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### 12.1.60 0x4000–0x5FFF Data Flash Access()

Accessing data flash (DF) is only supported by the *ManufacturerBlockAccess()* by addressing the physical address.

To write to the DF, send the starting address, followed by the DF data block. The DF data block is the intended revised DF data to be updated to DF. The size of the DF data block ranges from 1 byte to 32 bytes. All individual data must be sent in Little Endian.

Write to DF example:

Assuming: data1 locates at address 0x4000 and data2 locates at address 0x4002.

Both data1 and data2 are U2 type.

To update data1 and data2, send an SMBus block write with command = 0x44

block = starting address + DF data block

= 0x00 + 0x40 + data1\_LowByte + data1\_HighByte + data2\_LowByte + data2\_HighByte

To read the DF, send an SMBus block write to the *ManufacturerBlockAccess()*, followed by the starting address, then send an SMBus block read to the *ManufacturerBlockAccess()*. The return data contains the starting address followed by 32 bytes of DF data in Little Endian.

Read from DF example:

Taking the same assuming from the read DF example, to read DF,

a. Send SMBus write block with command 0x44, block = 0x00 + 0x40

b. Send SMBus read block with command 0x44

The returned block = a starting address + 32 bytes of DF data

= 0x00 + 0x40 + data1\_LowByte + data1\_HighByte + data2\_LowByte + data2\_HighByte....  
data32\_LowByte + data32\_HighByte

The gauge supports an auto-increment on the address during a DF read. This greatly reduces the time required to read out the entire DF. Continue with the read from the DF example. If another SMBus read block is sent with command 0x44, the gauge returns another 32 bytes of DF data, starting with address 0x4020.

### 12.1.61 ManufacturerAccess() 0xF080 Exit Calibration Output Mode

This command stops the output of calibration data to the *ManufacturerBlockAccess()* or *ManufacturerData()* command. Any other MAC command sent to the gauge will also stop the output of the calibration data.

| Status   | Condition                                                                                                         | Action                                                                                            |
|----------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Activate | <i>ManufacturerBlockAccess()</i> OR<br><i>ManufacturerData()</i> = 1 AND<br>0xF080 to <i>ManufacturerAccess()</i> | Stop output of ADC or CC data on <i>ManufacturerBlockAccess()</i> or<br><i>ManufacturerData()</i> |

### 12.1.62 ManufacturerAccess() 0xF081 Output CCADC Cal

This command instructs the device to output the raw values for calibration purposes on *ManufacturerBlockAccess()* or *ManufacturerData()*. All values are updated every 250 ms and the format of each value is 2's complement, MSB first.

| Status  | Condition                                                                          |
|---------|------------------------------------------------------------------------------------|
| Disable | <i>ManufacturingStatus()[CAL]</i> = 1 AND<br>0xF080 to <i>ManufacturerAccess()</i> |

**Action:** *OperationStatus()[CAL]* = 0, *[CAL\_OFFSET]* = 0

Stop output of ADC and CC data on *ManufacturerBlockAccess()* or *ManufacturerData()*

| Status | Condition                             |
|--------|---------------------------------------|
| Enable | 0xF081 to <i>ManufacturerAccess()</i> |

**Action:** *OperationStatus()[CAL]* = 1, *[CAL\_OFFSET]* = 0

Outputs the raw CC and AD values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the format of ZZYYaaAAAbbBBccCCddDDeeEEffFFggGGhhhHHiilJJjkkKKK:

| Value | Description                                                                                      |
|-------|--------------------------------------------------------------------------------------------------|
| ZZ    | Rolling 8-bit counter, increments when values are refreshed.                                     |
| YY    | Status, 1 when <i>ManufacturerAccess()</i> = 0xF081, 2 when <i>ManufacturerAccess()</i> = 0xF082 |
| AAaa  | Current (coulomb counter)                                                                        |
| BBbb  | Cell Voltage 1                                                                                   |
| CCcc  | Cell Voltage 2                                                                                   |
| DDdd  | Cell Voltage 3                                                                                   |
| EEee  | Cell Voltage 4                                                                                   |
| FFff  | PACK Voltage                                                                                     |
| GGgg  | BAT Voltage                                                                                      |
| HHhh  | Cell Current 1                                                                                   |
| IIii  | Cell Current 2                                                                                   |
| JJjj  | Cell Current 3                                                                                   |



| Value | Description    |
|-------|----------------|
| KKkk  | Cell Current 4 |

### 12.1.63 *ManufacturerAccess()* 0xF082 Output Shorted CCADC Cal

This command instructs the device to output the raw values for calibration purposes on *ManufacturerBlockAccess()* or *ManufacturerData()*. All values are updated every 250 ms and the format of each value is 2's complement, MSB first. This mode includes an internal short on the coulomb counter inputs for measuring offset.

| Status  | Condition                                                                          |
|---------|------------------------------------------------------------------------------------|
| Disable | <i>ManufacturingStatus()[CAL] = 1</i> AND<br>0xF080 to <i>ManufacturerAccess()</i> |

**Action:** *OperationStatus()[CAL] = 0*, *[CAL\_OFFSET] = 0*

Stop output of ADC and CC data on *ManufacturerBlockAccess()* or *ManufacturerData()*

| Status | Condition                             |
|--------|---------------------------------------|
| Enable | 0xF081 to <i>ManufacturerAccess()</i> |

**Action:** *OperationStatus()[CAL] = 1*, *[CAL\_OFFSET] = 1*

Outputs the raw CC and AD values on *ManufacturerBlockAccess()* or *ManufacturerData()* in the format of ZZZYaaAAbbBBccCCddDDeeEEffFFggGGhhHHiilJjJJkkKK:

| Value | Description                                                                                      |
|-------|--------------------------------------------------------------------------------------------------|
| ZZ    | Rolling 8-bit counter, increments when values are refreshed.                                     |
| YY    | Status, 1 when <i>ManufacturerAccess()</i> = 0xF081, 2 when <i>ManufacturerAccess()</i> = 0xF082 |
| AAaa  | Current (coulomb counter)                                                                        |
| BBbb  | Cell Voltage 1                                                                                   |
| CCcc  | Cell Voltage 2                                                                                   |
| DDdd  | Cell Voltage 3                                                                                   |
| EEee  | Cell Voltage 4                                                                                   |
| FFff  | PACK Voltage                                                                                     |
| GGgg  | BAT Voltage                                                                                      |
| HHhh  | Cell Current 1                                                                                   |
| IIii  | Cell Current 2                                                                                   |
| JJjj  | Cell Current 3                                                                                   |
| KKkk  | Cell Current 4                                                                                   |

## 12.2 0x01 *RemainingCapacityAlarm()*

This read/write word function sets a low capacity alarm threshold for the cell stack.

| SBS<br>Cmd | Name                            | Access |    |    | Proto-<br>col | Type | Min | Max | Default | Unit          |
|------------|---------------------------------|--------|----|----|---------------|------|-----|-----|---------|---------------|
|            |                                 | SE     | US | FA |               |      |     |     |         |               |
| 0x01       | <i>RemainingCapacityAlarm()</i> | R/W    |    |    | Word          | U2   | 0   | 700 | 300     | mAh<br>10 mWh |

**NOTE:** If *BatteryMode()[CAPM] = 0*, then the data reports in mAh.

If *BatteryMode()[CAPM] = 1*, then the data reports in 10 mWh.

## 12.3 0x02 RemainingTimeAlarm()

This read/write word function sets a low remaining time-to-fully discharge alarm threshold for the cell stack.

| SBS<br>Cmd | Name                        | Access |    |    | Proto-<br>col | Type | Min | Max | Default | Unit |
|------------|-----------------------------|--------|----|----|---------------|------|-----|-----|---------|------|
|            |                             | SE     | US | FA |               |      |     |     |         |      |
| 0x02       | <i>RemainingTimeAlarm()</i> | R/W    |    |    | Word          | U2   | 0   | 30  | 10      | min  |

## 12.4 0x03 BatteryMode()

This read/write word function sets various battery operating mode options.

| SBS<br>Cmd | Name                 | Access |    |    | Protocol | Type | Min    | Max    | Unit |
|------------|----------------------|--------|----|----|----------|------|--------|--------|------|
|            |                      | SE     | US | FA |          |      |        |        |      |
| 0x03       | <i>BatteryMode()</i> | R/W    |    |    | Word     | H2   | 0x0000 | 0xFFFF | —    |

|      |      |      |      |      |      |     |     |
|------|------|------|------|------|------|-----|-----|
| 15   | 14   | 13   | 12   | 11   | 10   | 9   | 8   |
| CAPM | CHGM | AM   | RSVD | RSVD | RSVD | PB  | CC  |
| 7    | 6    | 5    | 4    | 3    | 2    | 1   | 0   |
| CF   | RSVD | RSVD | RSVD | RSVD | RSVD | PBS | ICC |

### **CAPM (Bit 15):** CAPACITY Mode (R/W)

- 0 = Report in mA or mAh (default)
- 1 = Report in 10 mW or 10 mWh

### **CHGM (Bit 14):** CHARGER Mode (R/W)

- 0 = Enable *ChargingVoltage()* and *ChargingCurrent()* broadcasts to host and smart battery charger
- 1 = Disable *ChargingVoltage()* and *ChargingCurrent()* broadcasts to host and smart battery charger (default)

### **AM (Bit 13):** ALARM Mode (R/W)

- 0 = Enable AlarmWarning broadcasts to host and smart battery charger (default)
- 1 = Disable Alarm Warning broadcasts to host and smart battery charger

### **RSVD (Bits 12–10):** Reserved. Do not use.

### **PB (Bit 9):** Primary Battery

- 0 = Battery operating in its secondary role (default)
- 1 = Battery operating in its primary role

### **CC (Bit 8):** Charge Controller Enabled (R/W)

- 0 = Internal charge controller disabled (default)
- 1 = Internal charge controller enabled

### **CF (Bit 7):** Condition Flag (R)

- 0 = Battery OK
- 1 = Conditioning cycle requested

### **RSVD (Bits 6–2):** Reserved. Do not use.

### **PBS (Bit 1):** Primary Battery Support (R)

- 0 = Function not supported (default)
- 1 = Primary or Secondary Battery Support

**ICC (Bit 0):** Internal Charge Controller (R)  
0 = Function not supported (default)  
1 = Function supported

## 12.5 0x04 *AtRate()*

This read/write word function sets the value used in calculating *AtRateTimeToFull()* and *AtRateTimeToEmpty()*.

| SBS Cmd | Name            | Access |    |    | Protocol | Type | Min    | Max   | Default | Unit        |
|---------|-----------------|--------|----|----|----------|------|--------|-------|---------|-------------|
|         |                 | SE     | US | FA |          |      |        |       |         |             |
| 0x04    | <i>AtRate()</i> | R/W    |    |    | Word     | I2   | –32768 | 32767 | 0       | mA<br>10 mW |

**NOTE:** If *BatteryMode()[CAPM]* = 0, then the data reports in mA.  
If *BatteryMode()[CAPM]* = 1, then the data reports in 10 mW.

## 12.6 0x05 *AtRateTimeToFull()*

This word read function returns the remaining time-to-fully charge the battery stack.

| SBS Cmd | Name                      | Access |    |    | Protocol | Type | Min | Max   | Unit |
|---------|---------------------------|--------|----|----|----------|------|-----|-------|------|
|         |                           | SE     | US | FA |          |      |     |       |      |
| 0x05    | <i>AtRateTimeToFull()</i> | R      |    |    | Word     | U2   | 0   | 65535 | min  |

**NOTE:** 65535 indicates not being charged.

## 12.7 0x06 *AtRateTimeToEmpty()*

This word read function returns the remaining time-to-fully discharge the battery stack.

| SBS Cmd | Name                       | Access |    |    | Protocol | Type | Min | Max   | Unit |
|---------|----------------------------|--------|----|----|----------|------|-----|-------|------|
|         |                            | SE     | US | FA |          |      |     |       |      |
| 0x06    | <i>AtRateTimeToEmpty()</i> | R      |    |    | Word     | U2   | 0   | 65535 | min  |

**NOTE:** 65535 indicates not being charged.

## 12.8 0x07 *AtRateOK()*

This read-word function returns a Boolean value that indicates whether the battery can deliver *AtRate()* for at least 10 seconds.

| SBS Cmd | Name              | Access |    |    | Protocol | Type | Min | Max   | Unit |
|---------|-------------------|--------|----|----|----------|------|-----|-------|------|
|         |                   | SE     | US | FA |          |      |     |       |      |
| 0x07    | <i>AtRateOK()</i> | R      |    |    | Word     | U2   | 0   | 65535 | —    |

**NOTE:** 0 = False. The gauge *cannot* deliver energy for 10 s, based on the discharge rate indicated in *AtRate()*.

> than 0 = True. The gauge *can* deliver energy for 10 s, based on the discharge rate indicated in *AtRate()*.

## 12.9 0x08 Temperature()

This read-word function returns the temperature in units 0.1°K.

| SBS<br>Cmd | Name                 | Access |    |    | Protocol | Type | Min | Max   | Unit  |
|------------|----------------------|--------|----|----|----------|------|-----|-------|-------|
|            |                      | SE     | US | FA |          |      |     |       |       |
| 0x08       | <i>Temperature()</i> | R      |    |    | Word     | U2   | 0   | 65535 | 0.1°K |

## 12.10 0x09 Voltage()

This read-word function returns the sum of the measured cell voltages.

| SBS<br>Cmd | Name             | Access |    |    | Protocol | Type | Min | Max   | Unit |
|------------|------------------|--------|----|----|----------|------|-----|-------|------|
|            |                  | SE     | US | FA |          |      |     |       |      |
| 0x09       | <i>Voltage()</i> | R      |    |    | Word     | U2   | 0   | 65535 | mV   |

## 12.11 0x0A Current()

This read-word function returns the measured current from the coulomb counter.

| SBS<br>Cmd | Name             | Access |    |    | Protocol | Type | Min    | Max   | Unit |
|------------|------------------|--------|----|----|----------|------|--------|-------|------|
|            |                  | SE     | US | FA |          |      |        |       |      |
| 0x0A       | <i>Current()</i> | R      |    |    | Word     | I2   | –32767 | 32768 | mA   |

## 12.12 0x0B AverageCurrent()

| SBS<br>Cmd | Name                    | Access |    |    | Protocol | Type | Min    | Max   | Unit |
|------------|-------------------------|--------|----|----|----------|------|--------|-------|------|
|            |                         | SE     | US | FA |          |      |        |       |      |
| 0x0B       | <i>AverageCurrent()</i> | R      |    |    | Word     | I2   | –32767 | 32768 | mA   |

## 12.13 0x0C MaxError()

This read-word function returns the expected margin of error, in %, in the state-of-charge calculation with a range of 1 to 100%.

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min | Max | Unit |
|------------|-------------------|--------|----|----|----------|------|-----|-----|------|
|            |                   | SE     | US | FA |          |      |     |     |      |
| 0x0C       | <i>MaxError()</i> | R      |    |    | Word     | U1   | 0   | 100 | %    |

| Condition                                                       | Action                               |
|-----------------------------------------------------------------|--------------------------------------|
| Full device reset                                               | <i>MaxError()</i> = 100%             |
| RA-table only updated                                           | <i>MaxError()</i> = 5%               |
| QMax only updated                                               | <i>MaxError()</i> = 3%               |
| RA-table and QMax updated                                       | <i>MaxError()</i> = 1%               |
| Each <i>CycleCount()</i> increment after last valid QMax update | <i>MaxError()</i> increment by 0.05% |

| Condition                                                                                               | Action                                |
|---------------------------------------------------------------------------------------------------------|---------------------------------------|
| The <b>Configuration:Max Error Time Cycle Equivalent</b> period passed since the last valid QMax update | <i>MaxError()</i> increment by 0.05%. |

## 12.14 0x0D *RelativeStateOfCharge()*

This read-word function returns the predicted remaining battery capacity as a percentage of *FullChargeCapacity()*.

| SBS<br>Cmd | Name                           | Access |    |    | Protocol | Type | Min | Max | Unit |
|------------|--------------------------------|--------|----|----|----------|------|-----|-----|------|
|            |                                | SE     | US | FA |          |      |     |     |      |
| 0x0D       | <i>RelativeStateOfCharge()</i> | R      |    |    | Word     | U1   | 0   | 100 | %    |

## 12.15 0x0E *AbsoluteStateOfCharge()*

This read-word function returns the predicted remaining battery capacity as a percentage.

| SBS<br>Cmd | Name                           | Access |    |    | Protocol | Type | Min | Max | Unit |
|------------|--------------------------------|--------|----|----|----------|------|-----|-----|------|
|            |                                | SE     | US | FA |          |      |     |     |      |
| 0x0E       | <i>AbsoluteStateOfCharge()</i> | R      |    |    | Word     | U1   | 0   | 100 | %    |

## 12.16 0x0F *RemainingCapacity()*

This read-word function returns the predicted remaining battery capacity.

| SBS<br>Cmd | Name                       | Access |    |    | Protocol | Type | Min | Max   | Unit   |
|------------|----------------------------|--------|----|----|----------|------|-----|-------|--------|
|            |                            | SE     | US | FA |          |      |     |       |        |
| 0x0F       | <i>RemainingCapacity()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | mAh    |
|            |                            |        |    |    |          |      |     |       | 10 mWh |

---

**NOTE:** If *BatteryMode()[CAPM]* = 0, then the data reports in mAh.

If *BatteryMode()[CAPM]* = 1, then the data reports in 10 mWh.

---

## 12.17 0x10 *FullChargeCapacity()*

This read-word function returns the predicted battery capacity when fully charged.

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max   | Unit   |
|------------|-----------------------------|--------|----|----|----------|------|-----|-------|--------|
|            |                             | SE     | US | FA |          |      |     |       |        |
| 0x10       | <i>FullChargeCapacity()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | mAh    |
|            |                             |        |    |    |          |      |     |       | 10 mWh |

---

**NOTE:** If *BatteryMode()[CAPM]* = 0, then the data reports in mAh.

If *BatteryMode()[CAPM]* = 1, then the data reports in 10 mWh.

---

## 12.18 0x11 *RunTimeToEmpty()*

This read-word function returns the predicted remaining battery capacity based on the present rate of discharge.

| SBS<br>Cmd | Name                    | Access |    |    | Protocol | Type | Min | Max   | Unit |
|------------|-------------------------|--------|----|----|----------|------|-----|-------|------|
|            |                         | SE     | US | FA |          |      |     |       |      |
| 0x11       | <i>RunTimeToEmpty()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | min  |

---

**NOTE:** 65535 = Battery is not being discharged.

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## 12.19 0x12 *AverageTimeToEmpty()*

This read-word function returns the predicted remaining battery capacity based on *AverageCurrent()*.

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max   | Unit |
|------------|-----------------------------|--------|----|----|----------|------|-----|-------|------|
|            |                             | SE     | US | FA |          |      |     |       |      |
| 0x12       | <i>AverageTimeToEmpty()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | min  |

---

**NOTE:** 65535 = Battery is not being discharged.

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## 12.20 0x13 *AverageTimeToFull()*

This read-word function returns the predicted time-to-full charge based on *AverageCurrent()*.

| SBS<br>Cmd | Name                       | Access |    |    | Protocol | Type | Min | Max   | Unit |
|------------|----------------------------|--------|----|----|----------|------|-----|-------|------|
|            |                            | SE     | US | FA |          |      |     |       |      |
| 0x13       | <i>AverageTimeToFull()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | min  |

---

**NOTE:** 65535 = Battery is not being discharged.

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## 12.21 0x14 *ChargingCurrent()*

This read-word function returns the desired charging current.

| SBS<br>Cmd | Name                     | Access |    |    | Protocol | Type | Min | Max   | Unit |
|------------|--------------------------|--------|----|----|----------|------|-----|-------|------|
|            |                          | SE     | US | FA |          |      |     |       |      |
| 0x14       | <i>ChargingCurrent()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | mA   |

---

**NOTE:** 65535 = Request maximum current

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## 12.22 0x15 *ChargingVoltage()*

This read-word function returns the desired charging voltage.

| SBS<br>Cmd | Name                     | Access |    |    | Protocol | Type | Min | Max   | Unit |
|------------|--------------------------|--------|----|----|----------|------|-----|-------|------|
|            |                          | SE     | US | FA |          |      |     |       |      |
| 0x15       | <i>ChargingVoltage()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | mV   |

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**NOTE:** 65535 = Request maximum voltage

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## 12.23 0x16 *BatteryStatus()*

This read-word function returns various battery status information.

| SBS<br>Cmd | Name                   | Access |    |    | Protocol | Type | Min | Max |
|------------|------------------------|--------|----|----|----------|------|-----|-----|
|            |                        | SE     | US | FA |          |      |     |     |
| 0x16       | <i>BatteryStatus()</i> | R      | R  | R  | Word     | H2   | —   | —   |

|      |     |      |     |     |      |     |     |
|------|-----|------|-----|-----|------|-----|-----|
| 15   | 14  | 13   | 12  | 11  | 10   | 9   | 8   |
| OCA  | TCA | RSVD | OTA | TDA | RSVD | RCA | RTA |
| 7    | 6   | 5    | 4   | 3   | 2    | 1   | 0   |
| INIT | DSG | FC   | FD  | EC3 | EC2  | EC1 | EC0 |

**OCA (Bit 15):** Overcharged Alarm

1 = Detected

0 = Not Detected

**TCA (Bit 14):** Terminate Charge Alarm

1 = Detected

0 = Not Detected

**RSVD (Bit 13):** Undefined

**OTA (Bit 12):** Overtemperature Alarm

1 = Detected

0 = Not Detected

**TDA (Bit 11):** Terminate Discharge Alarm

1 = Detected

0 = Not Detected

**RSVD (Bit 10):** Undefined

**RCA (Bit 9):** Remaining Capacity Alarm

1 = *RemainingCapacity()* < *RemainingCapacityAlarm()*

0 = *RemainingCapacity()* ≥ *RemainingCapacityAlarm()*
**RTA (Bit 8):** Remaining Time Alarm

1 = *AverageTimeToEmpty()* < *RemainingTimeAlarm()*

0 = *AverageTimeToEmpty()* ≥ *RemainingTimeAlarm()*
**INIT (Bit 7):** Initialization

1 = Gauge initialization is complete.

0 = Initialization is in progress.

**DSG (Bit 6):** Discharging or Relax

1 = Battery is in DISCHARGE or RELAX mode.

0 = Battery is in CHARGE mode.

**FC (Bit 5):** Fully Charged

1 = Battery fully charged when *GaugingStatus()[FC]* = 1

0 = Battery not fully charged

**FD (Bit 4):** Fully Discharged

1 = Battery fully depleted

0 = Battery not depleted

**EC3,EC2,EC1,EC0 (Bits 3–0):** Error Code

0x0 = OK

0x1 = Busy

0x2 = Reserved Command  
 0x3 = Unsupported Command  
 0x4 = AccessDenied  
 0x5 = Overflow/Underflow  
 0x6 = BadSize  
 0x7 = UnknownError

## 12.24 0x17 CycleCount()

This read-word function returns the number of discharge cycles the battery has experienced. The default value is stored in the data flash value **Cycle Count**, which is updated in runtime.

| SBS<br>Cmd | Name         | Access |     |     | Protocol | Type | Min | Max   | Unit   |
|------------|--------------|--------|-----|-----|----------|------|-----|-------|--------|
|            |              | SE     | US  | FA  |          |      |     |       |        |
| 0x17       | CycleCount() | R      | R/W | R/W | Word     | U2   | 0   | 65535 | cycles |

## 12.25 0x18 DesignCapacity()

This read-word function returns the theoretical pack capacity. The default value is stored in the data flash value **Design Capacity mAh** or **Design Capacity cWh**.

| SBS<br>Cmd | Name             | Access |     |     | Protocol | Type | Min | Max   | Default | Unit   |
|------------|------------------|--------|-----|-----|----------|------|-----|-------|---------|--------|
|            |                  | SE     | US  | FA  |          |      |     |       |         |        |
| 0x18       | DesignCapacity() | R      | R/W | R/W | Word     | U2   | 0   | 65535 | 4400    | mAh    |
|            |                  |        |     |     |          |      |     |       | 6336    | 10 mWh |

**NOTE:** If *BatteryMode()[CAPM]* = 0, then the data reports in mAh.

If *BatteryMode()[CAPM]* = 1, then the data reports in 10 mWh.

## 12.26 0x19 DesignVoltage()

This read-word function returns the theoretical pack voltage. The default value is stored in data flash value **Design Voltage**.

| SBS<br>Cmd | Name            | Access |     |     | Protocol | Type | Min  | Max   | Default | Unit |
|------------|-----------------|--------|-----|-----|----------|------|------|-------|---------|------|
|            |                 | SE     | US  | FA  |          |      |      |       |         |      |
| 0x19       | DesignVoltage() | R      | R/W | R/W | Word     | U2   | 7000 | 18000 | 14400   | mV   |

## 12.27 0x1A SpecificationInfo()

| SBS<br>Cmd | Name                | Access |     |     | Protocol | Type | Min    | Max    |
|------------|---------------------|--------|-----|-----|----------|------|--------|--------|
|            |                     | SE     | US  | FA  |          |      |        |        |
| 0x1A       | SpecificationInfo() | R      | R/W | R/W | Word     | H2   | 0x0000 | 0xFFFF |

|         |         |         |         |          |          |          |          |
|---------|---------|---------|---------|----------|----------|----------|----------|
| 15      | 14      | 13      | 12      | 11       | 10       | 9        | 8        |
| IPScale | IPScale | IPScale | IPScale | VScale   | VScale   | VScale   | VScale   |
| 7       | 6       | 5       | 4       | 3        | 2        | 1        | 0        |
| Version | Version | Version | Version | Revision | Revision | Revision | Revision |



**IPScale (Bit 15–12):** IP Scale Factor

Not supported by the gas gauge  
MUST be set to 0, 0, 0, 0.

**VScale (Bits 11–8):** Voltage Scale Factor

Not supported by the gas gauge  
MUST be set to 0, 0, 0, 0.

**Version (Bits 7–4):** Version

0,0,0,1 = Version 1.0

0,0,1,1 = Version 1.1

0,0,1,1 = Version 1.1 with optional PEC support

**Revision (Bits 3–0):** Revision

0,0,0,1 = Version 1.0 and 1.1 (default)

## 12.28 0x1B ManufacturerDate()

This read-word function returns the pack's manufacturer date.

| SBS<br>Cmd | Name               | Access |     |     | Protocol | Type | Min | Max   | Default |
|------------|--------------------|--------|-----|-----|----------|------|-----|-------|---------|
|            |                    | SE     | US  | FA  |          |      |     |       |         |
| 0x1B       | ManufacturerDate() | R      | R/W | R/W | Word     | U2   |     | 65535 | 0       |

**NOTE:** ManufacturerDate() value in the following format: Day + Month\*32 + (Year–1980)\*256

## 12.29 0x1C SerialNumber()

This read-word function returns the assigned pack serial number.

| SBS<br>Cmd | Name           | Access |     |     | Protocol | Type | Min    | Max    | Default | Unit |
|------------|----------------|--------|-----|-----|----------|------|--------|--------|---------|------|
|            |                | SE     | US  | FA  |          |      |        |        |         |      |
| 0x1C       | SerialNumber() | R      | R/W | R/W | Word     | H2   | 0x0000 | 0xFFFF | 0x0001  |      |

## 12.30 0x20 ManufacturerName()

This read-block function returns the pack manufacturer's name.

| SBS<br>Cmd | Name               | Access |    |    | Protocol | Type  | Min | Max | Default     | Unit  |
|------------|--------------------|--------|----|----|----------|-------|-----|-----|-------------|-------|
|            |                    | SE     | US | FA |          |       |     |     |             |       |
| 0x20       | ManufacturerName() | R      | R  | R  | Block    | S20+1 | —   | —   | Texas Inst. | ASCII |

## 12.31 0x21 DeviceName()

This read-block function returns the assigned pack name.

| SBS<br>Cmd | Name         | Access |    |    | Protocol | Type  | Min | Max | Default | Unit  |
|------------|--------------|--------|----|----|----------|-------|-----|-----|---------|-------|
|            |              | SE     | US | FA |          |       |     |     |         |       |
| 0x21       | DeviceName() | R      | R  | R  | Block    | S20+1 | —   | —   | bq40z50 | ASCII |

## 12.32 0x22 DeviceChemistry()

This read-block function returns the battery chemistry used in the pack.

| SBS<br>Cmd | Name                     | Access |    |    | Protocol | Type | Min | Max | Default | Unit  |
|------------|--------------------------|--------|----|----|----------|------|-----|-----|---------|-------|
|            |                          | SE     | US | FA |          |      |     |     |         |       |
| 0x22       | <i>DeviceChemistry()</i> | R      | R  | R  | Block    | S4+1 | —   | —   | LION    | ASCII |

## 12.33 0x23 ManufacturerData()

This read-block function returns **ManufacturerInfo** by default. The command also returns a response to MAC command in order to maintain compatibility of the MAC system in bq30zxy family.

| SBS<br>Cmd | Name                      | Access |    |    | Protocol | Type  | Min | Max | Unit |
|------------|---------------------------|--------|----|----|----------|-------|-----|-----|------|
|            |                           | SE     | US | FA |          |       |     |     |      |
| 0x23       | <i>ManufacturerData()</i> | R      | R  | R  | Block    | Mixed | —   | —   | —    |

## 12.34 0x2F Authenticate()

This read/write block function provides SHA-1 authentication to send the challenge and read the response in the default mode. It is also used to input a new authentication key when the MAC *AuthenticationKey()* is used.

| SBS<br>Cmd | Name                  | Access |     |     | Protocol | Type  | Min | Max | Unit |
|------------|-----------------------|--------|-----|-----|----------|-------|-----|-----|------|
|            |                       | SE     | US  | FA  |          |       |     |     |      |
| 0x2F       | <i>Authenticate()</i> | R/W    | R/W | R/W | Block    | H20+1 | —   | —   | —    |

## 12.35 0x3C CellVoltage4()

This read-word function returns the Cell 4 voltage.

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min | Max   | Default | Unit |
|------------|-----------------------|--------|----|----|----------|------|-----|-------|---------|------|
|            |                       | SE     | US | FA |          |      |     |       |         |      |
| 0x3C       | <i>CellVoltage4()</i> | R      | R  | R  | Word     | U2   | —   | 65535 | 0       | mV   |

## 12.36 0x3D CellVoltage3()

This read-word function returns the Cell 3 voltage.

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min | Max   | Default | Unit |
|------------|-----------------------|--------|----|----|----------|------|-----|-------|---------|------|
|            |                       | SE     | US | FA |          |      |     |       |         |      |
| 0x3D       | <i>CellVoltage3()</i> | R      | R  | R  | Word     | U2   | —   | 65535 | 0       | mV   |

## 12.37 0x3E CellVoltage2()

This read-word function returns the Cell 2 voltage.

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min | Max   | Default | Unit |
|------------|-----------------------|--------|----|----|----------|------|-----|-------|---------|------|
|            |                       | SE     | US | FA |          |      |     |       |         |      |
| 0x3E       | <i>CellVoltage2()</i> | R      | R  | R  | Word     | U2   | —   | 65535 | 0       | mV   |

## 12.38 0x3F CellVoltage1()

This read-word function returns the Cell 1 voltage.

| SBS<br>Cmd | Name           | Access |    |    | Protocol | Type | Min | Max   | Default | Unit |
|------------|----------------|--------|----|----|----------|------|-----|-------|---------|------|
|            |                | SE     | US | FA |          |      |     |       |         |      |
| 0x3F       | CellVoltage1() | R      | R  | R  | Word     | U2   | —   | 65535 | 0       | mV   |

### 12.39 0x4A BTPDischargeSet()

This read/write word command updates the BTP set threshold for discharge mode for the next BTP interrupt, de-asserts the present BTP interrupt, and clears the *OperationStatus()*[BTP\_INT] bit.

| SBS<br>Cmd | Name              | Access |     |     | Format     | Size in<br>Bytes | Min | Max   | Default | Unit |
|------------|-------------------|--------|-----|-----|------------|------------------|-----|-------|---------|------|
|            |                   | SE     | US  | FA  |            |                  |     |       |         |      |
| 0x4A       | BTPDischargeSet() | R/W    | R/W | R/W | Signed Int | 2                | —   | 65535 | 150     | mAh  |

### 12.40 0x4B BTPChargeSet()

The read/write word command updates the BTP set threshold for charge mode for the next BTP interrupt, de-asserts the present BTP interrupt, and clears the *OperationStatus()*[BTP\_INT] bit.

| SBS<br>Cmd | Name           | Access |     |     | Format     | Size in<br>Bytes | Min | Max   | Default | Unit |
|------------|----------------|--------|-----|-----|------------|------------------|-----|-------|---------|------|
|            |                | SE     | US  | FA  |            |                  |     |       |         |      |
| 0x4B       | BTPChargeSet() | R/W    | R/W | R/W | Signed Int | 2                | —   | 65535 | 175     | mAh  |

### 12.41 0x4F State-of-Health (SoH)

This read word command returns the SoH information of the battery in percentage of design capacity and design energy.

### 12.42 0x50 SafetyAlert

This command returns the *SafetyAlert()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name          | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|---------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |               | SE     | US | FA |          |      |                |                |         |      |
| 0x50       | SafetyAlert() | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

### 12.43 0x51 SafetyStatus

This command returns the *SafetyStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name           | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|----------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |                | SE     | US | FA |          |      |                |                |         |      |
| 0x51       | SafetyStatus() | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

### 12.44 0x52 PFAAlert

This command returns the *PFAAlert()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name       | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |            | SE     | US | FA |          |      |                |                |         |      |
| 0x52       | PFAAlert() | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

## 12.45 0x53 PFStatus

This command returns the *PFStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|-------------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |                   | SE     | US | FA |          |      |                |                |         |      |
| 0x53       | <i>PFStatus()</i> | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

## 12.46 0x54 OperationStatus

This command returns the *OperationStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                     | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|--------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |                          | SE     | US | FA |          |      |                |                |         |      |
| 0x54       | <i>OperationStatus()</i> | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

## 12.47 0x55 ChargingStatus

This command returns the *ChargingStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                    | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|-------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |                         | SE     | US | FA |          |      |                |                |         |      |
| 0x55       | <i>ChargingStatus()</i> | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

## 12.48 0x56 GaugingStatus

This command returns the *GaugingStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                   | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |                        | SE     | US | FA |          |      |                |                |         |      |
| 0x56       | <i>GaugingStatus()</i> | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

## 12.49 0x57 ManufacturingStatus

This command returns the *ManufacturingStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                         | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit |
|------------|------------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|
|            |                              | SE     | US | FA |          |      |                |                |         |      |
| 0x57       | <i>ManufacturingStatus()</i> | —      | R  | R  | Block    | H4   | 0x000000<br>00 | 0xFFFFF<br>FFF | —       | —    |

## 12.50 0x58 AFE Register

This command returns a snapshot of the AFE register settings. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name          | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|---------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |               | SE     | US | FA |          |      |     |     |         |      |
| 0x58       | AFERegister() | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.51 0x59 TURBO\_POWER

TURBO\_POWER reports the maximal peak power value, MAX\_POWER. The gauge computes a new RAM value every second. *TURBO\_POWER()* is initialized to the result of the max power calculation at reset or power up.

| SBS<br>Cmd | Name          | Access |    |     | Protocol | Type | Min | Max | Default | Unit |
|------------|---------------|--------|----|-----|----------|------|-----|-----|---------|------|
|            |               | SE     | US | FA  |          |      |     |     |         |      |
| 0x59       | TURBO_POWER() | R      | R  | R/W | Word     |      |     |     |         | cW   |

**NOTE:** This computes and provides Turbo Power information based on the battery pack configuration.

## 12.52 0x5A TURBO\_FINAL

TURBO\_FINAL sets **Min Turbo Power**, which represents the minimal TURBO BOOST mode power level during active operation (for example, non-SLEEP).

| SBS<br>Cmd | Name          | Access |     |     | Protocol | Type | Min | Max | Default | Unit |
|------------|---------------|--------|-----|-----|----------|------|-----|-----|---------|------|
|            |               | SE     | US  | FA  |          |      |     |     |         |      |
| 0x5A       | TURBO_FINAL() | R/W    | R/W | R/W | Word     |      |     |     |         | cW   |

## 12.53 0x5B TURBO\_PACK\_R

TURBO\_PACK\_R sets the **PACK Resistance** value of the battery pack serial resistance, including resistance associated with FETs, traces, sense resistors, and so on *TURBO\_PACK\_R()* accesses to the data flash value **Pack Resistance**.

| SBS<br>Cmd | Name           | Access |     |     | Protocol | Type | Min | Max | Default | Unit |
|------------|----------------|--------|-----|-----|----------|------|-----|-----|---------|------|
|            |                | SE     | US  | FA  |          |      |     |     |         |      |
| 0x5B       | TURBO_PACK_R() | R/W    | R/W | R/W | Word     |      |     |     |         | mΩ   |

## 12.54 0x5C TURBO\_SYS\_R

TURBO\_SYS\_R sets the **System Resistance** value of the system serial resistance along the path from battery to system power converter input that includes FETs, traces, sense resistors, and so on *TURBO\_SYS\_R()* accesses to the data flash value **System Resistance**.

| SBS<br>Cmd | Name          | Access |     |     | Protocol | Type | Min | Max | Default | Unit |
|------------|---------------|--------|-----|-----|----------|------|-----|-----|---------|------|
|            |               | SE     | US  | FA  |          |      |     |     |         |      |
| 0x5C       | TURBO_SYS_R() | R/W    | R/W | R/W | Word     |      |     |     |         | mΩ   |

## 12.55 0x5D TURBO\_EDV

TURBO\_EDV sets the Minimal Voltage at the system power converter input at which the system will still operate. *TURBO\_EDV()* is written to the data flash value **Terminate Voltage**. Intended use is to write it once on first use to adjust for possible changes in system design from the time the battery pack was designed.

| SBS<br>Cmd | Name               | Access |     |     | Protocol | Type | Min | Max | Default | Unit |
|------------|--------------------|--------|-----|-----|----------|------|-----|-----|---------|------|
|            |                    | SE     | US  | FA  |          |      |     |     |         |      |
| 0x5D       | <i>MIN_SYS_V()</i> | R/W    | R/W | R/W | Word     |      |     |     |         | mV   |

## 12.56 0x5E TURBO\_CURRENT

The gauge computes a maximal discharge current supported by the cell design for a C-rate discharge pulse for 10 ms. This value is updated every 1 s for the system to read.

| SBS<br>Cmd | Name                   | Access |    |     | Protocol | Type | Min | Max | Default | Unit |
|------------|------------------------|--------|----|-----|----------|------|-----|-----|---------|------|
|            |                        | SE     | US | FA  |          |      |     |     |         |      |
| 0x5D       | <i>TURBO_CURRENT()</i> | R      | R  | R/W | Word     |      |     |     |         | mAh  |

**NOTE:** This computes a maximal discharge current supported by the cell design.

## 12.57 0x60 Lifetime Data Block 1

This command returns the first block of Lifetime Data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |
| 0x60       | <i>LifeTimeDataBlock1()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.58 0x61 Lifetime Data Block 2

This command returns the second block of Lifetime Data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |
| 0x61       | <i>LifeTimeDataBlock2()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.59 0x62 Lifetime Data Block 3

This command returns the third block of Lifetime Data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |
| 0x62       | <i>LifeTimeDataBlock3()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.60 0x63 Lifetime Data Block 4

This command returns the third block of Lifetime Data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |
| 0x63       | <i>LifeTimeDataBlock4()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.61 0x64 Lifetime Data Block 5

This command returns the third block of Lifetime Data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |
| 0x64       | <i>LifeTimeDataBlock5()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.62 0x70 ManufacturerInfo

This command returns manufacturer information. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                      | Access |     |     | Protocol | Type | Min | Max | Default | Unit |
|------------|---------------------------|--------|-----|-----|----------|------|-----|-----|---------|------|
|            |                           | SE     | US  | FA  |          |      |     |     |         |      |
| 0x70       | <i>ManufacturerInfo()</i> | R      | R/W | R/W | Block    | —    | —   | —   | —       | —    |

## 12.63 0x71 DStatus1

This command returns the Cell Voltages, Pack Voltage, Bat Voltage, Cell Currents, Cell Powers, Power, and Average Power. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                   | SE     | US | FA |          |      |     |     |         |      |
| 0x71       | <i>DStatus1()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.64 0x72 DStatus2

This command returns the internal temp sensor, TS1, TS2, TS3, TS4, Cell Temp, and FETTemp. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                   | SE     | US | FA |          |      |     |     |         |      |
| 0x72       | <i>DStatus2()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.65 0x73 GaugeStatus1

This command instructs the device to return Impedance Track related gauging information. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                       | SE     | US | FA |          |      |     |     |         |      |
| 0x73       | <i>GaugeStatus1()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

## 12.66 0x74 GaugeStatus2

This command instructs the device to return Impedance Track related gauging information. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                       | SE     | US | FA |          |      |     |     |         |      |
| 0x74       | <i>GaugeStatus2()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

### 12.67 0x75 GaugeStatus3

This command instructs the device to return Impedance Track related gauging information. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-----------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                       | SE     | US | FA |          |      |     |     |         |      |
| 0x75       | <i>GaugeStatus3()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

### 12.68 0x76 CBStatus

This command instructs the device to return cell balance time information. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|-------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                   | SE     | US | FA |          |      |     |     |         |      |
| 0x75       | <i>CBStatus()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

### 12.69 0x77 State-of-Health

This command instructs the device to return the state-of-health full charge capacity and energy. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                   | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                        | SE     | US | FA |          |      |     |     |         |      |
| 0x75       | <i>StateOfHealth()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |

### 12.70 0x78 FilteredCapacity

This command instructs the device to return the filtered capacity and energy even if *[SMOOTH]* = 0. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 12.1](#).

| SBS<br>Cmd | Name                      | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|------------|---------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|            |                           | SE     | US | FA |          |      |     |     |         |      |
| 0x75       | <i>FilteredCapacity()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |



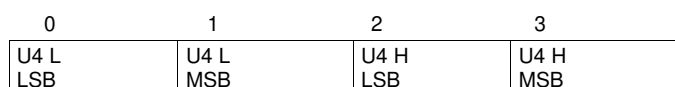
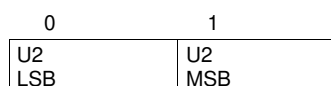
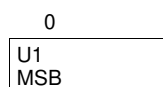


## **Data Flash Values**

### **13.1 Data Formats**

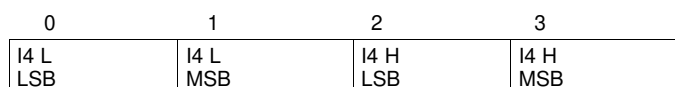
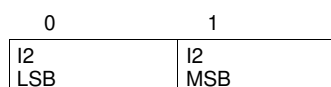
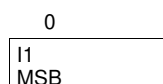
#### **13.1.1 Unsigned Integer**

Unsigned integers are stored without changes as 1-byte, 2-byte, or 4-byte values in Little Endian byte order.



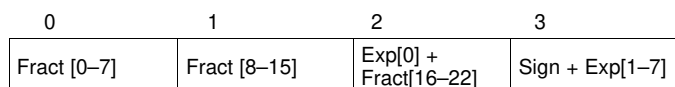
#### **13.1.2 Integer**

Integer values are stored in 2's-complement format in 1-byte, 2-byte, or 4-byte values in Little Endian byte order.



#### **13.1.3 Floating Point**

Floating point values are stored using the IEEE754 Single Precision 4-byte format in Little Endian byte order.



Where:

Exp: 8-bit exponent stored with an offset bias of 127. The values 00 and FF have unique meanings.

Fract: 23-bit fraction. If the exponent is > 0, then the mantissa is 1.fract. If the exponent is zero, then the mantissa is 0.fract.

The floating point value depends on the unique cases of the exponent:

- If the exponent is FF and the fraction is zero, this represents +/- infinity.
- If the exponent is FF and the fraction is non-zero this represents "not a number" (NaN).
- If the exponent is 00 then the value is a subnormal number represented by  $(-1)^{\text{sign}} \times 2^{-126} \times 0.\text{fraction}$ .
- Otherwise, the value is a normalized number represented by  $(-1)^{\text{sign}} \times 2^{(\text{exponent} - 127)} \times 1.\text{fraction}$ .

### 13.1.4 Hex

Bit register definitions are stored in unsigned integer format.

### 13.1.5 String

String values are stored with length byte first, followed by a number of data bytes defined with the length byte.

|        |       |     |       |
|--------|-------|-----|-------|
| 0      | 1     | ... | N     |
| Length | Data0 | ... | DataN |

## 13.2 Calibration

### 13.2.1 Voltage

| Class       | Subclass | Name      | Start  | Type | Min    | Max   | Default | Unit | Description        |
|-------------|----------|-----------|--------|------|--------|-------|---------|------|--------------------|
| Calibration | Voltage  | Cell Gain | 0x4000 | I2   | -32767 | 32767 | 12101   | —    | VC[n]-VC[n-1] gain |
| Calibration | Voltage  | PACK Gain | 0x4002 | U2   | 0      | 65535 | 49669   | —    | PACK-VSS gain      |
| Calibration | Voltage  | BAT Gain  | 0x4004 | U2   | 0      | 65535 | 48936   | —    | BAT-VSS gain       |

### 13.2.2 Current

| Class       | Subclass | Name          | Start Address | Type | Min       | Max       | Default     | Description          |
|-------------|----------|---------------|---------------|------|-----------|-----------|-------------|----------------------|
| Calibration | Current  | CC Gain       | 0x4006        | F4   | 1.00E-001 | 4.00E+000 | 3.58422     | Coulomb Counter Gain |
| Calibration | Current  | Capacity Gain | 0x400A        | F4   | 2.98E+004 | 1.19E+006 | 1069035.256 | Capacity Gain        |

### 13.2.3 Current Offset

#### 13.2.3.1 CC Offset

| Class       | Subclass       | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|-----------|---------------|------|--------|-------|---------|------|
| Calibration | Current Offset | CC Offset | 0x400E        | I2   | -32767 | 32767 | 0       | —    |

**Description:** Coulomb Counter Offset. This offset is used for *Current()* and *AverageCurrent()* measurement.

#### 13.2.3.2 Coulomb Counter Offset Samples

| Class       | Subclass       | Name                           | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------------|--------------------------------|---------------|------|-----|-------|---------|------|
| Calibration | Current Offset | Coulomb Counter Offset Samples | 0x4010        | U2   | 0   | 65535 | 64      | —    |

**Description:** Coulomb Counter Offset Samples is used for averaging.

### 13.2.3.3 Board Offset

| Class       | Subclass       | Name         | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|--------------|---------------|------|--------|-------|---------|------|
| Calibration | Current Offset | Board Offset | 0x4012        | I2   | –32768 | 32767 | 0       | —    |

**Description:** PCB board offset

### 13.2.4 CC Auto Config

| Class       | Subclass       | Name           | Start Address | Type | Min  | Max  | Default | Units |
|-------------|----------------|----------------|---------------|------|------|------|---------|-------|
| Calibration | Current Offset | CC Auto Config | 0x40C0        | H1   | 0x00 | 0x07 | 0x03    | Hex   |

|      |      |      |      |      |              |             |             |
|------|------|------|------|------|--------------|-------------|-------------|
| 7    | 6    | 5    | 4    | 3    | 2            | 1           | 0           |
| RSVD | RSVD | RSVD | RSVD | RSVD | OFFSET_TAKEN | AUTO_NESTON | AUTO_CAL_EN |

*SpecificationInformation()* values

**RSVD (Bits 7–3):** Reserved. Do not use.

**OFFSET\_TAKEN (Bit 2):** CC Auto offset is taken.

1 = CC Auto Offset has been measured.

0 = CC Auto Offset has not been measured.

**AUTO\_NESTON (Bit 1):** NEST Circuit ON

1 = When **[OFFSET\_TAKEN]** = 1, FW automatically controls the HW NEST circuit for best current and cell current measurements.

0 = HW NEST circuit is always on. Individual cell current measurement may have error relative to *Current()*, but the *Current()* accuracy is not impacted.

**AUTO\_CAL\_EN (Bit 0):** Auto CC offset calibration enable

1 = FW will perform auto CC calibration on entry into SLEEP mode. A min auto CC calibration interval is set to 10hr to prevent flash wear out. The result is saved to CC Auto Offset.

0 = Auto CC offset calibration is disabled.

### 13.2.5 CC Auto Offset

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|
| Calibration | Current Offset | CC Auto Offset | 0x40C1        | I2   | –10000 | 10000 | 0       |

**Description:** CC offset collected via CC Auto Calibration. This offset is used for cell current measurement and is different than CC Offset.

## 13.2.6 Temperature

### 13.2.6.1 Internal Temp Offset

| Class       | Subclass    | Name                 | Start Address | Type | Min  | Max | Default | Unit   |
|-------------|-------------|----------------------|---------------|------|------|-----|---------|--------|
| Calibration | Temperature | Internal Temp Offset | 0x4014        | I1   | –128 | 127 | 0       | 0.1 °C |

**Description:** Internal temperature sensor reading offset

### 13.2.6.2 External 1 Temp Offset

| Class       | Subclass    | Name                   | Start Address | Type | Min  | Max | Default | Unit   |
|-------------|-------------|------------------------|---------------|------|------|-----|---------|--------|
| Calibration | Temperature | External 1 Temp Offset | 0x4015        | I1   | –128 | 127 | 0       | 0.1 °C |

**Description:** TS1 temperature sensor reading offset

### 13.2.6.3 External 2 Temp Offset

| Class       | Subclass    | Name                   | Start Address | Type | Min  | Max | Default | Unit   |
|-------------|-------------|------------------------|---------------|------|------|-----|---------|--------|
| Calibration | Temperature | External 2 Temp Offset | 0x4016        | I1   | –128 | 127 | 0       | 0.1 °C |

**Description:** TS2 temperature sensor reading offset

### 13.2.6.4 External 3 Temp Offset

| Class       | Subclass    | Name                   | Start Address | Type | Min  | Max | Default | Unit   |
|-------------|-------------|------------------------|---------------|------|------|-----|---------|--------|
| Calibration | Temperature | External 3 Temp Offset | 0x4017        | I1   | –128 | 127 | 0       | 0.1 °C |

**Description:** TS3 temperature sensor reading offset

### 13.2.6.5 External 4 Temp Offset

| Class       | Subclass    | Name                   | Start Address | Type | Min  | Max | Default | Unit   |
|-------------|-------------|------------------------|---------------|------|------|-----|---------|--------|
| Calibration | Temperature | External 4 Temp Offset | 0x4018        | I1   | –128 | 127 | 0       | 0.1 °C |

**Description:** TS4 temperature sensor reading offset

## 13.2.7 Internal Temp Model

### 13.2.7.1 Int Gain

| Class       | Subclass            | Name     | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|---------------------|----------|---------------|------|--------|-------|---------|------|
| Calibration | Internal Temp Model | Int Gain | 0x45C0        | I2   | –32768 | 32767 | –12143  | —    |

**Description:** Internal temperature gain

### 13.2.7.2 Int Base Offset

| Class       | Subclass            | Name            | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|---------------------|-----------------|---------------|------|--------|-------|---------|------|
| Calibration | Internal Temp Model | Int Base Offset | 0x45C2        | I2   | –32768 | 32767 | 6232    | —    |

**Description:** Internal temperature base offset

### 13.2.7.3 Int Minimum AD

| Class       | Subclass            | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|---------------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Internal Temp Model | Int Minimum AD | 0x45C4        | I2   | –32768 | 32767 | 0       | —    |

**Description:** Minimum AD count used for calculation

### 13.2.7.4 Int Maximum Temp

| Class       | Subclass            | Name             | Start Address | Type | Min    | Max   | Default | Unit   |
|-------------|---------------------|------------------|---------------|------|--------|-------|---------|--------|
| Calibration | Internal Temp Model | Int Maximum Temp | 0x45C6        | I2   | –32768 | 32767 | 6232    | 0.1 °K |

**Description:** Maximum Temperature boundary

## 13.2.8 Cell Temp Model

### 13.2.8.1 Coefficient a1

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient a1 | 0x45C8        | I2   | –32768 | 32767 | –11130  | —    |

**Description:** Cell Temperature calculation polynomial a1

### 13.2.8.2 Coefficient a2

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient a2 | 0x45CA        | I2   | –32768 | 32767 | 19142   | —    |

**Description:** Cell Temperature calculation polynomial a2

### 13.2.8.3 Coefficient a3

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient a3 | 0x45CC        | I2   | –32768 | 32767 | –19262  | —    |

**Description:** Cell Temperature calculation polynomial a3

#### 13.2.8.4 Coefficient a4

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient a4 | 0x45CE        | I2   | –32768 | 32767 | 28203   | —    |

**Description:** Cell Temperature calculation polynomial a4

#### 13.2.8.5 Coefficient a5

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient a5 | 0x45D0        | I2   | –32768 | 32767 | 892     | —    |

**Description:** Cell Temperature calculation polynomial a5

#### 13.2.8.6 Coefficient b1

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient b1 | 0x45D2        | I2   | –32768 | 32767 | 328     | —    |

**Description:** Cell Temperature calculation polynomial b1

#### 13.2.8.7 Coefficient b2

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient b2 | 0x45D4        | I2   | –32768 | 32767 | –605    | —    |

**Description:** Cell Temperature calculation polynomial b2

#### 13.2.8.8 Coefficient b3

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient b3 | 0x45D6        | I2   | –32768 | 32767 | –2443   | —    |

**Description:** Cell Temperature calculation polynomial b3

#### 13.2.8.9 Coefficient b4

| Class       | Subclass        | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Coefficient b4 | 0x41D8        | I2   | –32768 | 32767 | 4969    | —    |

**Description:** Cell Temperature calculation polynomial b4

#### 13.2.8.10 Rc0

| Class       | Subclass        | Name | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Rc0  | 0x41DA        | I2   | –32768 | 32767 | 11703   | Ω    |

**Description:** Resistance at 25°C

### 13.2.8.11 Adc0

| Class       | Subclass        | Name | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Adc0 | 0x41DC        | I2   | –32768 | 32767 | 11703   | —    |

**Description:** ADC reading at 25°C

### 13.2.8.12 Rpad

| Class       | Subclass        | Name | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Rpad | 0x41DE        | I2   | –32768 | 32767 | 0       | Ω    |

**Description:** Pad Resistance (0 to use factory calibration)

### 13.2.8.13 Rint

| Class       | Subclass        | Name | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------------|------|---------------|------|--------|-------|---------|------|
| Calibration | Cell Temp Model | Rint | 0x41E0        | I2   | –32768 | 32767 | 0       | Ω    |

**Description:** Pull up resistor resistance (0 to use factory calibration)

## 13.2.9 FET Temp Model

### 13.2.9.1 Coefficient a1

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient a1 | 0x41E2        | I2   | –32768 | 32767 | –11130  | —    |

**Description:** FET Temperature calculation polynomial a1

### 13.2.9.2 Coefficient a2

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient a2 | 0x41E4        | I2   | –32768 | 32767 | 19142   | —    |

**Description:** FET Temperature calculation polynomial a2

### 13.2.9.3 Coefficient a3

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient a3 | 0x41E6        | I2   | –32768 | 32767 | –19262  | —    |

**Description:** FET Temperature calculation polynomial a3

### 13.2.9.4 Coefficient a4



| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient a4 | 0x41E8        | I2   | –32768 | 32767 | 28203   | —    |

**Description:** FET Temperature calculation polynomial a4

### 13.2.9.5 Coefficient a5

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient a5 | 0x41EA        | I2   | –32768 | 32767 | 892     | —    |

**Description:** FET Temperature calculation polynomial a5

### 13.2.9.6 Coefficient b1

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient b1 | 0x41EC        | I2   | –32768 | 32767 | 328     | —    |

**Description:** FET Temperature calculation polynomial b1

### 13.2.9.7 Coefficient b2

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient b2 | 0x41EE        | I2   | –32768 | 32767 | –605    | —    |

**Description:** FET Temperature calculation polynomial b2

### 13.2.9.8 Coefficient b3

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient b3 | 0x41F0        | I2   | –32768 | 32767 | –2443   | —    |

**Description:** FET Temperature calculation polynomial b3

### 13.2.9.9 Coefficient b4

| Class       | Subclass       | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|----------------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Coefficient b4 | 0x41F2        | I2   | –32768 | 32767 | 4969    | —    |

**Description:** FET Temperature calculation polynomial b4

### 13.2.9.10 Rc0

| Class       | Subclass       | Name | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Rc0  | 0x41F4        | I2   | –32768 | 32767 | 11703   | Ω    |

**Description:** Resistance at 25°C

### 13.2.9.11 Adc0

| Class       | Subclass       | Name | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------------|------|---------------|------|--------|-------|---------|------|
| Calibration | FET Temp Model | Adc0 | 0x41F6        | I2   | –32768 | 32767 | 11703   | —    |

**Description:** ADC reading at 25°C

### 13.2.9.12 Rpad

| Class       | Subclass       | Name | Start Address | Type | Min    | Max   | Default          | Unit |
|-------------|----------------|------|---------------|------|--------|-------|------------------|------|
| Calibration | FET Temp Model | Rpad | 0x41F8        | I2   | –32768 | 32767 | 0 <sup>(1)</sup> | Ω    |

<sup>(1)</sup> Setting this value to 0 causes the gauge to use the internal factory calibration default.

**Description:** Pad Resistance (0 to use factory calibration)

### 13.2.9.13 Rint

| Class       | Subclass       | Name | Start Address | Type | Min    | Max   | Default          | Unit |
|-------------|----------------|------|---------------|------|--------|-------|------------------|------|
| Calibration | FET Temp Model | Rint | 0x41FA        | I2   | –32768 | 32767 | 0 <sup>(1)</sup> | Ω    |

<sup>(1)</sup> Setting this value to 0 causes the gauge to use the internal factory calibration default.

**Description:** Pull up resistor resistance (0 to use factory calibration)

## 13.2.10 Current Deadband

### 13.2.10.1 Deadband

| Class       | Subclass         | Name     | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------------|----------|---------------|------|-----|-----|---------|------|
| Calibration | Current Deadband | Deadband | 0x4606        | U1   | 0   | 255 | 3       | mA   |

**Description:** Pack-based Deadband to report 0 mA

### 13.2.10.2 Coulomb Counter Deadband

| Class       | Subclass         | Name                     | Start Address | Type | Min | Max | Default | Unit   |
|-------------|------------------|--------------------------|---------------|------|-----|-----|---------|--------|
| Calibration | Current Deadband | Coulomb Counter Deadband | 0x4607        | U1   | 0   | 255 | 9       | 116 nV |

**Description:** Coulomb counter deadband to report 0 charge (This setting should not be modified.)

## 13.3 Settings

### 13.3.1 Configuration

#### 13.3.1.1 FET Options

| Class    | Subclass      | Name        | Start Address | Type | Min  | Max  | Default | Unit |
|----------|---------------|-------------|---------------|------|------|------|---------|------|
| Settings | Configuration | FET Options | 0x47C7        | H1   | 0x00 | 0xFF | 0x20    | Hex  |

| 7         | 6        | 5      | 4     | 3     | 2     | 1    | 0         |
|-----------|----------|--------|-------|-------|-------|------|-----------|
| PACK_FUSE | SLEEPCHG | CHGFET | CHGIN | CHGSU | OTFET | RSVD | PCHG_COMM |

**PACK\_FUSE (Bit 7):** Source of voltage to check for *Min Blow Fuse Voltage*

1 = Pack+ voltage

0 = Battery stack voltage

**SLEEPCHG (Bit 6):** CHG FET enabled during sleep

1 = CHG FET remains on during sleep

0 = CHG FET off during sleep (default)

**CHGFET (Bit 5):** FET action on setting of *BatteryStatus()* [FC]

1 = Charging and Precharging disabled, FET off

0 = FET active (default)

**CHGIN (Bit 4):** FET action in CHARGE INHIBIT mode

1 = Charging and Precharging disabled, FETs off

0 = FET active (default)

**CHGSU (Bit 3):** FET action in CHARGE SUSPEND mode

1 = Charging and Precharging disabled, FETs off

0 = FET active (default)

**OTFET (Bit 2):** FET action in OVERTEMPERATURE mode

1 = CHG and DSG FETs will be turned off for overtemperature conditions

0 = No FET action for overtemperature condition (default)

**RSVD (Bit 1):** Reserved. Do not use.

**PCHG\_COMM (Bit 0):** Precharge FET selection

1 = CHG FET

0 = PCHG FET (default)

### 13.3.1.2 SBS Gauging Configuration

| Class    | Subclass      | Name                      | Start Address | Type | Min  | Max  | Default | Unit |
|----------|---------------|---------------------------|---------------|------|------|------|---------|------|
| Settings | Configuration | SBS Gauging Configuration | 0x47C8        | H1   | 0x00 | 0xFF | 0x04    | Hex  |

| 7    | 6    | 5    | 4    | 3    | 2     | 1         | 0     |
|------|------|------|------|------|-------|-----------|-------|
| RSVD | RSVD | RSVD | RSVD | RSVD | LOCK0 | RSOC_HOLD | RSOCL |

**RSVD (Bit 7–3):** Reserved. Do not use.

**LOCK0 (Bit 2):** Keep *RemainingCapacity()* and *RelativeStateOfCharge()* jumping back during relaxation after 0 was reached during discharge.

1 = Enabled (default)

0 = Disabled

**RSOC\_HOLD (Bit 1):** Prevent RSOC from increasing during discharge

1 = RSOC not allowed to increase during discharge

0 = RSOC not limited (default)

**RSOCL (Bit 0):** *RelativeStateOfCharge()* and *RemainingCapacity()* behavior at end of charge

- 1 = Held at 99% until valid charge termination. On entering valid charge termination update to 100%
- 0 = Actual value shown (default)

### 13.3.1.3 SBS Configuration

| Class    | Subclass      | Name              | Start Address | Type | Min  | Max  | Default | Unit |
|----------|---------------|-------------------|---------------|------|------|------|---------|------|
| Settings | Configuration | SBS Configuration | 0x47C9        | H1   | 0x7F | 0xFF | 0x20    | Hex  |

|      |      |      |      |    |     |     |       |
|------|------|------|------|----|-----|-----|-------|
| 7    | 6    | 5    | 4    | 3  | 2   | 1   | 0     |
| RSVD | RSVD | BLT1 | BLT0 | XL | HPE | CPE | BCAST |

**RSVD (Bit 7):** Reserved. Do not use.

**RSVD (Bit 6):** Reserved. Do not use.

**BLT1 (Bit 5):** Bus low timeout

0,0 = No SBS bus low timeout

0,1 = 1-s SBS bus low timeout

1,0 = 2-s SBS bus low timeout (default)

1,1 = 3-s SBS bus low timeout

**BLT0 (Bit 4):** Bus low timeout

0,0 = No SBS bus low timeout

0,1 = 1-s SBS bus low timeout

1,0 = 2-s SBS bus low timeout (default)

1,1 = 3-s SBS bus low timeout

**XL (Bit 3):** Enable 400-kHz COM mode

1 = 400-kHz bus speed

0 = Normal SBS bus speed (default)

**HPE (Bit 2):** PEC on host communication

1 = Enabled

0 = Disabled (default)

**CPE (Bit 1):** PEC on charger broadcast

1 = Enabled

0 = Disabled (default)

**BCAST (Bit 0):** Enable alert and charging broadcast from device to host

1 = Enabled

0 = Disabled (default)

### 13.3.1.4 Power Config

| Class    | Subclass      | Name         | Start Address | Type | Min  | Max  | Default | Unit |
|----------|---------------|--------------|---------------|------|------|------|---------|------|
| Settings | Configuration | Power Config | 0x47CA        | H1   | 0x00 | 0x01 | 0x00    | Hex  |

|      |      |      |      |      |      |      |              |
|------|------|------|------|------|------|------|--------------|
| 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0            |
| RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | AUTO_SHIP_EN |

**RSVD (Bit 7–1):** Reserved. Do not use.

**AUTO\_SHIP\_EN (Bit 0):** *Automatically Shutdown for Shipment*

1 = Enable auto shutdown after the device is in SLEEP mode without communication for a set period of time.

0 = Disable auto shutdown feature

### 13.3.1.5 IO Config

| Class    | Subclass      | Name      | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|-----------|---------------|------|-----|------|---------|------|
| Settings | Configuration | IO Config | 0x47CB        | H1   | 0x0 | 0x03 | 0x00    | Hex  |

|      |      |      |      |      |      |         |        |
|------|------|------|------|------|------|---------|--------|
| 7    | 6    | 5    | 4    | 3    | 2    | 1       | 0      |
| RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | BTP_POL | BTP_EN |

**RSVD (Bit 7–2):** Reserved. Do not use.

**BTP\_POL (Bit 1):** *Control polarity of BTP pin*

1 = BTP pin is asserted high when BTP is triggered.

0 = BTP pin is asserted low when BTP is triggered (default).

**BTP\_EN (Bit 0):** *Enable assertion of BTP pin*

1 = Enable assertion of BTP pin when BTP is triggered.

0 = Disable assertion of BTP pin when BTP is triggered (default).

### 13.3.1.6 LED Configuration

| Class    | Subclass      | Name              | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|-------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | LED Configuration | 0x47E8        | H1   | 0x0 | 0xFF | 0x0D0   | Hex  |

|       |       |        |        |         |        |        |      |
|-------|-------|--------|--------|---------|--------|--------|------|
| 7     | 6     | 5      | 4      | 3       | 2      | 1      | 0    |
| LEDC1 | LEDC0 | LEDPF1 | LEDPF0 | LEDMODE | LEDCHG | LEDRCA | LEDR |

**LEDC1, LEDC0 (Bit 7–Bit 6):** LED Current sink

0,0 = 0.94 mA average LED current (default)

0, 1 = 1.87 mA average LED current

1, 0 = 2.81 mA average LED current

1, 1 = 3.75 mA average LED current

**LEDPF1, LEDPF0 (Bit 5–Bit 4):** LED Display PF Error Code

0,0 = PF Error Code not available

0, 1 = PF Error Code shown after SOC if DISP is held low for LED Hold Time (default)

1, 0 = PF Error Code not available

1, 1 = PF Error Code shown after SOC

**LEDMODE (Bit 3):** LED Display Capacity Selector

1 = Display ASOC/DC

0 = Display RSOC (default)

**LEDCHG (Bit 2):** LED Display During Charging

1 = Enabled

0 = Disabled

**LEDRCA (Bit 1):** Flashing of LED Display when [RCA] is set

1 = Enabled

0 = Disabled

**LEDR (Bit 0):** LED Display activation at Exit of Device Reset

1 = Enabled

0 = Disabled

### 13.3.1.7 SOC Flag Config A

| Class    | Subclass      | Name              | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|-------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | SOC Flag Config A | 0x480E        | H2   | 0x0 | 0xFF | 0xC8C   | Hex  |

|                 |           |          |        |                 |           |          |        |
|-----------------|-----------|----------|--------|-----------------|-----------|----------|--------|
| 15              | 14        | 13       | 12     | 11              | 10        | 9        | 8      |
| RSVD            | RSVD      | RSVD     | RSVD   | TCSETVCT        | FCSETVCT  | RSVD     | RSVD   |
| 7               | 6         | 5        | 4      | 3               | 2         | 1        | 0      |
| TCCLEAR<br>RSOC | TCSETRSOC | TCCLEARV | TCSETV | TDCLEAR<br>RSOC | TDSETRSOC | TDCLEARV | TDSETV |

**RSVD (Bit 15–12):** Reserved. Do not use.

**TCSETVCT (Bit 11):** Enable TC flag set by primary charge termination

1 = Enabled (default)

0 = Disabled

**FCSETVCT (Bit 10):** Enable FC flag set by primary charge termination

1 = Enabled (default)

0 = Disabled

**RSVD (Bit 9–8):** Reserved. Do not use.

**TCCLEARRSOC (Bit 7):** Enable TC flag clear by RSOC threshold

1 = Enabled (default)

0 = Disabled

**TCSETRSOC (Bit 6):** Enable TC flag set by RSOC threshold

1 = Enabled

0 = Disabled (default)

**TCCLEARV (Bit 5):** Enable TC flag clear by cell voltage threshold

1 = Enabled

0 = Disabled (default)

**TCSETV (Bit 4):** Enable TC flag set by cell voltage threshold

1 = Enabled

0 = Disabled (default)

**TDCLEARRSOC (Bit 3):** Enable TD flag clear by RSOC threshold

1 = Enabled (default)

0 = Disabled

**TDSETRSOC (Bit 2):** Enable TD flag set by RSOC threshold

- 1 = Enabled (default)
- 0 = Disabled

**TDCLEARV (Bit 1):** Enable TD flag clear by cell voltage threshold

- 1 = Enabled
- 0 = Disabled (default)

**TDSETV (Bit 0):** Enable TD flag set by cell voltage threshold

- 1 = Enabled
- 0 = Disabled (default)

### 13.3.1.8 SOC Flag Config B

| Class    | Subclass      | Name              | Start Address | Type | Min  | Max  | Default | Unit |
|----------|---------------|-------------------|---------------|------|------|------|---------|------|
| Settings | Configuration | SOC Flag Config B | 0x4810        | H1   | 0x00 | 0xFF | 0x8C    | Hex  |

|                 |           |          |        |                 |           |          |        |
|-----------------|-----------|----------|--------|-----------------|-----------|----------|--------|
| 7               | 6         | 5        | 4      | 3               | 2         | 1        | 0      |
| FCCLEAR<br>RSOC | FCSETRSOC | FCCLEARV | FCSETV | FDCLEAR<br>RSOC | FDSETRSOC | FDCLEARV | FDSETV |

**FCCLEARRSOC (Bit 7):** Enable FC flag clear by RSOC threshold

- 1 = Enabled (default)
- 0 = Disabled

**FCSETRSOC (Bit 6):** Enable FC flag set by RSOC threshold

- 1 = Enabled
- 0 = Disabled (default)

**FCCLEARV (Bit 5):** Enable FC flag clear by cell voltage threshold

- 1 = Enabled
- 0 = Disabled (default)

**FCSETV (Bit 4):** Enable FC flag set by cell voltage threshold

- 1 = Enabled
- 0 = Disabled (default)

**FDCLEARRSOC (Bit 3):** Enable FD flag clear by RSOC threshold

- 1 = Enabled (default)
- 0 = Disabled

**FDSETRSOC Bit 2:** Enable FD flag set by RSOC threshold

- 1 = Enabled (default)
- 0 = Disabled

**FDCLEARV (Bit 1):** Enable FD flag clear by cell voltage threshold

- 1 = Enabled
- 0 = Disabled (default)

**FDSETV (Bit 0):** Enable FD flag set by cell voltage threshold

- 1 = Enabled
- 0 = Disabled (default)

### 13.3.1.9 IT Gauging Configuration

| Class    | Subclass      | Name                     | Start Address | Type | Min | Max    | Default | Unit |
|----------|---------------|--------------------------|---------------|------|-----|--------|---------|------|
| Settings | Configuration | IT Gauging Configuration | 0x4829        | H2   | 0x0 | 0xFFFF | 0x5FE   | Hex  |

|               |                 |           |        |               |             |           |               |
|---------------|-----------------|-----------|--------|---------------|-------------|-----------|---------------|
| 15            | 14              | 13        | 12     | 11            | 10          | 9         | 8             |
| VOLT_CONSIST  | RELAX_SMOOTH_OK | TDELTA_V  | SMOOTH | RELAX_JUMP_OK | FF_NEAR_EDV | CELL_TERM | FAST_QMAX_FLD |
| 7             | 6               | 5         | 4      | 3             | 2           | 1         | 0             |
| FAST_QMAX_LRN | RSOC_CONV       | LFP_RELAX | DOD0EW | OCVFR         | RFACTSTEP   | CSYNC     | CCT           |

**VOLT\_CONSIST (Bit 15):** Voltage Consistency Check

- 1 = Enabled (default)
- 0 = Disabled

**RELAX\_SMOOTH\_OK (Bit 14):** Smooth RSOC during RELAX mode

- 1 = Enabled (default)
- 0 = Disabled

**TDELTA\_V (Bit 13):** TURBO Mode Delta Voltage

- 1 = Calculate **DeltaVoltage** that corresponds to the power spike defined in **Min Turbo Power**. Must set this flag to 1 to support TURBO mode.
- 0 = Use of **DeltaVoltage** learned as the maximal difference between instantaneous and average voltage (default).

**SMOOTH (Bit 12):** Smooth RSOC

- 1 = Smoothed *FullChargeCapacity()* and *RemainingCapacity()* is used (default).
- 0 = True *FullChargeCapacity()* and *RemainingCapacity()* is used.

**RELAX\_JUMP\_OK (Bit 11):** Allows RSOC jump during RELAX mode

- 1 = Enabled
- 0 = Disabled (default)

**FF\_NEAR\_EDV (Bit 10):** Fast Filter Near EDV

- 1 = **Near EDV Ra Param Filter** is used for Ra update in the **[RSOC\_CONV]** region (fast scaling region starts around 10% RSOC) (default).
- 0 = Regular **Resistance Parameter Filter** is used for Ra update.

**CELL\_TERM (Bit 9):** Cell Based Termination

- 1 = Cell based termination
- 0 = Stack voltage based termination (default)

**FAST\_QMAX\_FLD (Bit 8):** Fast Qmax Update in Field

- 1 = Enabled
- 0 = Disabled (default)

**FAST\_QMAX\_LRN (Bit 7):** Fast Qmax Update in Learning

- 1 = Enabled (default)
- 0 = Disabled

**RSOC\_CONV (Bit 6):** RSOC Convergence (Fast Scaling)

- 1 = Enabled (default)
- 0 = Disabled

**LFP\_RELAX (Bit 5):** Lithium Iron Phosphate Relax

- 1 = Enabled



0 = Disabled

**DOD0EW (Bit 4):** DOD0 error weighting

1 = Enabled

0 = Disabled

**OCVFR (Bit 3):** Open Circuit Voltage Flat Region

1 = Enabled

0 = Disabled

**RFACTSTEP (Bit 2):** Ra Factor Step

1 = Enabled (default).

0 = Disabled

**CSYNC (Bit 1):** Sync *RemainingCapacity()* with *FullChargeCapacity()* at valid charge termination

1 = Synchronized (default)

0 = Not synchronized

**CCT (Bit 0):** Cycle count threshold

1 = Use CC % of *FullChargeCapacity()*

0 = Use CC % of *DesignCapacity()* (default)

### 13.3.1.10 Charging Configuration

| Class    | Subclass      | Name                   | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|------------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | Charging Configuration | 0x490C        | H1   | 0x0 | 0x3F | 0x0     | Hex  |

|      |      |      |      |      |      |     |       |
|------|------|------|------|------|------|-----|-------|
| 7    | 6    | 5    | 4    | 3    | 2    | 1   | 0     |
| RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | CCC | CRATE |

**RSVD (Bits 7–2):** Reserved. Do not use.

**CCC (Bit 1)**

1 = Enable Charging Loss Compensation feature

0 = Charging Loss Compensation disabled (default)

**CRATE (Bit 0):** ChargeCurrent rate

1 = *ChargingCurrent()* adjusted based on *FullChargeCapacity()* / *DesignCapacity()*

0 = No adjustment to *ChargingCurrent()* (default)

### 13.3.1.11 Temperature Enable

| Class    | Subclass      | Name               | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|--------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | Temperature Enable | 0x494E        | H1   | 0x0 | 0x1F | 0x6     | Hex  |

|      |      |      |     |     |     |     |             |
|------|------|------|-----|-----|-----|-----|-------------|
| 7    | 6    | 5    | 4   | 3   | 2   | 1   | 0           |
| RSVD | RSVD | RSVD | TS4 | TS3 | TS2 | TS1 | Internal TS |

**RSVD (Bit 7–5):** Reserved. Do not use.

**TS4 (Bit 4):** Enable TS4

1 = Enable TS4 (default)

0 = Disable TS4

**TS3 (Bit 3):** Enable TS3

1 = Enable TS3 (default)

0 = Disable TS3

**TS2 (Bit 2):** Enable TS2

1 = Enable TS2 (default)

0 = Disable TS2

**TS1 (Bit 1):** Enable TS1

1 = Enable TS1 (default)

0 = Disable TS1

**Internal TS (Bit 0):** Enable internal TS

1 = Enable internal TS

0 = Disable internal TS (default)

### 13.3.1.12 Temperature Mode

| Class    | Subclass      | Name             | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | Temperature Mode | 0x494F        | H1   | 0x0 | 0x1F | 0x4     | Hex  |

|      |      |      |          |          |          |          |            |
|------|------|------|----------|----------|----------|----------|------------|
| 7    | 6    | 5    | 4        | 3        | 2        | 1        | 0          |
| RSVD | RSVD | RSVD | TS4 Mode | TS3 Mode | TS2 Mode | TS1 Mode | TSInt Mode |

**RSVD (Bit 7–5):** Reserved. Do not use.

**TS4 Mode (Bit 4):** Cell temp or FET temp

1 = FET temp (default)

0 = Cell temp

**TS3 Mode (Bit 3):** Cell temp or FET temp

1 = FET temp (default)

0 = Cell temp

**TS2 Mode (Bit 2):** Cell temp or FET temp

1 = FET temp (default)

0 = Cell temp

**TS1 Mode (Bit 1):** Cell temp or FET temp

1 = FET temp

0 = Cell temp (default)

**TSInt Mode (Bit 0):** Cell temp or FET temp

1 = FET temp

0 = Cell temp (default)

### 13.3.1.13 DA Configuration

| Class    | Subclass      | Name             | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | DA Configuration | 0x4950        | H1   | 0x0 | 0xFF | 0x12    | Hex  |

| 7     | 6     | 5         | 4     | 3               | 2  | 1   | 0   |
|-------|-------|-----------|-------|-----------------|----|-----|-----|
| FTEMP | CTEMP | EMSHUT_EN | SLEEP | IN_SYSTEM_SLEEP | NR | CC1 | CC0 |

**FTEMP (Bit 7):** FET Temperature protection source

1 = Average

0 = MAX (default)

**CTEMP (Bit 6):** Cell Temperature protection source

1 = Average

0 = MAX (default)

**EMSHUT\_EN (Bit 5):** Emergency Shutdown Enable

1 = Enable

0 = Disable

**SLEEP (Bit 4):** SLEEP mode

1 = Enable SLEEP mode (default)

0 = Disable SLEEP mode

**IN\_SYSTEM\_SLEEP (Bit 3):** In-system SLEEP mode

1 = Enable

0 = Disable (default)

**NR (Bit 2):** Use  $\overline{\text{PRES}}$  in system detection

1 = NON-REMOVABLE mode

0 = Use  $\overline{\text{PRES}}$ , REMOVABLE mode (default)

**CC1, CC0 (Bit 1,0):** Cell Count

1,1 = 4 cell

1,0 = 3 cell (default)

0,1 = 2 cell

0,0 = 1 cell

### 13.3.1.14 Balancing Configuration

| Class    | Subclass      | Name                    | Start Address | Type | Min | Max  | Default | Unit |
|----------|---------------|-------------------------|---------------|------|-----|------|---------|------|
| Settings | Configuration | Balancing Configuration | 0x4C40        | H1   | 0x0 | 0xFF | 0x1     | Hex  |

| 7    | 6    | 5    | 4    | 3    | 2   | 1   | 0  |
|------|------|------|------|------|-----|-----|----|
| RSVD | RSVD | RSVD | RSVD | RSVD | CBR | CBM | CB |

**RSVD (Bits 7–3):** Reserved. Do not use.

**CBR (Bit 2):** Cell balancing at rest

1 = Enable cell balancing at rest

0 = Disable cell balancing at rest (default)

**CBM (Bit 1):** Cell balancing method

1 = External cell balancing

0 = Internal cell balancing (default)

**CB (Bit 0):** Cell balancing

- 1 = Cell balancing enabled (default)  
0 = Cell balancing disabled

### 13.3.2 Fuse

#### 13.3.2.1 Permanent Fail Fuse A

| Class    | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|----------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Settings | Fuse     | Permanent Fail Fuse A | 0x47C0        | H1   | 0x0 | 0xFF | 0x0     | —    |

|     |      |      |     |      |      |     |     |
|-----|------|------|-----|------|------|-----|-----|
| 7   | 6    | 5    | 4   | 3    | 2    | 1   | 0   |
| QIM | SOTF | RSVD | SOT | SOCD | SOCC | SOV | SUV |

Fuse blow action for *PFStatus()* bits:

**QIM (Bit 7):** QMax Imbalance

- 1 = Enabled  
0 = Disabled (default)

**SOTF (Bit 6):** Safety Overtemperature FET

- 1 = Enabled  
0 = Disabled (default)

**RSVD (Bit 5):** Reserved. Do not use.

**SOT (Bit 4):** Safety Overtemperature

- 1 = Enabled  
0 = Disabled (default)

**SOCD (Bit 3):** Safety Overcurrent in Discharge

- 1 = Enabled  
0 = Disabled (default)

**SOCC (Bit 2):** Safety Overcurrent in Charge

- 1 = Enabled  
0 = Disabled (default)

**SOV (Bit 1):** Safety Cell Overvoltage

- 1 = Enabled  
0 = Disabled (default)

**SUV (Bit 0):** Safety Cell Undervoltage

- 1 = Enabled  
0 = Disabled (default)

#### 13.3.2.2 Permanent Fail Fuse B

| Class    | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|----------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Settings | Fuse     | Permanent Fail Fuse B | 0x47C1        | H1   | 0x0 | 0xFF | 0       | Hex  |

|      |      |      |      |      |    |     |    |
|------|------|------|------|------|----|-----|----|
| 7    | 6    | 5    | 4    | 3    | 2  | 1   | 0  |
| RSVD | RSVD | RSVD | VIMA | VIMR | CD | IMP | CB |

Fuse blow action for *PFStatus()* bits:

**RSVD (Bit 7–5):** Reserved. Do not use.

**VIMA (Bit 4):** Voltage Imbalance At Rest

1 = Enabled

0 = Disabled (default)

**VIMR (Bit 3):** Voltage Imbalance At Rest

1 = Enabled

0 = Disabled (default)

**CD (Bit 2):** Capacity Degradation

1 = Enabled

0 = Disabled (default)

**IMP (Bit 1):** Cell impedance

1 = Enabled

0 = Disabled (default)

**CB (Bit 0):** Cell balancing

1 = Enabled

0 = Disabled (default)

### 13.3.2.3 Permanent Fail Fuse C

| Class    | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|----------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Settings | Fuse     | Permanent Fail Fuse C | 0x47C2        | H1   | 0x0 | 0xFF | 0       | Hex  |

|     |      |      |      |      |      |       |       |
|-----|------|------|------|------|------|-------|-------|
| 7   | 6    | 5    | 4    | 3    | 2    | 1     | 0     |
| PTC | 2LVL | AFEC | AFER | FUSE | RSVD | DFETF | CFETF |

Fuse blow action for *PFStatus()* bits:

**PTC (Bit 7):** Permanent Fail flag Display

1 = Allow ***PFStatus[PTC]*** = 1 when PTC fault is triggered. Function should be enabled/disabled by the PTCEN pin connection.

0 = Disable the ***PFStatus[PTC]*** = 1 when PTC fault is triggered. Function should be enabled/disabled by the PTCEN pin connection.

**2LVL (Bit 6):** FUSE input indicating fuse trigger by external 2nd level protection

1 = Enabled

0 = Disabled (default)

**AFEC (Bit 5):** AFE Communication

1 = Enabled

0 = Disabled (default)

**AFER (Bit 4):** AFE Register

1 = Enabled

0 = Disabled (default)

**FUSE (Bit 3):** Fuse

1 = Enabled

0 = Disabled (default)

**RSVD (Bit 2):** Reserved. Do not use.

**DFETF (Bit 1):** Discharge FET

1 = Enabled

0 = Disabled (default)

**CFETF (Bit 0):** Charge FET

1 = Enabled

0 = Disabled (default)

#### 13.3.2.4 Permanent Fail Fuse D

| Class    | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|----------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Settings | Fuse     | Permanent Fail Fuse D | 0x47C3        | H1   | 0x0 | 0xFF | 0x0     | Hex  |

|     |     |     |     |      |     |         |     |
|-----|-----|-----|-----|------|-----|---------|-----|
| 15  | 14  | 13  | 12  | 11   | 10  | 9       | 8   |
| TS4 | TS3 | TS2 | TS1 | RSVD | DFW | OPNCELL | IFC |

Fuse blow action for *PFStatus()* bits:

**TS4 (Bit 15)**

1 = Enabled

0 = Disabled (default)

**TS3 (Bit 14)**

1 = Enabled

0 = Disabled (default)

**TS2 (Bit 13)**

1 = Enabled

0 = Disabled (default)

**TS1 (Bit 12)**

1 = Enabled

0 = Disabled (default)

**RSVD (Bit 11):** Reserved. Do not use.

**DFW (Bit 10):** DF wearout

1 = Enabled

0 = Disabled (default)

**OPNCELL (Bit 9):** Open Cell tab (tab to PCB)

1 = Enabled

0 = Disabled (default)

**IFC (Bit 8)**

1 = Enabled

0 = Disabled (default)

#### 13.3.2.5 Min Blow Fuse Voltage

| Class    | Subclass | Name                  | Start Address | Type | Min | Max   | Default |
|----------|----------|-----------------------|---------------|------|-----|-------|---------|
| Settings | Fuse     | Min Blow Fuse Voltage | 0x47C4        | I2   | 0   | 65535 | 3500    |

**Description:** Minimum voltage required to attempt fuse blow, pack based, FET failures bypass this requirement to blow the fuse.

### 13.3.2.6 Fuse Blow Timeout

| Class    | Subclass | Name                  | Start Address | Type | Min | Max | Default |
|----------|----------|-----------------------|---------------|------|-----|-----|---------|
| Settings | Fuse     | Min Blow Fuse Voltage | 0x47C6        | U1   | 0   | 255 | 30      |

**Description:** Minimum time to keep the fuse blow voltage high

### 13.3.3 BTP

#### 13.3.3.1 Init Discharge Set

| Class    | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Settings | BTP      | Init Discharge Set | 0x47CC        | I2   | 0   | 32767 | 150     | mA   |

**Description:** Initial value for *BTPDischargeSet()*

#### 13.3.3.2 Init Charge Set

| Class    | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Settings | BTP      | Init Discharge Set | 0x47C2        | I2   | 0   | 32767 | 175     | mA   |

**Description:** Initial value for *BTPChargeSet()*

### 13.3.4 Protection

#### 13.3.4.1 Protection Configuration

| Class    | Subclass   | Name                     | Start Address | Type | Min  | Max  | Default | Unit |
|----------|------------|--------------------------|---------------|------|------|------|---------|------|
| Settings | Protection | Protection Configuration | 0x4840        | H1   | 0x00 | 0x03 | 0x00    | Hex  |

| 7    | 6    | 5    | 4    | 3    | 2    | 1             | 0        |
|------|------|------|------|------|------|---------------|----------|
| RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | CUV_RECOV_CHG | SUV_MODE |

**RSVD (Bits 7–2):** Reserved. Do not use.

**CUV\_RECOV\_CHG (Bit 1):** Require charge to recover *SafetyStatus()*[CUV]

1 = Enabled (default)

0 = Disabled

**SUV\_MODE (Bit 0):** Copper Deposition check for *PFStatus()*[CUV]

1 = Enabled (default)

0 = Disabled

#### 13.3.4.2 Enabled Protections A

| Class    | Subclass   | Name                  | Start Address | Type | Min  | Max  | Default | Unit |
|----------|------------|-----------------------|---------------|------|------|------|---------|------|
| Settings | Protection | Enabled Protections A | 0x4841        | H1   | 0x00 | 0xFF | 0xFF    | Hex  |

| 7     | 6    | 5    | 4    | 3    | 2    | 1   | 0   |
|-------|------|------|------|------|------|-----|-----|
| AOLDL | AOLD | OCD2 | OCD1 | OCC2 | OCC1 | COV | CUV |

**AOLDL (Bit 7):** Overload in Discharge latch

1 = Enabled (default)

0 = Disabled

**AOLD (Bit 6):** Overload in Discharge

1 = Enabled (default)

0 = Disabled

**OCD2 (Bit 5):** Overcurrent in Discharge 2nd Tier

1 = Enabled (default)

0 = Disabled

**OCD1 (Bit 4):** Overcurrent in Discharge 1st Tier

1 = Enabled (default)

0 = Disabled

**OCC2 (Bit 3):** Overcurrent in Charge 2nd Tier

1 = Enabled (default)

0 = Disabled

**OCC1 (Bit 2):** Overcurrent in Charge 1st Tier

1 = Enabled (default)

0 = Disabled

**COV (Bit 1):** Cell Overvoltage

1 = Enabled (default)

0 = Disabled

**CUV (Bit 0):** Cell Undervoltage

1 = Enabled (default)

0 = Disabled

### 13.3.4.3 Enabled Protections B

| Class    | Subclass   | Name                  | Start Address | Type | Min  | Max  | Default | Unit |
|----------|------------|-----------------------|---------------|------|------|------|---------|------|
| Settings | Protection | Enabled Protections B | 0x4842        | H1   | 0x00 | 0xFF | 0xFF    | —    |

| 7    | 6    | 5   | 4   | 3     | 2    | 1     | 0    |
|------|------|-----|-----|-------|------|-------|------|
| RSVD | CUVC | OTD | OTC | ASCDL | ASCD | ASCCL | ASCC |

**RSVD (Bit 7):** Reserved. Do not use.

**CUVC (Bit 6):** I\*R compensated CUV

1 = Enabled (default)

0 = Disabled

**OTD (Bit 5):** Overtemperature in discharge

1 = Enabled (default)

0 = Disabled

**OTC (Bit 4):** Overtemperature in charge



- 1 = Enabled (default)
- 0 = Disabled

**ASCDL (Bit 3):** Short circuit in discharge latch

- 1 = Enabled (default)
- 0 = Disabled

**ASCD (Bit 2):** Short circuit in discharge

- 1 = Enabled (default)
- 0 = Disabled

**ASCCL (Bit 1):** Short circuit in charge latch

- 1 = Enabled (default)
- 0 = Disabled

**ASCC (Bit 0):** Short circuit in charge

- 1 = Enabled (default)
- 0 = Disabled

#### 13.3.4.4 Enabled Protections C

| Class    | Subclass   | Name                  | Start Address | Type | Min  | Max  | Default | Unit |
|----------|------------|-----------------------|---------------|------|------|------|---------|------|
| Settings | Protection | Enabled Protections C | 0x4843        | H1   | 0x00 | 0xFF | 0xFF    | Hex  |

|      |    |      |     |      |     |      |     |
|------|----|------|-----|------|-----|------|-----|
| 7    | 6  | 5    | 4   | 3    | 2   | 1    | 0   |
| CHGC | OC | RSVD | CTO | RSVD | PTO | HWDF | OTF |

**CHGC (Bit 7):** *ChargingCurrent()* higher than requested

- 1 = Enabled (default)
- 0 = Disabled

**OC (Bit 6):** Overcharge

- 1 = Enabled (default)
- 0 = Disabled

**RSVD (Bit 5):** Reserved. Do not use.

**CTO (Bit 4):** Charging timeout

- 1 = Enabled (default)
- 0 = Disabled

**RSVD (Bit 3):** Reserved. Do not use.

**PTO (Bit 2):** Pre-charging timeout

- 1 = Enabled (default)
- 0 = Disabled

**HWDF (Bit 1):** SBS Host watchdog timeout

- 1 = Enabled (default)
- 0 = Disabled

**OTF (Bit 0):** FET overtemperature

- 1 = Enabled (default)
- 0 = Disabled

### 13.3.4.5 Enabled Protections D

| Class    | Subclass   | Name                  | Start Address | Type | Min  | Max  | Default | Unit |
|----------|------------|-----------------------|---------------|------|------|------|---------|------|
| Settings | Protection | Enabled Protections D | 0x4844        | H1   | 0x00 | 0xFF | 0xFF    | Hex  |

|      |      |      |      |     |     |       |      |
|------|------|------|------|-----|-----|-------|------|
| 7    | 6    | 5    | 4    | 3   | 2   | 1     | 0    |
| RSVD | RSVD | RSVD | RSVD | UTD | UTC | PCHGV | CHGV |

**RSVD (Bits 7–4):** Reserved. Do not use.

**UTD (Bit 3):** Under temperature while not charging

1 = Enabled (default)

0 = Disabled

**UTC (Bit 2):** Under temperature while charging

1 = Enabled (default)

0 = Disabled

**PCHGV (Bit 1):** *ChargingVoltage()* higher than requested in precharge

1 = Enabled (default)

0 = Disabled

**CHGV (Bit 0):** *ChargingVoltage()* higher than requested

1 = Enabled (default)

0 = Disabled

## 13.3.5 Permanent Failure

### 13.3.5.1 Enabled PF A

| Class    | Subclass          | Name         | Start Address | Type | Min  | Max  | Default | Unit |
|----------|-------------------|--------------|---------------|------|------|------|---------|------|
| Settings | Permanent Failure | Enabled PF A | 0x48BE        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

|     |      |      |     |       |       |     |     |
|-----|------|------|-----|-------|-------|-----|-----|
| 7   | 6    | 5    | 4   | 3     | 2     | 1   | 0   |
| QIM | SOTF | RSVD | SOT | S OCD | S OCC | SOV | SUV |

**QIM (Bit 7):** QMax Imbalance

1 = Enabled (default)

0 = Disabled

**OTF (Bit 6):** Overtemperature FET

1 = Enabled (default)

0 = Disabled

**RSVD (Bit 5):** Reserved. Do not use.

**PF\_OTCE (Bit 4):** Overtemperature

1 = Enabled (default)

0 = Disabled

**RSVD (Bits 3–2):** Reserved. Do not use.

**SOT (Bit 4):** Safety Overtemperature

- 1 = Enabled
- 0 = Disabled (default)

**SOCD (Bit 3):** Safety Overcurrent in Discharge

- 1 = Enabled
- 0 = Disabled (default)

**SOCC (Bit 2):** Safety Overcurrent in Charge

- 1 = Enabled
- 0 = Disabled (default)

**SOV (Bit 1):** Safety Cell Overvoltage

- 1 = Enabled
- 0 = Disabled (default)

**SUV (Bit 0):** Safety Cell Undervoltage

- 1 = Enabled
- 0 = Disabled (default)

**13.3.5.2 Enabled PF B**

| Class    | Subclass          | Name         | Start Address | Type | Min  | Max  | Default | Unit |
|----------|-------------------|--------------|---------------|------|------|------|---------|------|
| Settings | Permanent Failure | Enabled PF B | 0x48BF        | H1   | 0x00 | 0xFF | 0x00    | —    |

|      |      |      |      |      |    |     |    |
|------|------|------|------|------|----|-----|----|
| 7    | 6    | 5    | 4    | 3    | 2  | 1   | 0  |
| RSVD | RSVD | RSVD | VIMA | VIMR | CD | IMP | CB |

**RSVD (Bits 7–5):** Reserved. Do not use.

**VIMA (Bit 4):** Voltage imbalance At Rest

- 1 = Enabled (default)
- 0 = Disabled

**VIMR (Bit 3):** Voltage imbalance At Rest

- 1 = Enabled (default)
- 0 = Disabled

**CD (Bit 2):** Capacity Degradation

- 1 = Enabled (default)
- 0 = Disabled

**IMP (Bit 1):** Cell impedance

- 1 = Enabled (default)
- 0 = Disabled

**CB (Bit 0):** Cell balancing

- 1 = Enabled (default)
- 0 = Disabled

**13.3.5.3 Enabled PF C**

| Class    | Subclass          | Name         | Start Address | Type | Min  | Max  | Default | Unit |
|----------|-------------------|--------------|---------------|------|------|------|---------|------|
| Settings | Permanent Failure | Enabled PF C | 0x48C0        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

| 7   | 6    | 5    | 4    | 3    | 2    | 1    | 0     |
|-----|------|------|------|------|------|------|-------|
| PTC | 2LVL | AFEC | AFER | FUSE | RSVD | DFET | CFETF |

**PTC (Bit 7):** Permanent Fail Flag Display

1 = Allow **PFStatus[PTC]** = 1 when PTC fault is triggered. Function should be enabled/disabled by the PTCEN pin connection.

0 = Disable the **PFStatus[PTC]** = 1 when PTC fault is triggered. Function should be enabled/disabled by the PTCEN pin connection.

**2LVL (Bit 6):** FUSE input indicating fuse trigger by external 2nd level protection

1 = Enabled (default)

0 = Disabled

**AFEC (Bit 5):** AFE Communication

1 = Enabled (default)

0 = Disabled

**AFER (Bit 4):** AFE Register

1 = Enabled (default)

0 = n/a

**FUSE (Bit 3):** Fuse

1 = Enabled (default)

0 = Disabled

**RSVD (Bit 2):** Reserved. Do not use.

**DFET (Bit 1):** Discharge FET

1 = Enabled (default)

0 = Disabled

**CFETF (Bit 0):** Charge FET

1 = Enabled (default)

0 = Disabled

### 13.3.5.4 Enabled PF D

| Class    | Subclass          | Name         | Start Address | Type | Min  | Max  | Default | Unit |
|----------|-------------------|--------------|---------------|------|------|------|---------|------|
| Settings | Permanent Failure | Enabled PF D | 0x48C1        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

| 7   | 6   | 5   | 4   | 3    | 2    | 1       | 0    |
|-----|-----|-----|-----|------|------|---------|------|
| TS4 | TS3 | TS2 | TS1 | RSVD | RSVD | OPNCELL | RSVD |

**TS4 (Bit 7)**

1 = Enabled (default)

0 = Disabled

**TS3 (Bit 6)**

1 = Enabled (default)

0 = Disabled

**TS2 (Bit 5)**

1 = Enabled (default)

0 = Disabled

**TS1 (Bit 4)**

- 1 = Enabled (default)
- 0 = Disabled

**RSVD (Bits 3–2):** Reserved. Do not use.

**OPNCELL (Bit 1):** Open Cell tab (tab to PCB)

- 1 = Enabled (default)
- 0 = Disabled

**RSVD (Bit 0):** Reserved. Do not use.

### 13.3.6 AFE

#### 13.3.6.1 AFE Protection Control

| Class         | Subclass | Name                   | Start Address | Type | Min  | Max  | Default | Unit |
|---------------|----------|------------------------|---------------|------|------|------|---------|------|
| Configuration | AFE      | AFE Protection Control | 0x4952        | H1   | 0x00 | 0xFF | 0x70    | Hex  |

|        |        |        |        |      |      |        |      |
|--------|--------|--------|--------|------|------|--------|------|
| 7      | 6      | 5      | 4      | 3    | 2    | 1      | 0    |
| RSTRIM | RSTRIM | RSTRIM | RSTRIM | RSVD | RSVD | SCDDx2 | RSNS |

**RSTRIM (Bits 7–4):** *Unsupport* function. Should leave the default setting 0x7. Changing this setting may cause an error to the AFE current protection accuracy.

**RSVD (Bits 3–2):** Reserved. Do not use.

**SCDDx2 (Bit 1):** Double SCD Delay Times

- 1 = 2 × SCD delay times
- 0 = Normal SCD delay times (default)

**RSNS (Bit 0):** AOLD, ASCC, ASCD1, ASCD2 Thresholds

- 1 = Normal AFE Protection Thresholds
- 0 = 0.5 × AFE Protection Thresholds (default)

### 13.3.7 ZVCHG Exit Threshold

| Class         | Subclass | Name                 | Start Address | Type | Min | Max    | Default | Unit |
|---------------|----------|----------------------|---------------|------|-----|--------|---------|------|
| Configuration | AFE      | ZVCHG Exit Threshold | 0x4958        | I2   | 0x0 | 0xFFFF | 0x0000  | mV   |

**Description:** *Voltage()* threshold where the gauge will exit ZVCHG mode when CFET is used for precharging.

## 13.4 Manufacturing

### 13.4.1 Manufacturing Status Init

| Class    | Subclass      | Name                      | Start Address | Type | Min | Max    | Default | Unit |
|----------|---------------|---------------------------|---------------|------|-----|--------|---------|------|
| Settings | Manufacturing | Manufacturing Status Init | 0x4580        | H2   | 0x0 | 0xFFFF | 0x0000  | Hex  |

|        |       |       |        |          |      |        |         |
|--------|-------|-------|--------|----------|------|--------|---------|
| 15     | 14    | 13    | 12     | 11       | 10   | 9      | 8       |
| RSVD   | RSVD  | RSVD  | RSVD   | RSVD     | RSVD | LED_EN | FUSE_EN |
| 7      | 6     | 5     | 4      | 3        | 2    | 1      | 0       |
| BBR_EN | PF_EN | LF_EN | FET_EN | GAUGE_EN | RSVD | RSVD   | RSVD    |

**RSVD (Bits 15–10):** Reserved. Do not use.

**LED\_EN (Bit 9):** LED Display

1 = Enabled

0 = Disabled

**FUSE\_EN (Bit 8):** FUSE action

1 = Enabled

0 = Disabled (default)

**BBR\_EN (Bit 7):** Black Box Recorder

1 = Enabled

0 = Disabled (default)

**PF\_EN (Bit 6):** Permanent Fail

1 = Enabled

0 = Disabled (default)

**LF\_EN (Bit 5):** Lifetime Data Collection

1 = Enabled

0 = Disabled

**FET\_EN (Bit 4):** FET action

1 = Enabled

0 = Disabled (default)

**GAUGE\_EN (Bit 3):** Gauging

1 = Enabled

0 = Disabled (default)

**RSVD (Bits 2–0):** Reserved. Do not use.

## 13.5 Advanced Charging Algorithm

### 13.5.1 Temperature Ranges

#### 13.5.1.1 T1 Temp

| Class                        | Subclass           | Name    | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | T1 Temp | 0x490D        | I1   | –128 | 127 | 0       | °C   |

**Description:** T1 low temperature range lower limit

#### 13.5.1.2 T2 Temp

| Class                        | Subclass           | Name    | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | T2 Temp | 0x490E        | I1   | -128 | 127 | 12      | °C   |

**Description:** T2 low temperature range to standard temperature range

### 13.5.1.3 T5 Temp

| Class                        | Subclass           | Name    | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | T5 Temp | 0x490F        | I1   | -128 | 127 | 20      | °C   |

**Description:** T5 recommended temperature range lower limit

### 13.5.1.4 T6 Temp

| Class                        | Subclass           | Name    | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | T6 Temp | 0x4910        | I1   | -128 | 127 | 25      | °C   |

**Description:** T6 recommended temperature range upper limit

### 13.5.1.5 T3 Temp

| Class                        | Subclass           | Name    | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | T3 Temp | 0x4911        | I1   | -128 | 127 | 30      | °C   |

**Description:** T3 standard temperature range to high temperature range

### 13.5.1.6 T4 Temp

| Class                        | Subclass           | Name    | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | T4 Temp | 0x4912        | I1   | -128 | 127 | 55      | °C   |

**Description:** T4 high temperature range upper limit

### 13.5.1.7 Hysteresis

| Class                        | Subclass           | Name            | Start Address | Type | Min  | Max | Default | Unit |
|------------------------------|--------------------|-----------------|---------------|------|------|-----|---------|------|
| Advanced Charging Algorithms | Temperature Ranges | Hysteresis Temp | 0x4913        | I1   | -128 | 127 | 1       | °C   |

**Description:** Temperature Hysteresis, applied when temperature is decreasing.

## 13.5.2 Low Temp Charging

### 13.5.2.1 Voltage

| Class                        | Subclass          | Name    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|---------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Low Temp Charging | Voltage | 0x4914        | I2   | 0   | 32767 | 4000    | mV   |

**Description:** Low temperature range *ChargingVoltage()*

### 13.5.2.2 Current Low

| Class                        | Subclass          | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Low Temp Charging | Current Low | 0x4916        | I2   | 0   | 32767 | 132     | mA   |

**Description:** Low temperature range low voltage range *ChargingCurrent()*

### 13.5.2.3 Current Med

| Class                        | Subclass          | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Low Temp Charging | Current Med | 0x4918        | I2   | 0   | 32767 | 352     | mA   |

**Description:** Low temperature range medium voltage range *ChargingCurrent()*

### 13.5.2.4 Current High

| Class                        | Subclass          | Name         | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|--------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Low Temp Charging | Current High | 0x491A        | I2   | 0   | 32767 | 264     | mA   |

**Description:** Low temperature range high voltage range *ChargingCurrent()*

## 13.5.3 Standard Temp Charging

### 13.5.3.1 Voltage

| Class                        | Subclass               | Name    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|------------------------|---------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Standard Temp Charging | Voltage | 0x491C        | I2   | 0   | 32767 | 4200    | mV   |

**Description:** Standard temperature range *ChargingVoltage()*

### 13.5.3.2 Current Low

| Class                        | Subclass               | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|------------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Standard Temp Charging | Current Low | 0x491E        | I2   | 0   | 32767 | 1980    | mA   |

**Description:** Standard temperature range low voltage range *ChargingCurrent()*

### 13.5.3.3 Current Med



| Class                        | Subclass               | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|------------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Standard Temp Charging | Current Med | 0x4920        | I2   | 0   | 32767 | 4004    | mA   |

**Description:** Standard temperature range medium voltage range *ChargingCurrent()*

### 13.5.3.4 Current High

| Class                        | Subclass               | Name         | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|------------------------|--------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Standard Temp Charging | Current High | 0x4922        | I2   | 0   | 32767 | 2992    | mA   |

**Description:** Standard temperature range high voltage range *ChargingCurrent()*

## 13.5.4 High Temp Charging

### 13.5.4.1 Voltage

| Class                        | Subclass           | Name    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------|---------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | High Temp Charging | Voltage | 0x4924        | I2   | 0   | 32767 | 4000    | mV   |

**Description:** High temperature range *ChargingVoltage()*

### 13.5.4.2 Current Low

| Class                        | Subclass           | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | High Temp Charging | Current Low | 0x4926        | I2   | 0   | 32767 | 1012    | mA   |

**Description:** High temperature range low voltage range *ChargingCurrent()*

### 13.5.4.3 Current Med

| Class                        | Subclass           | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | High Temp Charging | Current Med | 0x4928        | I2   | 0   | 32767 | 1980    | mA   |

**Description:** High temperature range medium voltage range *ChargingCurrent()*

### 13.5.4.4 Current High

| Class                        | Subclass           | Name         | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------|--------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | High Temp Charging | Current High | 0x492A        | I2   | 0   | 32767 | 1496    | mA   |

**Description:** High temperature range high voltage range *ChargingCurrent()*

### 13.5.5 Rec Temp Charging

#### 13.5.5.1 Voltage

| Class                        | Subclass          | Name    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|---------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Rec Temp Charging | Voltage | 0x492C        | I2   | 0   | 32767 | 4100    | mV   |

**Description:** Recommended temperature range *ChargingVoltage()*

#### 13.5.5.2 Current Low

| Class                        | Subclass          | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Rec Temp Charging | Current Low | 0x492E        | I2   | 0   | 32767 | 2508    | mA   |

**Description:** Recommended temperature range low voltage range *ChargingCurrent()*

#### 13.5.5.3 Current Med

| Class                        | Subclass          | Name        | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|-------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Rec Temp Charging | Current Med | 0x4930        | I2   | 0   | 32767 | 4488    | mA   |

**Description:** Recommended temperature range medium voltage range *ChargingCurrent()*

#### 13.5.5.4 Current High

| Class                        | Subclass          | Name         | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|-------------------|--------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Rec Temp Charging | Current High | 0x4932        | I2   | 0   | 32767 | 3520    | mA   |

**Description:** Recommended temperature range high voltage range *ChargingCurrent()*

### 13.5.6 Pre-Charging

| Class                        | Subclass | Name    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|----------|---------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | PCHG     | Current | 0x4934        | I2   | 0   | 32767 | 88      | mA   |

**Description:** Precharge *ChargingCurrent()*

### 13.5.7 Maintenance Charging

| Class                        | Subclass | Name    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|----------|---------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | MCHG     | Current | 0x4936        | I2   | 0   | 32767 | 44      | mA   |

**Description:** Maintenance *ChargingCurrent()*

## 13.5.8 Voltage Range

### 13.5.8.1 Precharge Start Voltage

| Class                        | Subclass      | Name                    | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|---------------|-------------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Voltage Range | Precharge Start Voltage | 0x4938        | I2   | 0   | 32767 | 2500    | mV   |

**Description:** Min cell voltage to enter PRECHARGE mode

### 13.5.8.2 Charging Voltage Low

| Class                        | Subclass      | Name                 | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|---------------|----------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Voltage Range | Charging Voltage Low | 0x493A        | I2   | 0   | 32767 | 2900    | mV   |

**Description:** Precharge Voltage range to Charging Voltage Low range

### 13.5.8.3 Charging Voltage Med

| Class                        | Subclass      | Name                 | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|---------------|----------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Voltage Range | Charging Voltage Med | 0x493C        | I2   | 0   | 32767 | 3600    | mV   |

**Description:** Charging Voltage Low range to Charging Voltage Med range

### 13.5.8.4 Charging Voltage High

| Class                        | Subclass      | Name                  | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|---------------|-----------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Voltage Range | Charging Voltage High | 0x493E        | I2   | 0   | 32767 | 4000    | mV   |

**Description:** Charging Voltage Med to Charging Voltage High range

### 13.5.8.5 Charging Voltage Hysteresis

| Class                        | Subclass      | Name                        | Start Address | Type | Min | Max | Default | Unit |
|------------------------------|---------------|-----------------------------|---------------|------|-----|-----|---------|------|
| Advanced Charging Algorithms | Voltage Range | Charging Voltage Hysteresis | 0x4940        | U1   | 0   | 255 | 0       | mV   |

**Description:** Charging Voltage Hysteresis applied when voltage is decreasing

## 13.5.9 Termination Config

### 13.5.9.1 Charge Term Taper Current

| Class                        | Subclass           | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------|---------------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Termination Config | Charge Term Taper Current | 0x4941        | I2   | 0   | 32767 | 250     | mA   |

**Description:** Valid Charge Termination taper current qualifier threshold

### 13.5.9.2 Charge Term Voltage

| Class                        | Subclass           | Name                | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------|---------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Termination Config | Charge Term Voltage | 0x4945        | I2   | 0   | 32767 | 75      | mV   |

**Description:** Valid Charge Termination delta voltage qualifier, max cell-based

## 13.5.10 Charging Rate of Change

### 13.5.10.1 Current Rate

| Class                        | Subclass                | Name         | Start Address | Type | Min | Max | Default | Unit    |
|------------------------------|-------------------------|--------------|---------------|------|-----|-----|---------|---------|
| Advanced Charging Algorithms | Charging Rate of Change | Current Rate | 0x4948        | U1   | 1   | 255 | 1       | steps/s |

**Description:** Number of steps to add between any two *ChargingCurrent()* settings

### 13.5.10.2 Voltage Rate

| Class                        | Subclass                | Name         | Start Address | Type | Min | Max | Default | Unit    |
|------------------------------|-------------------------|--------------|---------------|------|-----|-----|---------|---------|
| Advanced Charging Algorithms | Charging Rate of Change | Voltage Rate | 0x4949        | U1   | 1   | 255 | 1       | steps/s |

**Description:** Number of steps to add between any two *ChargingVoltage()* settings

## 13.5.11 Charge Loss Compensation

### 13.5.11.1 CCC Current Threshold

| Class                        | Subclass                 | Name                  | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------------|-----------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Charge Loss Compensation | CCC Current Threshold | 0x494A        | I2   | 0   | 32767 | 3520    | mA   |

**Description:** CONSTANT CURRENT CHARGE mode *ChargingCurrent()* threshold to activate Charge Loss Compensation

### 13.5.11.2 CCC Voltage Threshold

| Class                        | Subclass                 | Name                  | Start Address | Type | Min | Max   | Default | Unit |
|------------------------------|--------------------------|-----------------------|---------------|------|-----|-------|---------|------|
| Advanced Charging Algorithms | Charge Loss Compensation | CCC Voltage Threshold | 0x494C        | I2   | 0   | 32767 | 4200    | mV   |

**Description:** CONSTANT CURRENT CHARGE mode max *ChargingVoltage()* increase limit

## 13.5.12 Cell Balancing Config

### 13.5.12.1 Balance Time per mAh Cell 1

| Class                        | Subclass              | Name                        | Start Address | Type | Min | Max   | Default | Unit  |
|------------------------------|-----------------------|-----------------------------|---------------|------|-----|-------|---------|-------|
| Advanced Charging Algorithms | Cell Balancing Config | Balance Time per mAh Cell 1 | 0x4C41        | U2   | 0   | 65535 | 367     | s/mAh |

**Description:** Required balance time per mAh for Cell 1. For information on how to calculate balancing time, see [Section 7.1](#).

### 13.5.12.2 Balance Time per mAh Cell 2–4

| Class                        | Subclass              | Name                          | Start Address | Type | Min | Max   | Default | Unit  |
|------------------------------|-----------------------|-------------------------------|---------------|------|-----|-------|---------|-------|
| Advanced Charging Algorithms | Cell Balancing Config | Balance Time per mAh Cell 2–4 | 0x4C43        | U2   | 0   | 65535 | 514     | s/mAh |

**Description:** Required balance time per mAh for cells 2 to 4. For information on how to calculate balancing time, see [Section 7.1](#).

### 13.5.12.3 Min Start Balance Delta

| Class                        | Subclass              | Name                    | Start Address | Type | Min | Max | Default | Unit |
|------------------------------|-----------------------|-------------------------|---------------|------|-----|-----|---------|------|
| Advanced Charging Algorithms | Cell Balancing Config | Min Start Balance Delta | 0x4C45        | U1   | 0   | 255 | 3       | mV   |

**Description:** Minimum cell voltage delta to start cell balancing during **Relax Balance Interval** checks. This condition is checked in RELAX mode and so it only applies if cell balancing at rest is enabled.

### 13.5.12.4 Relax Balance Interval

| Class                        | Subclass              | Name                   | Start Address | Type | Min | Max        | Default | Unit |
|------------------------------|-----------------------|------------------------|---------------|------|-----|------------|---------|------|
| Advanced Charging Algorithms | Cell Balancing Config | Relax Balance Interval | 0x4C46        | U4   | 0   | 4294967295 | 18000   | s    |

**Description:** Interval during RELAX mode to check for cell imbalance. This parameter applies to cell balancing at rest only.

### 13.5.12.5 Min RSOC for Balancing

| Class                        | Subclass              | Name                   | Start Address | Type | Min | Max | Default | Unit |
|------------------------------|-----------------------|------------------------|---------------|------|-----|-----|---------|------|
| Advanced Charging Algorithms | Cell Balancing Config | Min RSOC for Balancing | 0x4C4A        | U1   | 0   | 100 | 80      | %    |

**Description:** Minimum *RelativeStateOfCharge()* threshold for cell balancing. This condition is checked during relaxation and so it only applies if cell balancing at rest is enabled.

## 13.6 Power

### 13.6.1 Power

| Class | Subclass | Name                 | Start Address | Type | Min | Max   | Default | Unit |
|-------|----------|----------------------|---------------|------|-----|-------|---------|------|
| Power | Power    | Valid Update Voltage | 0x47D0        | I2   | 0   | 32767 | 3500    | mV   |

**Description:** Min stack voltage threshold for Flash update

## 13.6.2 Shutdown

### 13.6.2.1 Shutdown Voltage

| Class | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit |
|-------|----------|------------------|---------------|------|-----|-------|---------|------|
| Power | Shutdown | Shutdown Voltage | 0x47D2        | I2   | 0   | 32767 | 1750    | mV   |

**Description:** Cell-based shutdown voltage trip threshold

### 13.6.2.2 Shutdown Time

| Class | Subclass | Name          | Start Address | Type | Min | Max | Default | Unit |
|-------|----------|---------------|---------------|------|-----|-----|---------|------|
| Power | Shutdown | Shutdown Time | 0x47D4        | U2   | 0   | 255 | 10      | s    |

**Description:** Cell-based shutdown voltage trip delay

### 13.6.2.3 Charger Present Threshold

| Class | Subclass | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|-------|----------|---------------------------|---------------|------|-----|-------|---------|------|
| Power | Shutdown | Charger Present Threshold | 0x47D5        | I2   | 0   | 32767 | 3000    | mV   |

**Description:** Pack pin charger present detect threshold

## 13.6.3 Sleep

### 13.6.3.1 Sleep Current

| Class | Subclass | Name          | Start Address | Type | Min | Max   | Default | Unit |
|-------|----------|---------------|---------------|------|-----|-------|---------|------|
| Power | Sleep    | Sleep Current | 0x47D7        | I2   | 0   | 32767 | 10      | mA   |

**Description:** |*Current()*| threshold to enter SLEEP mode. If this parameter is set to 0, then the **deadband** will effectively become the Sleep Current setting because any current blow the **deadband** will set the *Current()* = 0 mA.

### 13.6.3.2 Bus Timeout

| Class | Subclass | Name        | Start Address | Type | Min | Max | Default | Unit |
|-------|----------|-------------|---------------|------|-----|-----|---------|------|
| Power | Sleep    | Bus Timeout | 0x47D9        | U1   | 0   | 255 | 5       | s    |

**Description:** Bus low or no communication time to enter SLEEP mode

### 13.6.3.3 Voltage Time

| Class | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------|----------|--------------|---------------|------|-----|-----|---------|------|
| Power | Sleep    | Voltage Time | 0x47DE        | U1   | 0   | 255 | 5       | s    |

**Description:** *Voltage()* sampling period in SLEEP mode

### 13.6.3.4 Current Time

| Class | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------|----------|--------------|---------------|------|-----|-----|---------|------|
| Power | Sleep    | Current Time | 0x47DF        | U1   | 0   | 255 | 20      | s    |

**Description:** *Current()* sampling period in SLEEP mode

### 13.6.3.5 Wake Comparator

| Class | Subclass | Name            | Start Address | Type | Min  | Max  | Default | Unit |
|-------|----------|-----------------|---------------|------|------|------|---------|------|
| Power | Sleep    | Wake Comparator | 0x47E0        | H1   | 0x00 | 0xFF | 0x00    | —    |

|      |      |      |      |     |     |      |      |
|------|------|------|------|-----|-----|------|------|
| 7    | 6    | 5    | 4    | 3   | 2   | 1    | 0    |
| RSVD | RSVD | RSVD | RSVD | WK1 | WK0 | RSVD | RSVD |

**RSVD (Bits 7–4):** Reserved. Do not use.

**WK1, WK0 (Bits 3–2):** Wake Comparator Threshold

1,1 =  $\pm 5$  mV

1,0 =  $\pm 2.5$  mV

0,1 =  $\pm 1.25$  mV

0,0 =  $\pm 0.625$  mV

**RSVD (Bits 1–0):** Reserved. Do not use.

## 13.6.4 Ship

### 13.6.4.1 FET Off Time

| Class | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------|----------|--------------|---------------|------|-----|-----|---------|------|
| Power | Ship     | FET Off Time | 0x47E1        | U1   | 0   | 127 | 10      | s    |

**Description:** Delay time to turn off FETs prior to entering SHUTDOWN mode. This setting should not be longer than the **Ship Delay** setting.

### 13.6.4.2 Delay

| Class | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------|----------|-------|---------------|------|-----|-----|---------|------|
| Power | Ship     | Delay | 0x47E2        | U1   | 0   | 254 | 20      | s    |

**Description:** Delay time to enter SHUTDOWN mode after FETs are turned off.

### 13.6.4.3 Auto Ship Time

| Class | Subclass | Name           | Start Address | Type | Min | Max   | Default | Unit |
|-------|----------|----------------|---------------|------|-----|-------|---------|------|
| Power | Ship     | Auto Ship time | 0x47E3        | U2   | 0   | 65535 | 1440    | min  |

**Description:** The device will automatically enter SHUTDOWN mode after staying in SLEEP mode without communication for this amount of time when **Power Config[AUTO\_SHIP\_EN] = 1**.

## 13.6.5 Power Off

| Class | Subclass  | Name    | Start Address | Type | Min | Max   | Default | Unit |
|-------|-----------|---------|---------------|------|-----|-------|---------|------|
| Power | Power Off | Timeout | 0x47E5        | U2   | 0   | 65535 | 30      | min  |

**Description:** Timeout to exit the Emergency Shutdown condition

### 13.6.6 Manual FET Control

| Class | Subclass           | Name      | Start Address | Type | Min | Max | Default | Unit |
|-------|--------------------|-----------|---------------|------|-----|-----|---------|------|
| Power | Manual FET Control | MFC Delay | 0x47E7        | U1   | 0   | 255 | 60      | min  |

**Description:** Delay time to turn off FETs through MFC

## 13.7 LED Support

### 13.7.1 LED Config

#### 13.7.1.1 LED Flash Period

| Class       | Subclass   | Name             | Start Address | Type | Min | Max   | Default | Unit        |
|-------------|------------|------------------|---------------|------|-----|-------|---------|-------------|
| LED Support | LED Config | LED Flash Period | 0x47E9        | U2   | 32  | 65535 | 512     | 488 $\mu$ s |

**Description:** LED Flashing period for alarm display

#### 13.7.1.2 LED Blink Period

| Class       | Subclass   | Name             | Start Address | Type | Min | Max   | Default | Unit        |
|-------------|------------|------------------|---------------|------|-----|-------|---------|-------------|
| LED Support | LED Config | LED Blink Period | 0x47EB        | U2   | 32  | 65535 | 1024    | 488 $\mu$ s |

**Description:** LED Blinking period for state-of-charge display

#### 13.7.1.3 LED Delay

| Class       | Subclass   | Name      | Start Address | Type | Min | Max   | Default | Unit        |
|-------------|------------|-----------|---------------|------|-----|-------|---------|-------------|
| LED Support | LED Config | LED Delay | 0x47ED        | U2   | 16  | 65535 | 100     | 488 $\mu$ s |

**Description:** Delay time from LED to LED for state-of-charge display

#### 13.7.1.4 LED Hold Time

| Class       | Subclass   | Name          | Start Address | Type | Min | Max | Default | Unit   |
|-------------|------------|---------------|---------------|------|-----|-----|---------|--------|
| LED Support | LED Config | LED Hold Time | 0x47EF        | U1   | 1   | 63  | 16      | 0.25 s |

**Description:** LED display active time

#### 13.7.1.5 CHG Flash Alarm

| Class       | Subclass   | Name            | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|-----------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | CHG Flash Alarm | 0x4046        | I1   | 0   | 100 | 10      | %    |



**Description:** *RelativeStateOfCharge()* alarm threshold during charging

### 13.7.1.6 CHG Thresh 1

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | CHG Thresh 1 | 0x47F0        | I1   | 0   | 100 | 0       | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED1 during charging

### 13.7.1.7 CHG Thresh 2

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | CHG Thresh 2 | 0x47F1        | I1   | 0   | 100 | 20      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED2 during charging

### 13.7.1.8 CHG Thresh 3

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | CHG Thresh 3 | 0x47F2        | I1   | 0   | 100 | 40      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED3 during charging

### 13.7.1.9 CHG Thresh 4

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | CHG Thresh 4 | 0x47F3        | I1   | 0   | 100 | 60      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED4 during charging

### 13.7.1.10 CHG Thresh 5

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | CHG Thresh 5 | 0x47F4        | I1   | 0   | 100 | 80      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED5 during charging

### 13.7.1.11 DSG Flash Alarm

| Class       | Subclass   | Name            | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|-----------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | DSG Flash Alarm | 0x47F5        | I1   | 0   | 100 | 10      | %    |

**Description:** *RelativeStateOfCharge()* alarm threshold during discharging

### 13.7.1.12 DSG Thresh 1

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | DSG Thresh 1 | 0x47F6        | I1   | 0   | 100 | 0       | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED1 during discharging

### 13.7.1.13 DSG Thresh 2

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | DSG Thresh 2 | 0x47F7        | I1   | 0   | 100 | 20      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED2 during discharging

### 13.7.1.14 DSG Thresh 3

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | DSG Thresh 3 | 0x47F8        | I1   | 0   | 100 | 40      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED3 during discharging

### 13.7.1.15 DSG Thresh 4

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | DSG Thresh 4 | 0x47F9        | I1   | 0   | 100 | 60      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED4 during discharging

### 13.7.1.16 DSG Thresh 5

| Class       | Subclass   | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|------------|--------------|---------------|------|-----|-----|---------|------|
| LED Support | LED Config | DSG Thresh 5 | 0x47FA        | I1   | 0   | 100 | 80      | %    |

**Description:** *RelativeStateOfCharge()* threshold for LED5 during discharging

## 13.8 System Data

### 13.8.1 Manufacturer Info

| Class       | Subclass          | Name             | Start Address | Type | Min | Max | Default                          | Units |
|-------------|-------------------|------------------|---------------|------|-----|-----|----------------------------------|-------|
| System Data | Manufacturer Data | ManufacturerInfo | 0x4040        | S33  | —   | —   | abcdefghijklmnopqrstuvwxyz012345 | —     |

**Description:** *ManufacturerInfo()* value

### 13.8.2 Static DF Signature

| Class       | Subclass  | Name                | Start Address | Type | Min | Max    | Default | Units |
|-------------|-----------|---------------------|---------------|------|-----|--------|---------|-------|
| System Data | Integrity | Static DF Signature | 0x4061        | H2   | 0x0 | 0x7FFF | 0x0     | Hex   |

**Description:** Static data flash signature. Use MAC *StaticDFSsignature()* (with MSB set to 0) to initialize this value.

### 13.8.3 Static Chem DF

| Class       | Subclass  | Name                     | Start Address | Type | Min | Max    | Default | Units |
|-------------|-----------|--------------------------|---------------|------|-----|--------|---------|-------|
| System Data | Integrity | Static Chem DF Signature | 0x4063        | H2   | 0x0 | 0x7FFF | 0x0     | Hex   |

**Description:** Static chemistry data signature. Use MAC *StaticChemDFSsignature()* (with MSB set to 0) to initialize this value.

### 13.8.4 All DF Signature

| Class       | Subclass  | Name             | Start Address | Type | Min | Max    | Default | Units |
|-------------|-----------|------------------|---------------|------|-----|--------|---------|-------|
| System Data | Integrity | All DF Signature | 0x4065        | H2   | 0x0 | 0x7FFF | 0x0     | Hex   |

**Description:** Static data flash signature. Use MAC *AllDFSsignature()* (with MSB set to 0) to initialize this value.

## 13.9 Lifetimes

### 13.9.1 Voltage

#### 13.9.1.1 Cell 1 Max Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 1 Max Voltage | 0x4380        | I2   | 0   | 32767 | 0       | mV   |

**Description:** Maximum reported cell voltage 1

#### 13.9.1.2 Cell 2 Max Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 2 Max Voltage | 0x4382        | I2   | 0   | 32767 | 0       | mV   |

**Description:** Maximum reported cell voltage 2

#### 13.9.1.3 Cell 3 Max Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 3 Max Voltage | 0x4384        | I2   | 0   | 32767 | 0       | mV   |

**Description:** Maximum reported cell voltage 3

#### 13.9.1.4 Cell 4 Max Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 4 Max Voltage | 0x4386        | I2   | 0   | 32767 | 0       | mV   |

**Description:** Maximum reported cell voltage 4

#### 13.9.1.5 Cell 1 Min Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 1 Min Voltage | 0x4388        | I2   | 0   | 32767 | 32767   | mV   |

**Description:** Minimum reported cell voltage 1

#### 13.9.1.6 Cell 2 Min Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 2 Min Voltage | 0x438A        | I2   | 0   | 32767 | 32767   | mV   |

**Description:** Minimum reported cell voltage 2

### 13.9.1.7 Cell 3 Min Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 3 Min Voltage | 0x438C        | I2   | 0   | 32767 | 32767   | mV   |

**Description:** Minimum reported cell voltage 3

### 13.9.1.8 Cell 4 Min Voltage

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Cell 4 Min Voltage | 0x438E        | I2   | 0   | 32767 | 32767   | mV   |

**Description:** Minimum reported cell voltage 4

### 13.9.1.9 Max Delta Cell Voltage

| Class     | Subclass | Name                   | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Voltage  | Max Delta Cell Voltage | 0x4390        | I2   | 0   | 32767 | 0       | mV   |

**Description:** Maximum reported delta between cell voltages 1..4

## 13.9.2 Current

### 13.9.2.1 Max Charge Current

| Class     | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Current  | Max Charge Current | 0x4392        | I2   | 0   | 32767 | 0       | mA   |

**Description:** Maximum reported *Current()* in charge direction

### 13.9.2.2 Max Discharge Current

| Class     | Subclass | Name                  | Start Address | Type | Min    | Max | Default | Unit |
|-----------|----------|-----------------------|---------------|------|--------|-----|---------|------|
| Lifetimes | Current  | Max Discharge Current | 0x4394        | I2   | -32768 | 0   | 0       | mA   |

**Description:** Maximum reported *Current()* in discharge direction

### 13.9.2.3 Max Avg Dsg Current

| Class     | Subclass | Name                | Start Address | Type | Min    | Max | Default | Unit |
|-----------|----------|---------------------|---------------|------|--------|-----|---------|------|
| Lifetimes | Current  | Max Avg Dsg Current | 0x4396        | I2   | -32768 | 0   | 0       | mA   |

**Description:** Maximum reported *AverageCurrent()* in discharge direction

### 13.9.2.4 Max Avg Dsg Power

| Class     | Subclass | Name              | Start Address | Type | Min    | Max | Default | Unit |
|-----------|----------|-------------------|---------------|------|--------|-----|---------|------|
| Lifetimes | Current  | Max Avg Dsg Power | 0x4398        | I2   | –32768 | 0   | 0       | cW   |

**Description:** Maximum reported Power in discharge direction

## 13.9.3 Temperature

### 13.9.3.1 Max Temp Cell

| Class     | Subclass    | Name          | Start Address | Type | Min  | Max | Default | Unit |
|-----------|-------------|---------------|---------------|------|------|-----|---------|------|
| Lifetimes | Temperature | Max Temp Cell | 0x439A        | I1   | –128 | 127 | –128    | °C   |

**Description:** Maximum reported cell temperature

### 13.9.3.2 Min Temp Cell

| Class     | Subclass    | Name          | Start Address | Type | Min  | Max | Default | Unit |
|-----------|-------------|---------------|---------------|------|------|-----|---------|------|
| Lifetimes | Temperature | Min Temp Cell | 0x439B        | I1   | –128 | 127 | 127     | °C   |

**Description:** Minimum reported cell temperature

### 13.9.3.3 Max Delta Cell Temp

| Class     | Subclass    | Name                | Start Address | Type | Min  | Max | Default | Unit |
|-----------|-------------|---------------------|---------------|------|------|-----|---------|------|
| Lifetimes | Temperature | Max Delta Cell Temp | 0x439C        | I1   | –128 | 127 | 0       | °C   |

**Description:** Maximum reported temperature delta for TSx inputs configured as cell temperature

### 13.9.3.4 Max Temp Int Sensor

| Class     | Subclass    | Name                | Start Address | Type | Min  | Max | Default | Unit |
|-----------|-------------|---------------------|---------------|------|------|-----|---------|------|
| Lifetimes | Temperature | Max Temp Int Sensor | 0x439D        | I1   | –128 | 127 | –128    | °C   |

**Description:** Maximum reported internal temperature sensor temperature

### 13.9.3.5 Min Temp Int Sensor

| Class     | Subclass    | Name                | Start Address | Type | Min  | Max | Default | Unit |
|-----------|-------------|---------------------|---------------|------|------|-----|---------|------|
| Lifetimes | Temperature | Min Temp Int Sensor | 0x439E        | I1   | –128 | 127 | 127     | °C   |

**Description:** Minimum reported internal temperature sensor temperature

### 13.9.3.6 Max Temp Fet

| Class     | Subclass    | Name         | Start Address | Type | Min  | Max | Default | Unit |
|-----------|-------------|--------------|---------------|------|------|-----|---------|------|
| Lifetimes | Temperature | Max Temp Fet | 0x439F        | I1   | –128 | 127 | –128    | °C   |

**Description:** Maximum reported FET temperature

## 13.9.4 Safety Events

### 13.9.4.1 No Of COV Events

| Class     | Subclass      | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of COV Events | 0x43A0        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[COV]* events

### 13.9.4.2 Last COV Event

| Class     | Subclass      | Name           | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last COV Event | 0x43A2        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()[COV]* event in *CycleCount()* cycles

### 13.9.4.3 No Of CUV Events

| Class     | Subclass      | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of CUV Events | 0x43A4        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[CUV]* events

### 13.9.4.4 Last CUV Event

| Class     | Subclass      | Name           | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last CUV Event | 0x43A6        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()[CUV]* event in *CycleCount()* cycles

### 13.9.4.5 No Of OCD1 Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OCD1 Events | 0x43A8        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[OCD1]* events

### 13.9.4.6 Last OCD1 Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OCD1 Event | 0x43AA        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()[OCD1]* event in *CycleCount()* cycles

### 13.9.4.7 No Of OCD2 Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OCD2 Events | 0x43AC        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[OCD2]* events

#### 13.9.4.8 Last OCD2 Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OCD2 Event | 0x43AE        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()[OCD2]* event in *CycleCount()* cycles

#### 13.9.4.9 No Of OCC1 Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OCC1 Events | 0x43B0        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[OCC1]* events

#### 13.9.4.10 Last OCC1 Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OCC1 Event | 0x43B2        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()[OCC1]* event in *CycleCount()* cycles

#### 13.9.4.11 No Of OCC2 Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OCC2 Events | 0x43B4        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[OCC2]* events

#### 13.9.4.12 Last OCC2 Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OCC2 Event | 0x43B6        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()[OCC2]* event in *CycleCount()* cycles

#### 13.9.4.13 No Of AOLD Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of AOLD Events | 0x43B8        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()[OLD]* events

#### 13.9.4.14 Last AOLD Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last AOLD Event | 0x43BA        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()*[OLD] event in *CycleCount()* cycles

#### 13.9.4.15 No Of ASCD Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of ASCD Events | 0x43BC        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()*[SCD] events

#### 13.9.4.16 Last ASCD Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last ASCD Event | 0x43BE        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()*[SCD] event in *CycleCount()* cycles

#### 13.9.4.17 No Of ASCC Events

| Class     | Subclass      | Name              | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of ASCC Events | 0x43C0        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()*[SCC] events

#### 13.9.4.18 Last ASCC Event

| Class     | Subclass      | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last ASCC Event | 0x43C2        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()*[SCC] event in *CycleCount()* cycles

#### 13.9.4.19 No Of OTC Events

| Class     | Subclass      | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OTC Events | 0x43C4        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()*[OTC] events

#### 13.9.4.20 Last OTC Event

| Class     | Subclass      | Name           | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OTC Event | 0x43C6        | U2   | 0   | 32767 | 0       | cycles |



**Description:** Last *SafetyStatus()*[*OTC*] event in *CycleCount()* cycles

#### 13.9.4.21 No Of OTD Events

| Class     | Subclass      | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OTD Events | 0x43C8        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()*[*OTD*] events

#### 13.9.4.22 Last OTD Event

| Class     | Subclass      | Name           | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OTD Event | 0x43CA        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()*[*OTD*] event in *CycleCount()* cycles

#### 13.9.4.23 No Of OTF Events

| Class     | Subclass      | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | No Of OTF Events | 0x43CC        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *SafetyStatus()*[*OTF*] events

#### 13.9.4.24 Last OTF Event

| Class     | Subclass      | Name           | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|---------------|----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Safety Events | Last OTF Event | 0x43CE        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *SafetyStatus()*[*OTF*] event in *CycleCount()* cycles

### 13.9.5 Charging Events

#### 13.9.5.1 No Valid Charge Term

| Class     | Subclass        | Name                 | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|-----------------|----------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Charging Events | No Valid Charge Term | 0x43D0        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of valid charge termination events

#### 13.9.5.2 Last Valid Charge Term

| Class     | Subclass        | Name                   | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|-----------------|------------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Charging Events | Last Valid Charge Term | 0x43D2        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last valid charge termination in *CycleCount()* cycles

## 13.9.6 Gauging Events

### 13.9.6.1 No Of Qmax Updates

| Class     | Subclass       | Name               | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|----------------|--------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Gauging Events | No Of Qmax Updates | 0x43D4        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *GaugingStatus()*[QMax] toggles

### 13.9.6.2 Last Qmax Update

| Class     | Subclass       | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|----------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Gauging Events | Last Qmax Update | 0x43D6        | U2   | 0   | 32767 | 0       | cycles |

**Description:** The *CycleCount()* cycles made at the last event of *GaugingStatus()*[QMax] update

### 13.9.6.3 No Of Ra Updates

| Class     | Subclass       | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|----------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Gauging Events | No Of Ra Updates | 0x43D8        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *GaugingStatus()*[RX] toggles

### 13.9.6.4 Last Ra Update

| Class     | Subclass       | Name           | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|----------------|----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Gauging Events | Last Ra Update | 0x43DA        | U2   | 0   | 32767 | 0       | cycles |

**Description:** Last *GaugingStatus()*[RX] toggle in *CycleCount()* cycles

### 13.9.6.5 No Of Ra Disable

| Class     | Subclass       | Name             | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|----------------|------------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Gauging Events | No Of Ra Disable | 0x43DC        | U2   | 0   | 32767 | 0       | events |

**Description:** Total number of *GaugingStatus()*[R\_DIS] = 1 event

### 13.9.6.6 Last Ra Disable

| Class     | Subclass       | Name            | Start Address | Type | Min | Max   | Default | Unit   |
|-----------|----------------|-----------------|---------------|------|-----|-------|---------|--------|
| Lifetimes | Gauging Events | Last Ra Disable | 0x43DE        | U2   | 0   | 32767 | 0       | cycles |

**Description:** The *CycleCount()* cycles of the last update event of *GaugingStatus()*[R\_DIS] = 1

## 13.9.7 Power Events

| Class     | Subclass     | Name            | Start Address | Type | Min | Max | Default | Unit   |
|-----------|--------------|-----------------|---------------|------|-----|-----|---------|--------|
| Lifetimes | Power Events | No of Shutdowns | 0x43E0        | U1   | 0   | 255 | 0       | events |

**Description:** Total number of shutdown events

## 13.9.8 Cell Balancing

### 13.9.8.1 CB Time Cell 1

| Class     | Subclass       | Name           | Start Address | Type | Min | Max | Default | Unit |
|-----------|----------------|----------------|---------------|------|-----|-----|---------|------|
| Lifetimes | Cell Balancing | CB Time Cell 1 | 0x43E4        | U1   | 0   | 255 | 0       | 2 h  |

**Description:** Total performed cell balancing bypass time Cell 0

### 13.9.8.2 CB Time Cell 2

| Class     | Subclass       | Name           | Start Address | Type | Min | Max | Default | Unit |
|-----------|----------------|----------------|---------------|------|-----|-----|---------|------|
| Lifetimes | Cell Balancing | CB Time Cell 2 | 0x43E5        | U1   | 0   | 255 | 0       | 2 h  |

**Description:** Total performed cell balancing bypass time Cell 1

### 13.9.8.3 CB Time Cell 3

| Class     | Subclass       | Name           | Start Address | Type | Min | Max | Default | Unit |
|-----------|----------------|----------------|---------------|------|-----|-----|---------|------|
| Lifetimes | Cell Balancing | CB Time Cell 3 | 0x43E6        | U1   | 0   | 255 | 0       | 2 h  |

**Description:** Total performed cell balancing bypass time Cell 2

### 13.9.8.4 CB Time Cell 4

| Class     | Subclass       | Name           | Start Address | Type | Min | Max | Default | Unit |
|-----------|----------------|----------------|---------------|------|-----|-----|---------|------|
| Lifetimes | Cell Balancing | CB Time Cell 4 | 0x43E7        | U1   | 0   | 255 | 0       | 2 h  |

**Description:** Total performed cell balancing bypass time Cell 3

## 13.9.9 Time

### 13.9.9.1 Total Firmware Runtime

| Class     | Subclass | Name                   | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Total Firmware Runtime | 0x43E8        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime between resets

### 13.9.9.2 Time Spent in UT

| Class     | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in UT | 0x43EA        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent below T1

### 13.9.9.3 Time Spent in LT

| Class     | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in LT | 0x43EC        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent between T1 and T2

#### 13.9.9.4 Time Spent in STL

| Class     | Subclass | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|-------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in STL | 0x43EE        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent between T2 and T5

#### 13.9.9.5 Time Spent in RT

| Class     | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in RT | 0x43F0        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent between T5 and T6

#### 13.9.9.6 Time Spent in STH

| Class     | Subclass | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|-------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in STH | 0x43F2        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent between T6 and T3

#### 13.9.9.7 Time Spent in HT

| Class     | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in HT | 0x43F4        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent between T3 and T4

#### 13.9.9.8 Time Spent in OT

| Class     | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit |
|-----------|----------|------------------|---------------|------|-----|-------|---------|------|
| Lifetimes | Time     | Time Spent in OT | 0x43F6        | U2   | 0   | 65535 | 0       | 2 h  |

**Description:** Total firmware runtime spent above T6

### 13.10 Protections

#### 13.10.1 CUV—Cell Undervoltage

##### 13.10.1.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|-----|-------|---------|------|
| Protections | CUV      | Threshold | 0x4845        | I2   | 0   | 32767 | 2500    | mV   |

**Description:** Cell undervoltage trip threshold

##### 13.10.1.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | CUV      | Delay | 0x4847        | U1   | 0   | 255 | 2       | s    |

**Description:** Cell undervoltage trip delay

### 13.10.1.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|----------|---------------|------|-----|-------|---------|------|
| Protections | CUV      | Recovery | 0x4848        | I2   | 0   | 32767 | 3000    | mV   |

**Description:** Cell undervoltage recovery threshold

## 13.10.2 CUV—Cell Undervoltage

### 13.10.2.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|-----|-------|---------|------|
| Protections | CUVC     | Threshold | 0x484A        | I2   | 0   | 32767 | 2400    | mV   |

**Description:** Cell undervoltage trip threshold

### 13.10.2.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | CUVC     | Delay | 0x484C        | U1   | 0   | 255 | 2       | s    |

**Description:** Cell undervoltage trip delay

### 13.10.2.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|----------|---------------|------|-----|-------|---------|------|
| Protections | CUVC     | Recovery | 0x484D        | I2   | 0   | 32767 | 3000    | mV   |

**Description:** Cell undervoltage recovery threshold

## 13.10.3 COV—Cell Overvoltage

### 13.10.3.1 Threshold Low Temp

| Class       | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Threshold Low Temp | 0x484F        | I2   | 0   | 32767 | 4300    | mV   |

**Description:** Cell overvoltage low temperature range trip threshold

### 13.10.3.2 Threshold Standard Tem

| Class       | Subclass | Name                    | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Threshold Standard Temp | 0x4851        | I2   | 0   | 32767 | 4300    | mV   |

**Description:** Cell overvoltage standard temperature range trip threshold

### 13.10.3.3 Threshold High Temp

| Class       | Subclass | Name                | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Threshold High Temp | 0x4853        | I2   | 0   | 32767 | 4300    | mV   |

**Description:** Cell overvoltage high temperature range trip threshold

#### 13.10.3.4 Threshold Rec Temp

| Class       | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Threshold Rec Temp | 0x4855        | I2   | 0   | 32767 | 4300    | mV   |

**Description:** Cell overvoltage recommended temperature range trip threshold

#### 13.10.3.5 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | COV      | Delay | 0x4857        | U1   | 0   | 255 | 2       | s    |

**Description:** Cell overvoltage trip delay

#### 13.10.3.6 Recovery Low Temp

| Class       | Subclass | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Recovery Low Temp | 0x4858        | I2   | 0   | 32767 | 3900    | mV   |

**Description:** Cell overvoltage low temperature range recovery threshold

#### 13.10.3.7 Recovery Standard Temp

| Class       | Subclass | Name                   | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|------------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Recovery Standard Temp | 0x485A        | I2   | 0   | 32767 | 3900    | mV   |

**Description:** Cell overvoltage standard temperature recovery range threshold

#### 13.10.3.8 Recovery High Temp

| Class       | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Recovery High Temp | 0x485C        | I2   | 0   | 32767 | 3900    | mV   |

**Description:** Cell overvoltage high temperature range recovery threshold

#### 13.10.3.9 Recovery Rec Temp

| Class       | Subclass | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-------|---------|------|
| Protections | COV      | Recovery Rec Temp | 0x485E        | I2   | 0   | 32767 | 3900    | mV   |

**Description:** Cell overvoltage recommended temperature range recovery threshold

### 13.10.4 OCC1—Overcurrent In Charge 1

#### 13.10.4.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | OCC1     | Threshold | 0x4860        | I2   | –32768 | 32767 | 6000    | mA   |

**Description:** Overcurrent in Charge 1 trip threshold

#### 13.10.4.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OCC1     | Delay | 0x4862        | U1   | 0   | 255 | 6       | s    |

**Description:** Overcurrent in Charge 1 trip delay

### 13.10.5 OCC2—Overcurrent In Charge 2

#### 13.10.5.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | OCC2     | Threshold | 0x4863        | I2   | –32768 | 32767 | 8000    | mA   |

**Description:** Overcurrent in Charge 2 trip threshold

#### 13.10.5.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OCC2     | Delay | 0x4865        | U1   | 0   | 255 | 3       | s    |

**Description:** Overcurrent in Charge 2 trip delay

### 13.10.6 OCC—Overcurrent In Charge Recovery

#### 13.10.6.1 Recovery Threshold

| Class       | Subclass | Name               | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|--------|-------|---------|------|
| Protections | OCC      | Recovery Threshold | 0x4866        | I2   | –32768 | 32767 | –200    | mA   |

**Description:** Overcurrent in Charge 1 and 2 recovery threshold

#### 13.10.6.2 Recovery Delay

| Class       | Subclass | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-----|---------|------|
| Protections | OCC      | Recovery Delay | 0x4868        | U1   | 0   | 255 | 5       | s    |

**Description:** Overcurrent in Charge 1 and 2 recovery delay

### 13.10.7 OCD1—Overcurrent In Discharge 1

#### 13.10.7.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | OCD1     | Threshold | 0x4869        | I2   | –32768 | 32767 | –6000   | mA   |

**Description:** Overcurrent in Discharge 1 trip threshold

### 13.10.7.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OCD1     | Delay | 0x486B        | U1   | 0   | 255 | 6       | s    |

**Description:** Overcurrent in Discharge 1 trip delay

## 13.10.8 OCD2—Overcurrent In Discharge 2

### 13.10.8.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | OCD2     | Threshold | 0x486C        | I2   | –32768 | 32767 | –8000   | mA   |

**Description:** Overcurrent in Discharge 2 trip threshold

### 13.10.8.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OCD2     | Delay | 0x486E        | U1   | 0   | 255 | 3       | s    |

**Description:** Overcurrent in Discharge 2 trip delay

## 13.10.9 OCD—Overcurrent In Discharge Recovery

### 13.10.9.1 Recovery Threshold

| Class       | Subclass | Name               | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|--------|-------|---------|------|
| Protections | OCD      | Recovery Threshold | 0x486F        | I2   | –32768 | 32767 | 200     | mA   |

**Description:** Overcurrent in Discharge 1 and 2 recovery threshold

### 13.10.9.2 Recovery Delay

| Class       | Subclass | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-----|---------|------|
| Protections | OCD      | Recovery Delay | 0x4871        | U1   | 0   | 255 | 5       | s    |

**Description:** Overcurrent in Discharge 1 and 2 recovery delay

## 13.10.10 AOLD—Overload in Discharge

### 13.10.10.1 Latch Limit

| Class       | Subclass | Name        | Start Address | Type | Min | Max | Default | Unit   |
|-------------|----------|-------------|---------------|------|-----|-----|---------|--------|
| Protections | AOLD     | Latch Limit | 0x4872        | U1   | 0   | 255 | 0       | counts |



**Description:** Overload latch counter trip threshold

### 13.10.10.2 Counter Dec Delay

| Class       | Subclass | Name              | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-----|---------|------|
| Protections | AOLD     | Counter Dec Delay | 0x4873        | U1   | 0   | 255 | 10      | s    |

**Description:** Overload latch counter decrement delay

### 13.10.10.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------|---------------|------|-----|-----|---------|------|
| Protections | AOLD     | Recovery | 0x4874        | U1   | 0   | 255 | 5       | s    |

**Description:** Overload recovery time

### 13.10.10.4 Reset

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | AOLD     | Reset | 0x4875        | U1   | 0   | 255 | 15      | s    |

**Description:** Overload latch reset time

### 13.10.10.5 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------|---------------|------|-----|------|---------|------|
| Protections | AOLD     | Threshold | 0x4953        | H1   | 0x0 | 0xFF | 0xF4    | Hex  |

**Description:** *AOLD:Threshold* Setting

Bits 7–4: OLDD: AOLD delay time

Bits 3–0: OLDV: AOLD threshold

## 13.10.11 ASCC—Short Circuit In Charge

### 13.10.11.1 Latch Limit

| Class       | Subclass | Name        | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------------|---------------|------|-----|-----|---------|------|
| Protections | ASCC     | Latch Limit | 0x4876        | U1   | 0   | 255 | 0       | —    |

**Description:** Short Circuit in Charge Latch counter trip threshold

### 13.10.11.2 Counter Dec Delay

| Class       | Subclass | Name              | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-----|---------|------|
| Protections | ASCC     | Counter Dec Delay | 0x4877        | U1   | 0   | 255 | 10      | s    |

**Description:** Short Circuit in Charge counter decrement delay

### 13.10.11.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------|---------------|------|-----|-----|---------|------|
| Protections | ASCC     | Recovery | 0x4878        | U1   | 0   | 255 | 5       | s    |

**Description:** Short Circuit in Charge recovery time

### 13.10.11.4 Reset

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | ASCC     | Reset | 0x4879        | U1   | 0   | 255 | 15      | s    |

**Description:** Short Circuit in Charge latch reset time

### 13.10.11.5 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------|---------------|------|-----|------|---------|------|
| Protections | ASCC     | Threshold | 0x4954        | H1   | 0x0 | 0xFF | 0x77    | Hex  |

**Description:** *ASCC:Threshold* Setting

Bits 7–4: SCCD: SCC delay time

Bit 3: Reserved

Bits 2–0: SCCV: SCC threshold

## 13.10.12 ASCD—Short Circuit in Discharge

### 13.10.12.1 Latch Limit

| Class       | Subclass | Name        | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------------|---------------|------|-----|-----|---------|------|
| Protections | ASCD     | Latch Limit | 0x487A        | U1   | 0   | 255 | 0       | —    |

**Description:** Short Circuit in Discharge Latch counter trip threshold

### 13.10.12.2 Counter Dec Delay

| Class       | Subclass | Name              | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-----|---------|------|
| Protections | ASCD     | Counter Dec Delay | 0x487B        | U1   | 0   | 255 | 10      | s    |

**Description:** Short Circuit in Discharge counter decrement delay

### 13.10.12.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------|---------------|------|-----|-----|---------|------|
| Protections | ASCD     | Recovery | 0x487C        | U1   | 0   | 255 | 5       | s    |

**Description:** Short Circuit in Discharge recovery time

### 13.10.12.4 Reset

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | ASCD     | Reset | 0x487D        | U1   | 0   | 255 | 15      | s    |

**Description:** Short Circuit in Discharge latch reset time

### 13.10.12.5 Thresholds 1 and 2

| Class       | Subclass | Name        | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-------------|---------------|------|-----|------|---------|------|
| Protections | ASCD     | Threshold 1 | 0x4955        | H1   | 0x0 | 0xFF | 0x77    | Hex  |
| Protections | ASCD     | Threshold 2 | 0x4956        | H1   | 0x0 | 0xFF | 0xE7    | Hex  |

**Threshold 1 Description:** *ASCD:Threshold 1* Setting

Bits 7–4: SCD1D–SCD1 delay time

Bit 3: Reserved

Bits 2–0: SCD1V: SCD1 threshold

**Threshold 2 Description:** *ASCD:Threshold 2* Setting

Bits 7–4: SCD2D–SCD2 delay time

Bit 3: Reserved

Bits 2–0: SCD2V: SCD2 threshold

### 13.10.13 OTC—Overtemperature in Charge

#### 13.10.13.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|-----------|---------------|------|------|------|---------|-------|
| Protections | OTC      | Threshold | 0x487E        | I2   | –400 | 1500 | 550     | 0.1°C |

**Description:** Overtemperature in Charge trip threshold

#### 13.10.13.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OTC      | Delay | 0x4880        | U1   | 0   | 255 | 2       | s    |

**Description:** Overtemperature in Charge Cell trip delay

#### 13.10.13.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|----------|---------------|------|------|------|---------|-------|
| Protections | OTC      | Recovery | 0x4881        | I2   | –400 | 1500 | 500     | 0.1°C |

**Description:** Overtemperature in Charge Cell recovery threshold

### 13.10.14 OTD—Overtemperature in Discharge

#### 13.10.14.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|-----------|---------------|------|------|------|---------|-------|
| Protections | OTD      | Threshold | 0x4883        | I2   | –400 | 1500 | 600     | 0.1°C |

**Description:** Overtemperature in Discharge trip threshold

### 13.10.14.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OTD      | Delay | 0x4885        | U1   | 0   | 255 | 2       | s    |

**Description:** Overtemperature in Discharge trip delay

### 13.10.14.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|----------|---------------|------|------|------|---------|-------|
| Protections | OTD      | Recovery | 0x4886        | I2   | −400 | 1500 | 550     | 0.1°C |

**Description:** Overtemperature in Discharge recovery threshold

## 13.10.15 OTF—Overtemperature FET

### 13.10.15.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|-----------|---------------|------|------|------|---------|-------|
| Protections | OTF      | Threshold | 0x4888        | I2   | −400 | 1500 | 800     | 0.1°C |

**Description:** Overtemperature FET trip threshold

### 13.10.15.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | OTF      | Delay | 0x488A        | U1   | 0   | 255 | 2       | s    |

**Description:** Overtemperature FET trip delay

### 13.10.15.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|----------|---------------|------|------|------|---------|-------|
| Protections | OTF      | Recovery | 0x488B        | I2   | −400 | 1500 | 650     | 0.1°C |

**Description:** Overtemperature FET recovery threshold

## 13.10.16 UTC—Under Temperature in Charge

### 13.10.16.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|-----------|---------------|------|------|------|---------|-------|
| Protections | UTC      | Threshold | 0x488D        | I2   | −400 | 1500 | 0       | 0.1°C |

**Description:** Undertemperature in Charge trip threshold

### 13.10.16.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | UTC      | Delay | 0x488F        | U1   | 0   | 255 | 2       | s    |

**Description:** Undertemperature in Charge Cell trip delay

### 13.10.16.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|----------|---------------|------|------|------|---------|-------|
| Protections | UTC      | Recovery | 0x4890        | I2   | –400 | 1500 | 50      | 0.1°C |

**Description:** Undertemperature in Charge Cell recovery threshold

### 13.10.17 UTD—Under Temperature in Discharge

#### 13.10.17.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|-----------|---------------|------|------|------|---------|-------|
| Protections | UTD      | Threshold | 0x4892        | I2   | –400 | 1500 | 0       | 0.1°C |

**Description:** Under Temperature in Discharge trip threshold

#### 13.10.17.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | UTD      | Delay | 0x4894        | U1   | 0   | 255 | 2       | s    |

**Description:** Under Temperature in Discharge trip delay

#### 13.10.17.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit  |
|-------------|----------|----------|---------------|------|------|------|---------|-------|
| Protections | UTD      | Recovery | 0x4895        | I2   | –400 | 1500 | 50      | 0.1°C |

**Description:** Under Temperature in Discharge recovery threshold

### 13.10.18 HWD—Host Watchdog

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | HWD      | Delay | 0x4897        | U1   | 0   | 255 | 10      | s    |

**Description:** SBS Host watchdog trip delay

### 13.10.19 PTO—Precharge mode Time Out

#### 13.10.19.1 Charge Threshold

| Class       | Subclass | Name             | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|------------------|---------------|------|--------|-------|---------|------|
| Protections | PTO      | Charge Threshold | 0x4898        | I2   | –32768 | 32767 | 2000    | mA   |

**Description:** Precharge Timeout Current Threshold

#### 13.10.19.2 Suspend Threshold

| Class       | Subclass | Name              | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-------------------|---------------|------|--------|-------|---------|------|
| Protections | PTO      | Suspend Threshold | 0x489A        | I2   | –32768 | 32767 | 1800    | mA   |

**Description:** Precharge Timeout Suspend Threshold

### 13.10.19.3 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-------|---------|------|
| Protections | PTO      | Delay | 0x489C        | U2   | 0   | 65535 | 1800    | s    |

**Description:** Precharge Timeout trip delay

### 13.10.19.4 Reset

| Class       | Subclass | Name  | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-------|---------------|------|--------|-------|---------|------|
| Protections | PTO      | Reset | 0x489E        | I2   | –32768 | 32767 | 2       | mA   |

**Description:** Precharge Timeout Reset Threshold

## 13.10.20 CTO—Fast Charge Mode Time Out

### 13.10.20.1 Charge Threshold

| Class       | Subclass | Name             | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|------------------|---------------|------|--------|-------|---------|------|
| Protections | CTO      | Charge Threshold | 0x48A0        | I2   | –32768 | 32767 | 2500    | mA   |

**Description:** Fast-Charge Timeout Current Threshold

### 13.10.20.2 Suspend Threshold

| Class       | Subclass | Name              | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-------------------|---------------|------|--------|-------|---------|------|
| Protections | CTO      | Suspend Threshold | 0x48A2        | I2   | –32768 | 32767 | 2000    | mA   |

**Description:** Fast-Charge Timeout Suspend Threshold

### 13.10.20.3 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-------|---------|------|
| Protections | CTO      | Delay | 0x48A4        | U2   | 0   | 65535 | 54000   | s    |

**Description:** Fast-Charge Timeout trip delay

### 13.10.20.4 Reset

| Class       | Subclass | Name  | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-------|---------|------|
| Protections | CTO      | Reset | 0x48A6        | I2   | 0   | 32767 | 2       | mA   |

**Description:** Fast-Charge Timeout Reset Threshold

## 13.10.21 OC—Overcharge

### 13.10.21.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | OC       | Threshold | 0x48A8        | I2   | –32768 | 32767 | 300     | mAh  |

**Description:** Overcharge trip threshold

### 13.10.21.2 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|----------|---------------|------|--------|-------|---------|------|
| Protections | OC       | Recovery | 0x48AA        | I2   | –32768 | 32767 | 2       | mAh  |

**Description:** Overcharge recovery threshold

### 13.10.21.3 RSOC Recovery

| Class       | Subclass | Name          | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|---------------|---------------|------|-----|-----|---------|------|
| Protections | OC       | RSOC Recovery | 0x48AC        | U1   | 0   | 100 | 90      | %    |

**Description:** Overcharge *RelativeStateOfCharge()* recovery threshold

## 13.10.22 CHGV—ChargingVoltage

### 13.10.22.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | CHGV     | Threshold | 0x48AD        | I2   | –32768 | 32767 | 500     | mV   |

**Description:** *ChargingVoltage()* delta trip threshold

### 13.10.22.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | CHGV     | Delay | 0x48AF        | U1   | 0   | 255 | 30      | s    |

**Description:** *ChargingVoltage()* delta trip delay

### 13.10.22.3 Recovery

| Class       | Subclass | Name     | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|----------|---------------|------|--------|-------|---------|------|
| Protections | CHGV     | Recovery | 0x48B0        | I2   | –32768 | 32767 | –500    | mV   |

**Description:** *ChargingVoltage()* delta recovery threshold

## 13.10.23 CHGC—ChargingCurrent

### 13.10.23.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | CHGC     | Threshold | 0x48B2        | I2   | –32768 | 32767 | 500     | mA   |

**Description:** *ChargingCurrent()* delta trip threshold

### 13.10.23.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | CHGC     | Delay | 0x48B4        | U1   | 0   | 255 | 2       | s    |

**Description:** *ChargingCurrent()* delta trip delay

### 13.10.23.3 Recovery Threshold

| Class       | Subclass | Name               | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|--------|-------|---------|------|
| Protections | CHGC     | Recovery Threshold | 0x48B5        | I2   | –32768 | 32767 | 100     | mA   |

**Description:** *ChargingCurrent()* delta recovery threshold

### 13.10.23.4 Recovery Delay

| Class       | Subclass | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-----|---------|------|
| Protections | CHGC     | Recovery Delay | 0x48B7        | U1   | 0   | 255 | 2       | s    |

**Description:** *ChargingCurrent()* delta recovery delay

## 13.10.24 PCHGC—Pre-ChargingCurrent

### 13.10.24.1 Threshold

| Class       | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Protections | PCHGC    | Threshold | 0x48B8        | I2   | –32768 | 32767 | 50      | mA   |

**Description:** *Pre-ChargingCurrent()* trip threshold

### 13.10.24.2 Delay

| Class       | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------|---------------|------|-----|-----|---------|------|
| Protections | PCHGC    | Delay | 0x48BA        | U1   | 0   | 255 | 2       | s    |

**Description:** *Pre-ChargingCurrent()* trip delay

### 13.10.24.3 Recovery Threshold

| Class       | Subclass | Name               | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|--------|-------|---------|------|
| Protections | PCHGC    | Recovery Threshold | 0x48BB        | I2   | –32768 | 32767 | 10      | mA   |

**Description:** *Pre-ChargingCurrent()* recovery threshold

### 13.10.24.4 Recovery Delay

| Class       | Subclass | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-----|---------|------|
| Protections | PCHGC    | Recovery Delay | 0x48BD        | U1   | 0   | 255 | 2       | s    |

**Description:** *Pre-ChargingCurrent()* recovery delay



## 13.11 Permanent Fail

### 13.11.1 SUV—Safety Cell Undervoltage

#### 13.11.1.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-------|---------|------|
| Permanent Fail | SUV      | Threshold | 0x48C2        | I2   | 0   | 32767 | 2200    | mV   |

**Description:** Safety Cell Undervoltage trip threshold

#### 13.11.1.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | SUV      | Delay | 0x48C4        | U1   | 0   | 255 | 5       | s    |

**Description:** Safety Cell Undervoltage trip delay

### 13.11.2 SOV—Safety Cell Overvoltage

#### 13.11.2.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-------|---------|------|
| Permanent Fail | SOV      | Threshold | 0x48C5        | I2   | 0   | 32767 | 4500    | mV   |

**Description:** Safety Cell Overvoltage trip threshold

#### 13.11.2.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | SOV      | Delay | 0x48C7        | U1   | 0   | 255 | 5       | s    |

**Description:** Safety Cell Overvoltage trip delay

### 13.11.3 SOCC—Safety Overcurrent in Charge

#### 13.11.3.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|----------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Permanent Fail | SOCC     | Threshold | 0x48C8        | I2   | –32768 | 32767 | 10000   | mA   |

**Description:** Safety Overcurrent in Charge trip threshold

#### 13.11.3.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | SOCC     | Delay | 0x48CA        | U1   | 0   | 255 | 5       | s    |

**Description:** Safety Overcurrent in Charge trip delay

### 13.11.4 SOCD—Safety Overcurrent in Discharge

#### 13.11.4.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min    | Max   | Default | Unit |
|----------------|----------|-----------|---------------|------|--------|-------|---------|------|
| Permanent Fail | SOCD     | Threshold | 0x48CB        | I2   | –32768 | 32767 | –10000  | mA   |

**Description:** Safety Overcurrent in Discharge trip threshold

#### 13.11.4.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | SOCD     | Delay | 0x48CD        | U1   | 0   | 255 | 5       | s    |

**Description:** Safety Overcurrent in Discharge trip delay

### 13.11.5 SOT—Overtemperature Cell

#### 13.11.5.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|----------------|----------|-----------|---------------|------|------|------|---------|-------|
| Permanent Fail | SOT      | Threshold | 0x48CE        | I2   | –400 | 1500 | 650     | 0.1°C |

**Description:** Overtemperature Cell trip threshold

#### 13.11.5.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | SOT      | Delay | 0x48D0        | U1   | 0   | 255 | 5       | s    |

**Description:** Overtemperature Cell trip delay

### 13.11.6 SOTF—Overtemperature FET

#### 13.11.6.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit  |
|----------------|----------|-----------|---------------|------|------|------|---------|-------|
| Permanent Fail | SOTF     | Threshold | 0x48D1        | I2   | –400 | 1500 | 1000    | 0.1°C |

**Description:** Overtemperature FET trip threshold

#### 13.11.6.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | SOTF     | Delay | 0x48D3        | U1   | 0   | 255 | 5       | s    |

**Description:** Overtemperature FET trip delay

### 13.11.7 Open Thermistor—NTC Thermistor Failure

#### 13.11.7.1 Threshold

| Class          | Subclass        | Name      | Start Address | Type | Min | Max   | Default | Unit   |
|----------------|-----------------|-----------|---------------|------|-----|-------|---------|--------|
| Permanent Fail | Open Thermistor | Threshold | 0x48D4        | I2   | 0   | 32767 | 2232    | 0.1 °K |

**Description:** Temperature threshold for open thermistor

#### 13.11.7.2 Delay

| Class          | Subclass        | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|-----------------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | Open Thermistor | Delay | 0x48D6        | U1   | 0   | 255 | 5       | s    |

**Description:** Trip delay for open thermistor

#### 13.11.7.3 FET Delta

| Class          | Subclass        | Name      | Start Address | Type | Min | Max  | Default | Unit   |
|----------------|-----------------|-----------|---------------|------|-----|------|---------|--------|
| Permanent Fail | Open Thermistor | FET Delta | 0x48D7        | I2   | 0   | −400 | 1500    | 0.1 °K |

**Description:** Delta from internal temperature to enable Open Thermistor check for FET thermistors

#### 13.11.7.4 Cell Delta

| Class          | Subclass        | Name       | Start Address | Type | Min | Max  | Default | Unit   |
|----------------|-----------------|------------|---------------|------|-----|------|---------|--------|
| Permanent Fail | Open Thermistor | Cell Delta | 0x48D9        | I2   | 0   | −400 | 1500    | 0.1 °K |

**Description:** Delta from internal temperature to enable Open Thermistor check for cell thermistors

### 13.11.8 QIM—QMax Imbalance

#### 13.11.8.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit  |
|----------------|----------|-----------|---------------|------|-----|-------|---------|-------|
| Permanent Fail | QIM      | Threshold | 0x48DB        | I2   | 0   | 32767 | 100     | 0.10% |

**Description:** QMax Imbalance trip threshold

#### 13.11.8.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit    |
|----------------|----------|-------|---------------|------|-----|-----|---------|---------|
| Permanent Fail | QIM      | Delay | 0x48DD        | U1   | 0   | 255 | 2       | updates |

**Description:** QMax Imbalance trip delay

### 13.11.9 CB—Cell Balance

#### 13.11.9.1 Max Threshold

| Class          | Subclass | Name          | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|-------|---------|------|
| Permanent Fail | CB       | Max Threshold | 0x48DE        | I2   | 0   | 32767 | 120     | 2 h  |

**Description:** Cell Balance max trip threshold

### 13.11.9.2 Delta Threshold

| Class          | Subclass | Name            | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-----------------|---------------|------|-----|-----|---------|------|
| Permanent Fail | CB       | Delta Threshold | 0x48E0        | U1   | 0   | 255 | 20      | 2 h  |

**Description:** Cell Balance cell delta trip threshold

### 13.11.9.3 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit   |
|----------------|----------|-------|---------------|------|-----|-----|---------|--------|
| Permanent Fail | CB       | Delay | 0x48E1        | U1   | 0   | 255 | 2       | cycles |

**Description:** Cell Balance trip delay

## 13.11.10 VIMR—Voltage Imbalance At Rest

### 13.11.10.1 Check Voltage

| Class          | Subclass | Name          | Start Address | Type | Min | Max  | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|------|---------|------|
| Permanent Fail | VIMR     | Check Voltage | 0x48E2        | I2   | 0   | 5000 | 3500    | mV   |

**Description:** Voltage Imbalance At Rest Check Voltage

### 13.11.10.2 Check Current

| Class          | Subclass | Name          | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|-------|---------|------|
| Permanent Fail | VIMR     | Check Current | 0x48E4        | I2   | 0   | 32767 | 10      | mA   |

**Description:** Voltage Imbalance At Rest Check Current

### 13.11.10.3 Delta Threshold

| Class          | Subclass | Name            | Start Address | Type | Min | Max  | Default | Unit |
|----------------|----------|-----------------|---------------|------|-----|------|---------|------|
| Permanent Fail | VIMR     | Delta Threshold | 0x48E6        | I2   | 0   | 5000 | 200     | mV   |

**Description:** Voltage Imbalance At Rest trip threshold

### 13.11.10.4 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | VIMR     | Delay | 0x48E8        | U1   | 0   | 255 | 5       | s    |

**Description:** Voltage Imbalance At Rest Check trip delay

### 13.11.10.5 Duration

| Class          | Subclass | Name     | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|----------|---------------|------|-----|-------|---------|------|
| Permanent Fail | VIMR     | Duration | 0x48E9        | U2   | 0   | 65535 | 100     | s    |

**Description:** Voltage Imbalance At Rest Check Duration

### 13.11.11 VIMA—Voltage Imbalance Active

#### 13.11.11.1 Check Voltage

| Class          | Subclass | Name          | Start Address | Type | Min | Max  | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|------|---------|------|
| Permanent Fail | VIMA     | Check Voltage | 0x48EB        | I2   | 0   | 5000 | 3700    | mV   |

**Description:** Voltage Imbalance active Check Voltage

#### 13.11.11.2 Check Current

| Class          | Subclass | Name          | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|-------|---------|------|
| Permanent Fail | VIMA     | Check Current | 0x48ED        | I2   | 0   | 32767 | 50      | mA   |

**Description:** Voltage Imbalance active Check Current

#### 13.11.11.3 Delta Threshold

| Class          | Subclass | Name            | Start Address | Type | Min | Max  | Default | Unit |
|----------------|----------|-----------------|---------------|------|-----|------|---------|------|
| Permanent Fail | VIMA     | Delta Threshold | 0x48EF        | I2   | 0   | 5000 | 300     | mV   |

**Description:** Voltage Imbalance active trip threshold

#### 13.11.11.4 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | VIMA     | Delay | 0x48F1        | U1   | 0   | 255 | 5       | s    |

**Description:** Voltage Imbalance active check trip Delay

### 13.11.12 IMP—Impedance Imbalance

#### 13.11.12.1 Delta Threshold

| Class          | Subclass | Name            | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|-----------------|---------------|------|-----|-------|---------|------|
| Permanent Fail | IMP      | Delta Threshold | 0x48F2        | I2   | 0   | 32767 | 300     | %    |

**Description:** Impedance Imbalance delta threshold

#### 13.11.12.2 Max Threshold

| Class          | Subclass | Name          | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|-------|---------|------|
| Permanent Fail | IMP      | Max Threshold | 0x48F4        | I2   | 0   | 32767 | 400     | %    |

**Description:** Impedance Imbalance max threshold

#### 13.11.12.3 Ra Update Counts

| Class          | Subclass | Name             | Start Address | Type | Min | Max | Default | Unit   |
|----------------|----------|------------------|---------------|------|-----|-----|---------|--------|
| Permanent Fail | IMP      | Ra Update Counts | 0x48F6        | U1   | 0   | 255 | 2       | counts |

**Description:** Impedance Imbalance trip delay

### 13.11.13 CD—Capacity Degradation

#### 13.11.13.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-------|---------|------|
| Permanent Fail | CD       | Threshold | 0x48F7        | I2   | 0   | 32767 | 4200    | mAh  |

**Description:** Capacity Degradation threshold

#### 13.11.13.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit   |
|----------------|----------|-------|---------------|------|-----|-----|---------|--------|
| Permanent Fail | CD       | Delay | 0x48F9        | U1   | 0   | 255 | 2       | cycles |

**Description:** Capacity Degradation trip delay

### 13.11.14 CFET—CHG FET Failure

#### 13.11.14.1 OFF Threshold

| Class          | Subclass | Name          | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|---------------|---------------|------|-----|-----|---------|------|
| Permanent Fail | CFET     | OFF Threshold | 0x48FA        | I2   | 0   | 500 | 5       | mA   |

**Description:** CHG FET OFF current trip threshold

#### 13.11.14.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | CFET     | Delay | 0x48FC        | U1   | 0   | 255 | 5       | s    |

**Description:** CHG FET OFF trip delay

### 13.11.15 DFET—DFET Failure

#### 13.11.15.1 OFF Threshold

| Class          | Subclass | Name          | Start Address | Type | Min  | Max | Default | Unit |
|----------------|----------|---------------|---------------|------|------|-----|---------|------|
| Permanent Fail | DFET     | OFF Threshold | 0x48FD        | I2   | –500 | 0   | –5      | mA   |

**Description:** DSG FET OFF current trip threshold

#### 13.11.15.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | DFET     | Delay | 0x48FF        | U1   | 0   | 255 | 5       | s    |

**Description:** DSG FET OFF trip delay

### 13.11.16 FUSE—FUSE Failure

#### 13.11.16.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-----|---------|------|
| Permanent Fail | FUSE     | Threshold | 0x4900        | I2   | 0   | 255 | 5       | mA   |

**Description:** FUSE activation fail trip threshold

### 13.11.16.2 Delay

| Class          | Subclass | Name  | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-------|---------------|------|-----|-----|---------|------|
| Permanent Fail | FUSE     | Delay | 0x4902        | U1   | 0   | 255 | 5       | s    |

**Description:** FUSE activation fail trip delay

## 13.11.17 AFER—AFE Register

### 13.11.17.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-----|---------|------|
| Permanent Fail | AFER     | Threshold | 0x4903        | U1   | 0   | 255 | 100     | —    |

**Description:** AFE Register comparison fail trip threshold

### 13.11.17.2 Delay Period

| Class          | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|--------------|---------------|------|-----|-----|---------|------|
| Permanent Fail | AFER     | Delay Period | 0x4904        | U1   | 0   | 255 | 5       | s    |

**Description:** AFE Register comparison counter decrement period

### 13.11.17.3 Compare Period

| Class          | Subclass | Name           | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|----------------|---------------|------|-----|-----|---------|------|
| Permanent Fail | AFER     | Compare Period | 0x4905        | U1   | 0   | 255 | 5       | s    |

**Description:** AFE Register comparison compare period

## 13.11.18 AFEC—AFE Communication

### 13.11.18.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-----|---------|------|
| Permanent Fail | AFEC     | Threshold | 0x4906        | U1   | 0   | 255 | 100     | —    |

**Description:** AFE Communication fail trip threshold

### 13.11.18.2 Delay Period

| Class          | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|--------------|---------------|------|-----|-----|---------|------|
| Permanent Fail | AFEC     | Delay Period | 0x4907        | U1   | 0   | 255 | 5       | s    |

**Description:** AFE Communication counter decrement period

## 13.11.19 2LVL—2nd Level OV

| Class          | Subclass | Name      | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-----|---------|------|
| Permanent Fail | 2LVL     | Threshold | 0x4908        | U1   | 0   | 255 | 5       | s    |

**Description:** 2nd Level Protector trip detection delay

### 13.11.20 OPNCELL—Open Cell Connection

#### 13.11.20.1 Threshold

| Class          | Subclass | Name      | Start Address | Type | Min | Max   | Default | Unit |
|----------------|----------|-----------|---------------|------|-----|-------|---------|------|
| Permanent Fail | OPNCELL  | Threshold | 0x4909        | I2   | 0   | 32767 | 5000    | mV   |

**Description:** Open Cell Tab Connection trip threshold

#### 13.11.20.2 Delay Period

| Class          | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|----------------|----------|--------------|---------------|------|-----|-----|---------|------|
| Permanent Fail | OPNCELL  | Delay Period | 0x490B        | U1   | 0   | 255 | 5       | s    |

**Description:** Open Cell Tab Connection trip delay

### 13.12 PF Status

The data in this class is saved at the time of the PF event.

#### 13.12.1 Device Status Data

##### 13.12.1.1 Safety Alert A

| Class     | Subclass           | Name           | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Alert A | 0x4440        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

##### 13.12.1.2 Safety Status A

| Class     | Subclass           | Name            | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Status A | 0x4441        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

##### 13.12.1.3 Safety Alert B

| Class     | Subclass           | Name           | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Alert B | 0x4442        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

##### 13.12.1.4 Safety Status B



| Class     | Subclass           | Name            | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Status B | 0x4443        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

#### 13.12.1.5 Safety Alert C

| Class     | Subclass           | Name           | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Alert C | 0x4444        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

#### 13.12.1.6 Safety Status C

| Class     | Subclass           | Name            | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Status C | 0x4445        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

#### 13.12.1.7 Safety Alert D

| Class     | Subclass           | Name           | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Alert D | 0x4446        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

#### 13.12.1.8 Safety Status D

| Class     | Subclass           | Name            | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Safety Status D | 0x4447        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated safety flags since PF event

#### 13.12.1.9 PF Alert A

| Class     | Subclass           | Name       | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Alert A | 0x4448        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

#### 13.12.1.10 PF Status A

| Class     | Subclass           | Name        | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Status A | 0x4449        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.11 PF Alert B

| Class     | Subclass           | Name       | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Alert B | 0x444A        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.12 PF Status B

| Class     | Subclass           | Name        | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Status B | 0x444B        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.13 PF Alert C

| Class     | Subclass           | Name       | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Alert C | 0x444C        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.14 PF Status C

| Class     | Subclass           | Name        | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Status C | 0x444D        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.15 PF Alert D

| Class     | Subclass           | Name       | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Alert D | 0x444E        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.16 PF Status D

| Class     | Subclass           | Name        | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | PF Status D | 0x444F        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Accumulated PF flags since PF event

### 13.12.1.17 Fuse Flag

| Class     | Subclass           | Name      | Start Address | Type | Min | Max   | Default | Units |
|-----------|--------------------|-----------|---------------|------|-----|-------|---------|-------|
| PF Status | Device Status Data | Fuse Flag | 0x4450        | H2   | 0x0 | 0xFFf | 0x0     | Hex   |

**Description:** Flag set to indicate fuse blow

### 13.12.1.18 Operation Status A

| Class     | Subclass           | Name               | Start Address | Type | Min | Max    | Default | Units |
|-----------|--------------------|--------------------|---------------|------|-----|--------|---------|-------|
| PF Status | Device Status Data | Operation Status A | 0x4452        | H2   | 0x0 | 0xFFFF | 0x0     | Hex   |

**Description:** *OperationStatus()* data at the time of the PF event

### 13.12.1.19 Operation Status B

| Class     | Subclass           | Name               | Start Address | Type | Min | Max    | Default | Units |
|-----------|--------------------|--------------------|---------------|------|-----|--------|---------|-------|
| PF Status | Device Status Data | Operation Status B | 0x4454        | H2   | 0x0 | 0xFFFF | 0x0     | Hex   |

**Description:** *OperationStatus()* data at the time of the PF event

### 13.12.1.20 Temp Range

| Class     | Subclass           | Name       | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Temp Range | 0x4456        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** Temperature range status at the time of the PF event. The temperature range information returned by *ChargingStatus()*

### 13.12.1.21 Charging Status A

| Class     | Subclass           | Name              | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-------------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Charging Status A | 0x4457        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** The charging status at the time of the PF event. See [Section 12.48](#) for the bit definitions.

|     |      |    |    |    |    |    |    |
|-----|------|----|----|----|----|----|----|
| 7   | 6    | 5  | 4  | 3  | 2  | 1  | 0  |
| VCT | MCHG | SU | IN | HV | MV | LV | PV |

### 13.12.1.22 Charging Status B

| Class     | Subclass           | Name              | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|-------------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Charging Status B | 0x4458        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** The charging status at the time of the PF event. See [Section 12.48](#) for the bit definitions.

|     |      |      |      |      |     |     |     |
|-----|------|------|------|------|-----|-----|-----|
| 7   | 6    | 5    | 4    | 3    | 2   | 1   | 0   |
| VCT | RSVD | RSVD | RSVD | RSVD | CCC | CVR | CCR |

### 13.12.1.23 Gauging Status

| Class     | Subclass           | Name           | Start Address | Type | Min | Max  | Default | Units |
|-----------|--------------------|----------------|---------------|------|-----|------|---------|-------|
| PF Status | Device Status Data | Gauging Status | 0x4459        | H1   | 0x0 | 0xFF | 0x0     | Hex   |

**Description:** The gauging status at the time of the PF event.

|    |     |     |        |     |     |    |    |
|----|-----|-----|--------|-----|-----|----|----|
| 7  | 6   | 5   | 4      | 3   | 2   | 1  | 0  |
| CF | DSG | EDV | BAL_EN | TCA | TDA | FC | FD |

### 13.12.1.24 IT Status

| Class     | Subclass           | Name      | Start Address | Type | Min | Max    | Default | Units |
|-----------|--------------------|-----------|---------------|------|-----|--------|---------|-------|
| PF Status | Device Status Data | IT Status | 0x445A        | H2   | 0x0 | 0xFFFF | 0x0     | Hex   |

**Description:** The Impedance Track status at the time of the PF event. See [Section 12.48](#) for the bit definitions.

|      |      |             |       |      |      |      |     |
|------|------|-------------|-------|------|------|------|-----|
| 15   | 14   | 13          | 12    | 11   | 10   | 9    | 8   |
| RSVD | RSVD | SLPQ<br>MAX | QEN   | VOK  | RDIS | RSVD |     |
| 7    | 6    | 5           | 4     | 3    | 2    | 1    | 0   |
| RSVD | RSVD | RSVD        | OCVFR | LDMD | RX   | QMAX | VDQ |

## 13.12.2 Device Voltage Data

### 13.12.2.1 Cell 1 Voltage

| Class     | Subclass            | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|----------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Voltage Data | Cell 1 Voltage | 0x445C        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** Cell 1 voltage

### 13.12.2.2 Cell 2 Voltage

| Class     | Subclass            | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|----------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Voltage Data | Cell 2 Voltage | 0x445E        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** Cell 2 voltage

### 13.12.2.3 Cell 3 Voltage

| Class     | Subclass            | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|----------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Voltage Data | Cell 3 Voltage | 0x4460        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** Cell 3 voltage

### 13.12.2.4 Cell 4 Voltage

| Class     | Subclass            | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|----------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Voltage Data | Cell 4 Voltage | 0x4462        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** Cell 4 voltage

### 13.12.2.5 Battery Direct Voltage

| Class     | Subclass            | Name                   | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|------------------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Voltage Data | Battery Direct Voltage | 0x4464        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** Battery voltage

### 13.12.2.6 Pack Voltage

| Class     | Subclass            | Name         | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|--------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Voltage Data | Pack Voltage | 0x4466        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** Pack pin voltage

### 13.12.3 Device Current Data

| Class     | Subclass            | Name    | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|---------|---------------|------|--------|-------|---------|------|
| PF Status | Device Current Data | Current | 0x4468        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** *Current()*

### 13.12.4 Device Temperature Data

#### 13.12.4.1 Internal Temperature

| Class     | Subclass                | Name                 | Start Address | Type | Min    | Max   | Default | Unit  |
|-----------|-------------------------|----------------------|---------------|------|--------|-------|---------|-------|
| PF Status | Device Temperature Data | Internal Temperature | 0x446A        | I2   | –32768 | 32767 | 0       | 0.1°K |

**Description:** Internal temperature sensor temperature

#### 13.12.4.2 External 1 Temperature

| Class     | Subclass                | Name                   | Start Address | Type | Min    | Max   | Default | Unit  |
|-----------|-------------------------|------------------------|---------------|------|--------|-------|---------|-------|
| PF Status | Device Temperature Data | External 1 Temperature | 0x446C        | I2   | –32768 | 32767 | 0       | 0.1°K |

**Description:** External TS1 temperature

#### 13.12.4.3 External 2 Temperature

| Class     | Subclass                | Name                   | Start Address | Type | Min    | Max   | Default | Unit  |
|-----------|-------------------------|------------------------|---------------|------|--------|-------|---------|-------|
| PF Status | Device Temperature Data | External 2 Temperature | 0x446E        | I2   | –32768 | 32767 | 0       | 0.1°K |

**Description:** External TS2 temperature

#### 13.12.4.4 External 3 Temperature

| Class     | Subclass                | Name                   | Start Address | Type | Min    | Max   | Default | Unit  |
|-----------|-------------------------|------------------------|---------------|------|--------|-------|---------|-------|
| PF Status | Device Temperature Data | External 3 Temperature | 0x4470        | I2   | –32768 | 32767 | 0       | 0.1°K |

**Description:** External TS3 temperature

### 13.12.4.5 External 4 Temperature

| Class     | Subclass                | Name                   | Start Address | Type | Min    | Max   | Default | Unit  |
|-----------|-------------------------|------------------------|---------------|------|--------|-------|---------|-------|
| PF Status | Device Temperature Data | External 4 Temperature | 0x4472        | I2   | –32768 | 32767 | 0       | 0.1°K |

**Description:** External TS4 temperature

## 13.12.5 Device Gauging Data

### 13.12.5.1 Cell 1DOD0

| Class     | Subclass            | Name       | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Gauging Data | Cell 1DOD0 | 0x4474        | I2   | –32768 | 32767 | 0       | —    |

**Description:** Cell 1 depth of discharge

### 13.12.5.2 Cell 2 DOD0

| Class     | Subclass            | Name        | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|-------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Gauging Data | Cell 2 DOD0 | 0x4476        | I2   | –32768 | 32767 | 0       | —    |

**Description:** Cell 2 depth of discharge

### 13.12.5.3 Cell 3 DOD0

| Class     | Subclass            | Name        | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|-------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Gauging Data | Cell 3 DOD0 | 0x4478        | I2   | –32768 | 32767 | 0       | —    |

**Description:** Cell 3 depth of discharge

### 13.12.5.4 Cell 4 DOD0

| Class     | Subclass            | Name        | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|-------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Gauging Data | Cell 4 DOD0 | 0x447A        | I2   | –32768 | 32767 | 0       | —    |

**Description:** Cell 4 depth of discharge

### 13.12.5.5 Passed Charge

| Class     | Subclass            | Name          | Start Address | Type | Min    | Max   | Default | Unit |
|-----------|---------------------|---------------|---------------|------|--------|-------|---------|------|
| PF Status | Device Gauging Data | Passed Charge | 0x447C        | I2   | –32768 | 32767 | 0       | mAh  |

**Description:** Passed charge since last QMax update

### 13.12.6 AFE Regs

#### 13.12.6.1 AFE Interrupt Status

| Class     | Subclass | Name                 | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE Interrupt Status | 0x447E        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE Interrupt Status Register Contents

#### 13.12.6.2 AFE FET Status

| Class     | Subclass | Name           | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE FET Status | 0x447F        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE FET Status Register Contents

#### 13.12.6.3 AFE RXIN

| Class     | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE RXIN | 0x4480        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE Rxin Register Contents

#### 13.12.6.4 AFE Latch Status

| Class     | Subclass | Name             | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE Latch Status | 0x4481        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE Latch Status Register Contents

#### 13.12.6.5 AFE Interrupt Enable

| Class     | Subclass | Name                 | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE Interrupt Enable | 0x4482        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE Interrupt Enable Register Contents

#### 13.12.6.6 AFE FET Control

| Class     | Subclass | Name            | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-----------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE FET Control | 0x4483        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE FET Control Register Contents

#### 13.12.6.7 AFE RXIEN

| Class     | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE RXIEN | 0x4484        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE RXIEN Register Contents

### 13.12.6.8 AFE RLOUT

| Class     | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE RLOUT | 0x4485        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE RLOUT Register Contents

### 13.12.6.9 AFE RHOUT

| Class     | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE RHOUT | 0x4486        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE RHOUT Register Contents

### 13.12.6.10 AFE RHINT

| Class     | Subclass | Name      | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE RHINT | 0x4487        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE RHINT Register Contents

### 13.12.6.11 AFE Cell Balance

| Class     | Subclass | Name             | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE Cell Balance | 0x4488        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE Cell Balance Register Contents

### 13.12.6.12 AFE AD/CC Control

| Class     | Subclass | Name              | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE AD/CC Control | 0x4489        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE AD/CC Control Register Contents

### 13.12.6.13 AFE ADC Mux

| Class     | Subclass | Name        | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE ADC Mux | 0x448A        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE ADC Mux Register Contents



### 13.12.6.14 AFE LED Output

| Class     | Subclass | Name           | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE LED Output | 0x448B        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE LED Output Register Contents

### 13.12.6.15 AFE State Control

| Class     | Subclass | Name              | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|-------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE State Control | 0x448C        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE State Control Register Contents

### 13.12.6.16 AFE LED/Wake Control

| Class     | Subclass | Name                 | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE LED/Wake Control | 0x448D        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE LED/Wake Control Register Contents

### 13.12.6.17 AFE Protection Control

| Class     | Subclass | Name                   | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|------------------------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE Protection Control | 0x448E        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE Protection Control Register Contents

### 13.12.6.18 AFE OCD

| Class     | Subclass | Name    | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|---------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE OCD | 0x448F        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE OCD Register Contents

### 13.12.6.19 AFE SCC

| Class     | Subclass | Name    | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|---------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE SCC | 0x4490        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE SCC Register Contents

### 13.12.6.20 AFE SCD1

| Class     | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE SCD1 | 0x4491        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE SCD1 Register Contents

**13.12.6.21 AFE SCD2**

| Class     | Subclass | Name     | Start Address | Type | Min  | Max  | Default | Unit |
|-----------|----------|----------|---------------|------|------|------|---------|------|
| PF Status | AFE Regs | AFE SCD2 | 0x4492        | H1   | 0x00 | 0xFF | 0x00    | Hex  |

**Description:** AFE SCD2 Register Contents

**13.13 Black Box**
**13.13.1 Safety Status**

| Class     | Subclass      | Name                   | Start Address | Type | Min | Max  | Default | Unit | Description                      |
|-----------|---------------|------------------------|---------------|------|-----|------|---------|------|----------------------------------|
| Black Box | Safety Status | 1st Status Status A    | 0x4400        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 1st Status Status B    | 0x4401        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 1st Safety Status C    | 0x4402        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 1st Safety Status D    | 0x4403        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 1st Time to Next Event | 0x4404        | U1   | 0   | 255  | 0       | s    | Time from 1st event to 2nd event |
| Black Box | Safety Status | 2nd Status Status A    | 0x4405        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 2nd Status Status B    | 0x4406        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 2nd Safety Status C    | 0x4407        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 2nd Safety Status D    | 0x4408        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 2nd Time to Next Event | 0x4409        | U1   | 0   | 255  | 0       | s    | Time from 2nd event to 3rd event |
| Black Box | Safety Status | 3rd Status Status A    | 0x440A        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 3rd Status Status B    | 0x440B        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 3rd Safety Status C    | 0x440C        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 3rd Safety Status D    | 0x440D        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>SafetyStatus()</i> data       |
| Black Box | Safety Status | 3rd Time to Next Event | 0x440E        | U1   | 0   | 255  | 0       | s    | Time since 3rd event             |

**13.13.2 PF Status**

| Class     | Subclass  | Name            | Start Address | Type | Min | Max  | Default | Unit | Description            |
|-----------|-----------|-----------------|---------------|------|-----|------|---------|------|------------------------|
| Black Box | PF Status | 1st PF Status A | 0x440F        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data |
| Black Box | PF Status | 1st PF Status B | 0x4410        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data |
| Black Box | PF Status | 1st PF Status C | 0x4411        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data |
| Black Box | PF Status | 1st PF Status D | 0x4412        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data |

| Class     | Subclass  | Name                   | Start Address | Type | Min | Max  | Default | Unit | Description                      |
|-----------|-----------|------------------------|---------------|------|-----|------|---------|------|----------------------------------|
| Black Box | PF Status | 1st Time to Next Event | 0x4413        | U1   | 0   | 255  | 0       | s    | Time from 1st event to 2nd event |
| Black Box | PF Status | 2nd PF Status A        | 0x4414        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 2nd PF Status B        | 0x4415        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 2nd PF Status C        | 0x4416        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 2nd PF Status D        | 0x4417        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 2nd Time to Next Event | 0x4418        | U1   | 0   | 255  | 0       | s    | Time from 2nd event to 3rd event |
| Black Box | PF Status | 3rd PF Status A        | 0x4419        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 3rd PF Status B        | 0x441A        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 3rd PF Status C        | 0x441B        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 3rd PF Status D        | 0x441C        | H1   | 0x0 | 0xFF | 0x0     | Hex  | <i>PFStatus()</i> data           |
| Black Box | PF Status | 3rd Time to Next Event | 0x441D        | U1   | 0   | 255  | 0       | s    | Time since 3rd event             |

## 13.14 Gas Gauging

### 13.14.1 Current Thresholds

#### 13.14.1.1 Dsg Current Threshold

| Class       | Subclass           | Name                  | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|--------------------|-----------------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | Current Thresholds | Dsg Current Threshold | 0x495B        | I2   | –32768 | 32767 | 100     | mA   |

**Description:** DISCHARGE mode *Current()* threshold

#### 13.14.1.2 Chg Current Threshold

| Class       | Subclass           | Name                  | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|--------------------|-----------------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | Current Thresholds | Chg Current Threshold | 0x495D        | I2   | –32768 | 32767 | 50      | mA   |

**Description:** CHARGE mode *Current()* threshold

#### 13.14.1.3 Quit Current

| Class       | Subclass           | Name         | Start Address | Type | Min | Max   | Default | Unit |
|-------------|--------------------|--------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | Current Thresholds | Quit Current | 0x495F        | I2   | 0   | 32767 | 10      | mA   |

**Description:** *|Current()|* threshold to enter rest mode

#### 13.14.1.4 Dsg Relax Time

| Class       | Subclass           | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|--------------------|----------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | Current Thresholds | Dsg Relax Time | 0x4961        | U1   | 0   | 255 | 1       | mA   |

**Description:** Discharge to relax timeout. When discharge is stopped, the device will exit the DISCHARGE mode after this time is passed.

### 13.14.1.5 Chg Relax Time

| Class       | Subclass           | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|--------------------|----------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | Current Thresholds | Chg Relax Time | 0x4962        | U1   | 0   | 255 | 60      | mA   |

**Description:** Charge to relax timeout. When charging is stopped, the device will exit the CHARGE mode after this time is passed.

## 13.14.2 Design

### 13.14.2.1 Design Capacity mAh

| Class       | Subclass | Name                | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | Design   | Design Capacity mAh | 0x4806        | I2   | 0   | 32767 | 4400    | mAh  |

**Description:** Design Capacity in mAh. This is reported by *DesignCapacity()* if **[CAPM]** = 0.

### 13.14.2.2 Design Capacity in cWh

| Class       | Subclass | Name                | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | Design   | Design Capacity cWh | 0x4808        | I2   | 0   | 32767 | 6336    | cWh  |

**Description:** Design Capacity in cWh. This is reported by *DesignCapacity()* if **[CAPM]** = 1.

### 13.14.2.3 Design Voltage

| Class       | Subclass | Name           | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | Design   | Design Voltage | 0x480A        | I2   | 0   | 32767 | 14400   | mV   |

**Description:** Design Voltage. This is reported by *DesignVoltage()*.

## 13.14.3 Cycle

| Class       | Subclass | Name                   | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | Cycle    | Cycle Count Percentage | 0x480C        | U1   | 0   | 100 | 90      | %    |

**Description:** This is a threshold to increment the **Cycle Count** if the accumulated discharge is more than this set percentage of *FullChargeCapacity()* (if **[CCT]** = 1) or *DesignCapacity()* (if **[CCT]** = 0). Note that a minimum of 10% of *DesignCapacity()* change of the accumulated discharge is required for cycle count increment. This is to prevent an erroneous cycle count increment due to extremely low *FullChargeCapacity()*.

## 13.14.4 FD

### 13.14.4.1 Set Voltage Threshold

| Class       | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | FD       | Set Voltage Threshold | 0x4811        | I2   | 0   | 5000 | 3000    | mV   |

**Description:** *GaugingStatus()*[FD] and *BatteryStatus()*[FD] cell voltage set threshold

#### 13.14.4.2 Clear Voltage Threshold

| Class       | Subclass | Name                    | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-------------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | FD       | Clear Voltage Threshold | 0x4813        | I2   | 0   | 5000 | 3100    | mV   |

**Description:** *GaugingStatus()*[FD] and *BatteryStatus()*[FD] cell voltage clear threshold

#### 13.14.4.3 Set RSOC % Threshold

| Class       | Subclass | Name                 | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | FD       | Set RSOC % Threshold | 0x4815        | U1   | 0   | 100 | 0       | %    |

**Description:** *GaugingStatus()*[FD] and *BatteryStatus()*[FD] *RelativeStateOfCharge()* set threshold

#### 13.14.4.4 Clear RSOC % Threshold

| Class       | Subclass | Name                   | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | FD       | Clear RSOC % Threshold | 0x4816        | U1   | 0   | 100 | 5       | %    |

**Description:** *GaugingStatus()*[FD] and *BatteryStatus()*[FD] *RelativeStateOfCharge()* clear threshold

### 13.14.5 FC

#### 13.14.5.1 Set Voltage Threshold

| Class       | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | FC       | Set Voltage Threshold | 0x4817        | I2   | 0   | 5000 | 4200    | mV   |

**Description:** *GaugingStatus()*[FC] and *BatteryStatus()*[FC] cell voltage set threshold

#### 13.14.5.2 Clear Voltage Threshold

| Class       | Subclass | Name                    | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-------------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | FC       | Clear Voltage Threshold | 0x4819        | I2   | 0   | 5000 | 4100    | mV   |

**Description:** *GaugingStatus()*[FC] and *BatteryStatus()*[FC] cell voltage clear threshold

#### 13.14.5.3 Set RSOC % Threshold

| Class       | Subclass | Name                 | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | FC       | Set RSOC % Threshold | 0x481B        | U1   | 0   | 100 | 100     | %    |

**Description:** *GaugingStatus()*[FC] and *BatteryStatus()*[FC] *RelativeStateOfCharge()* set threshold

#### 13.14.5.4 Clear RSOC % Threshold

| Class       | Subclass | Name                   | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | FC       | Clear RSOC % Threshold | 0x481C        | U1   | 0   | 100 | 95      | %    |

**Description:** *GaugingStatus()[FC]* and *BatteryStatus()[FC]* *RelativeStateOfCharge()* clear threshold

### 13.14.6 TD

*GaugingStatus()[TD]* is used to set *BatteryStatus()[TDA]* when in DISCHARGE mode.

#### 13.14.6.1 Set Voltage Threshold

| Class       | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | TD       | Set Voltage Threshold | 0x481D        | I2   | 0   | 5000 | 3200    | mV   |

**Description:** *GaugingStatus()[TD]* cell voltage set threshold

#### 13.14.6.2 Clear Voltage Threshold

| Class       | Subclass | Name                    | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-------------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | TD       | Clear Voltage Threshold | 0x481F        | I2   | 0   | 5000 | 3300    | mV   |

**Description:** *GaugingStatus()[TD]* cell voltage clear threshold

#### 13.14.6.3 Set RSOC % Threshold

| Class       | Subclass | Name                 | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | TD       | Set RSOC % Threshold | 0x4821        | U1   | 0   | 100 | 6       | %    |

**Description:** *GaugingStatus()[TD]* *RelativeStateOfCharge()* set threshold

#### 13.14.6.4 Clear RSOC % Threshold

| Class       | Subclass | Name                   | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | TD       | Clear RSOC % Threshold | 0x4822        | U1   | 0   | 100 | 8       | %    |

**Description:** *GaugingStatus()[TD]* *RelativeStateOfCharge()* clear threshold

### 13.14.7 TC

*GaugingStatus()[TC]* is used to set *BatteryStatus()[TCA]* when in CHARGE mode

#### 13.14.7.1 Set Voltage Threshold

| Class       | Subclass | Name                  | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | TC       | Set Voltage Threshold | 0x4823        | I2   | 0   | 5000 | 4200    | mV   |

**Description:** *GaugingStatus()[TC]* cell voltage set threshold

#### 13.14.7.2 Clear Voltage Threshold

| Class       | Subclass | Name                    | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-------------------------|---------------|------|-----|------|---------|------|
| Gas Gauging | TC       | Clear Voltage Threshold | 0x4825        | I2   | 0   | 5000 | 4100    | mV   |

**Description:** *GaugingStatus()[TC]* cell voltage clear threshold

### 13.14.7.3 Set RSOC % Threshold

| Class       | Subclass | Name                 | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | TC       | Set RSOC % Threshold | 0x4827        | U1   | 0   | 100 | 100     | %    |

**Description:** *GaugingStatus()[TC] RelativeStateOfCharge()* set threshold

### 13.14.7.4 Clear RSOC % Threshold

| Class       | Subclass | Name                   | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | TC       | Clear RSOC % Threshold | 0x4828        | U1   | 0   | 100 | 95      | %    |

**Description:** *GaugingStatus()[TC] RelativeStateOfCharge()* clear threshold

## 13.14.8 State

### 13.14.8.1 QMax

| Class       | Subclass | Name             | Start Address | Type | Min | Max   | Default | Unit | Description                               |
|-------------|----------|------------------|---------------|------|-----|-------|---------|------|-------------------------------------------|
| Gas Gauging | State    | QMax Cell 1      | 0x4304        | I2   | 0   | 32767 | 4400    | mAh  | QMax Cell 1                               |
| Gas Gauging | State    | QMax Cell 2      | 0x4306        | I2   | 0   | 32767 | 4400    | mAh  | QMax Cell 2                               |
| Gas Gauging | State    | QMax Cell 3      | 0x4308        | I2   | 0   | 32767 | 4400    | mAh  | QMax Cell 3                               |
| Gas Gauging | State    | QMax Cell 4      | 0x430A        | I2   | 0   | 32767 | 4400    | mAh  | QMax Cell 4                               |
| Gas Gauging | State    | QMax Pack        | 0x430C        | I2   | 0   | 32767 | 4400    | mAh  | QMax of the whole stack                   |
| Gas Gauging | State    | Qmax Cycle Count | 0x430E        | U2   | 0   | 65535 | 0       | —    | The <i>CycleCount()</i> when Qmax updated |

### 13.14.8.2 Update Status

| Class       | Subclass | Name          | Start Address | Type | Min  | Max  | Default | Unit |
|-------------|----------|---------------|---------------|------|------|------|---------|------|
| Gas Gauging | State    | Update Status | 0x4310        | H1   | 0x00 | 0x0E | 0x00    | —    |

|      |      |      |      |      |        |         |         |
|------|------|------|------|------|--------|---------|---------|
| 7    | 6    | 5    | 4    | 3    | 2      | 1       | 0       |
| RSVD | RSVD | RSVD | RSVD | QMax | Enable | Update1 | Update0 |

**RSVD (Bits 7–4):** Reserved. Do not use.

**QMax update in the field (Bit 3)**

1 = Updated

0 = Not updated

**Enable (Bit 2):** Impedance Track gauging and lifetime updating enable

1 = Enabled

0 = Disabled

**Update1, Update0 (Bits 1–0):** Update Status

0,0 = Impedance Track gauging and lifetime updating is disabled.

0,1 = QMax updated

1,0 = QMax and Ra table have been updated.

### 13.14.8.3 Cell 1–4 Chg Voltage at EoC

#### 13.14.8.3.1 Cell 1Chg Voltage at EoC

| Class       | Subclass | Name                     | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | State    | Cell 1Chg Voltage at EoC | 0x4311        | I2   | 0   | 32767 | 4200    | mV   |

**Description:** Cell 1 voltage value at end of charge

#### 13.14.8.3.2 Cell 2 Chg Voltage at EoC

| Class       | Subclass | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | State    | Cell 2 Chg Voltage at EoC | 0x4313        | I2   | 0   | 32767 | 4200    | mV   |

**Description:** Cell 2 voltage value at end of charge

#### 13.14.8.3.3 Cell 3 Chg Voltage at EoC

| Class       | Subclass | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | State    | Cell 3 Chg Voltage at EoC | 0x4315        | I2   | 0   | 32767 | 4200    | mV   |

**Description:** Cell 3 voltage value at end of charge

#### 13.14.8.3.4 Cell 4 Chg Voltage at EoC

| Class       | Subclass | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | State    | Cell 4 Chg Voltage at EoC | 0x4317        | I2   | 0   | 32767 | 4200    | mV   |

**Description:** Cell 4 voltage value at end of charge

### 13.14.8.4 Current at EoC

| Class       | Subclass | Name           | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | State    | Current at EoC | 0x4319        | I2   | 0   | 32767 | 250     | mA   |

**Description:** Current at end of charge

### 13.14.8.5 Average Last Run

#### 13.14.8.5.1 Avg I Last Run

| Class       | Subclass | Name           | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|----------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | State    | Avg I Last Run | 0x431B        | I2   | –32768 | 32767 | –2000   | mA   |



**Description:** Average current last discharge cycle

### 13.14.8.5.2 Avg P Last Run

| Class       | Subclass | Name           | Start Address | Type | Min    | Max   | Default | Unit  |
|-------------|----------|----------------|---------------|------|--------|-------|---------|-------|
| Gas Gauging | State    | Avg P Last Run | 0x431D        | I2   | –32768 | 32767 | –3022   | 10 mW |

**Description:** Average power last discharge cycle

### 13.14.8.6 Delta Voltage

| Class       | Subclass | Name          | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|---------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | State    | Delta Voltage | 0x431F        | I2   | –32768 | 32767 | 0       | mV   |

**Description:** *Voltage()* delta between normal and short load spikes to optimize run time calculation

### 13.14.8.7 Temp

#### 13.14.8.7.1 Temp k

| Class       | Subclass | Name   | Start Address | Type | Min | Max   | Default | Unit              |
|-------------|----------|--------|---------------|------|-----|-------|---------|-------------------|
| Gas Gauging | State    | Temp k | 0x4321        | I2   | 0   | 32767 | 100     | 0.1°C/<br>2560 mW |

**Description:** Initial Thermal model temperature factor

#### 13.14.8.7.2 Temp a

| Class       | Subclass | Name   | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------|---------------|------|-----|-------|---------|------|
| Gas Gauging | State    | Temp a | 0x4323        | I2   | 0   | 32767 | 1000    | —    |

**Description:** Initial Thermal model temperature

### 13.14.8.8 Max Avg Last Run

#### 13.14.8.8.1 Max Avg I Last Run

| Class       | Subclass | Name               | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | State    | Max Avg I Last Run | 0x4325        | I2   | –32768 | 32767 | –2000   | mA   |

**Description:** Max current last discharge cycle

#### 13.14.8.8.2 Max Avg P Last Run

| Class       | Subclass | Name               | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | State    | Max Avg P Last Run | 0x4327        | I2   | –32768 | 32767 | –3022   | cW   |

**Description:** Max power last discharge cycle

### 13.14.9 Cycle Count

| Class       | Subclass | Name        | Start Address | Type | Min | Max   | Default | Unit | Description |
|-------------|----------|-------------|---------------|------|-----|-------|---------|------|-------------|
| Gas Gauging | State    | Cycle Count | 0x4340        | U2   | 0   | 65535 | 0       | —    | Cycle Count |

**Description:** Value reported by *CycleCount()*. Updated by the gauge automatically based on **Cycle Count Percentage**.

### 13.14.10 IT Config

#### 13.14.10.1 Load Select

| Class       | Subclass | Name        | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Load Select | 0x4300        | U1   | 0   | 7   | 7       | —    |

**Description:** Defines Load compensation mode used by gauging algorithm

#### 13.14.10.2 Load Mode

| Class       | Subclass | Name      | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-----------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Load Mode | 0x4302        | U1   | 0   | 1   | 0       | —    |

**Description:** Defines unit used by gauging algorithm:

0 = Constant Current

1 = Constant Power

#### 13.14.10.3 Design Resistance

| Class       | Subclass | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Design Resistance | 0x482E        | I2   | 1   | 32767 | 42      | mΩ   |

**Description:** Averaged cell resistance at **Reference Grid** point. Automatically updated when Update Status is set to 0x6 by the gauge. To automatically update again, set Update Status to 0x4 or manually set when Update Status is set to 0x6.

#### 13.14.10.4 User Rate-mA

| Class       | Subclass | Name         | Start Address | Type | Min   | Max | Default | Unit |
|-------------|----------|--------------|---------------|------|-------|-----|---------|------|
| Gas Gauging | IT Cfg   | User Rate-mA | 0x4830        | I2   | –9000 | 0   | 0       | mA   |

**Description:** Discharge rate used for capacity calculation selected by **Load Select** = 6

#### 13.14.10.5 User Rate-cW

| Class       | Subclass | Name         | Start Address | Type | Min    | Max | Default | Unit |
|-------------|----------|--------------|---------------|------|--------|-----|---------|------|
| Gas Gauging | IT Cfg   | User Rate-cW | 0x4832        | I2   | –32768 | 0   | 0       | cW   |

**Description:** Discharge rate used for capacity calculation selected by **Load Select** = 6

#### 13.14.10.6 Reserve Cap-mAh

| Class       | Subclass | Name            | Start Address | Type | Min | Max  | Default | Unit |
|-------------|----------|-----------------|---------------|------|-----|------|---------|------|
| Gas Gauging | IT Cfg   | Reserve Cap-mAh | 0x4834        | I2   | 0   | 9000 | 0       | mAh  |

**Description:** Capacity reserved available when the gauging algorithm reports 0% *RelativeStateOfCharge()*

### 13.14.10.7 Reserve Cap-cWh

| Class       | Subclass | Name            | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Reserve Cap-cWh | 0x4836        | I2   | 0   | 32000 | 0       | cWh  |

**Description:** Capacity reserved available when the gauging algorithm reports 0% *RelativeStateOfCharge()*

### 13.14.10.8 Ra Filter

| Class       | Subclass | Name      | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|-----------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Ra Filter | 0x474E        | U2   | 0   | 999 | 500     | %    |

**Description:** Filter value used in Ra Updates and specifies what percentage of Ra update is from the new value (100% setting) vs. old value (setting). The recommended setting is 80% if the **[RSOC\_CONV]** feature is enabled. Otherwise, the setting should be 50% as default.

### 13.14.10.9 Ra Max Delta

| Class       | Subclass | Name         | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|--------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Ra Max Delta | 0x4751        | U1   | 0   | 255 | 15      | %    |

**Description:** Maximum value of allowed Ra change

### 13.14.10.10 Reference Grid

| Class       | Subclass | Name           | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Reference Grid | 0x4753        | U1   | 0   | 14  | 4       | —    |

**Description:** **Reference Grid** point used by Design Resistance. The default setting should be used if the **[RSOC\_CONV]** feature is enabled. Otherwise, grid point 11 should be used to ensure resistance updates fast enough at the grid where discharge termination occurs.

### 13.14.10.11 Resistance Parameter Filter

| Class       | Subclass | Name                        | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Resistance Parameter Filter | 0x4754        | U2   | 1   | 65535 | 65142   | —    |

**Description:** This is one of the filters used for resistance update. Reducing this filter setting can improve low temperature performance at high rates. The default setting is 41 s. It is recommended to keep this filter within the range of 4 s (that is, DF setting = 61680) up to the default 41 s (that is, DF setting = 65142). Examining the Term Voltage Delta setting and Fast Scale Start SOC should be done prior to twisting this parameter when trying to improve the RSOC performance.

The following is the formula to convert the DF setting into actual filter time constant: Filter time constant =  $[0.25 / (1 - (DF\_Value / 65536))] - 0.25$ .

### 13.14.10.12 Near EDV Ra Param Filter

| Class       | Subclass | Name                     | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Near EDV Ra Param Filter | 0x4756        | U2   | 1   | 65535 | 59220   | —    |

**Description:** Ra filter used in the fast scaling region if [FF\_NEAR\_EDV] = 1. Default value should be used.

### 13.14.10.13 Qmax Delta

| Class       | Subclass | Name       | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Qmax Delta | 0x477C        | U1   | 3   | 100 | 5       | %    |

**Description:** Maximum allowed Qmax change from its previous value. The Qmax change will be capped by this setting if the delta from the previous Qmax is larger than **Qmax Delta**. **Qmax Delta** is a percentage of Design Capacity.

### 13.14.10.14 Qmax Upper Bound

| Class       | Subclass | Name             | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Qmax Upper Bound | 0x477D        | U1   | 100 | 255 | 130     | %    |

**Description:** Maximum Qmax value over the lifetime of the pack. If the updated Qmax value is larger than this setting, the updated Qmax will be capped to **Qmax Upper Bound**. **Qmax Upper Bound** is a percentage of Design Capacity.

### 13.14.10.15 Term Voltage

| Class       | Subclass | Name         | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Term Voltage | 0x477E        | I2   | 0   | 32767 | 9000    | mV   |

**Description:** Min stack voltage to be used for capacity calculation

### 13.14.10.16 Term Voltage Delta

| Class       | Subclass | Name               | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|--------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Term Voltage Delta | 0x4780        | I2   | 0   | 32767 | 300     | mV   |

**Description:** Controls when the [RSOC\_CONV] feature becomes active. The recommended setting is 3.3 – **Term Voltage**/Number Cells.

The default setting is 300 mV, which is assuming a typical 3-V termination voltage per cell. If a different termination voltage is used, this parameter should be adjusted accordingly.

### 13.14.10.17 Term Min Cell V

| Class       | Subclass | Name            | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Term Min Cell V | 0x4782        | I2   | 0   | 32767 | 2800    | mV   |

**Description:** Minimum cell termination voltage when used when [CELL\_TERM] = 1. This is intended to allow the IT algorithm to reach 0% before CUV is triggered; therefore, this value should be set at or above **CUV:Threshold**.

### 13.14.10.18 Voltage Consistency Delta

| Class       | Subclass | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|---------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Voltage Consistency Delta | 0x4787        | I2   | 0   | 32767 | 300     | mV   |

**Description:** Use in voltage consistency check. See *[VOLTAGE\_CONSIST]* in the [Section 6.6](#) for details.

### 13.14.10.19 Fast Scale Start SOC

| Class       | Subclass | Name                 | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------|----------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | IT Cfg   | Fast Scale Start SOC | 0x479A        | U1   | 0   | 100 | 10      | %    |

**Description:** Control the start of convergence when *[RSOC\_CONV]* = 1 based on RSOC %. Raising this setting can improve RSOC drop at the end of discharge. However, the RSOC % chosen for this setting must keep after the sharp drop of the discharge curve (the keen of the discharge curve).

### 13.14.10.20 Pack Resistance

| Class       | Subclass | Name            | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | Pack Resistance | 0x4300        | I2   | 0   | 32767 | 30      | mΩ   |

**Description:** Pack side resistance value accessed using *TURBO\_PACK\_R()*

### 13.14.10.21 System Resistance

| Class       | Subclass | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-------------|----------|-------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | IT Cfg   | System Resistance | 0x4302        | I2   | 0   | 32767 | 0       | mΩ   |

**Description:** System side resistance value accessed using *TURBO\_SYS\_R()*

### 13.14.11 Smoothing

| Class       | Subclass  | Name              | Start Address | Type | Min | Max   | Default | Unit |
|-------------|-----------|-------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | Smoothing | Smooth Relax Time | 0x4838        | I2   | 1   | 32767 | 1000    | s    |

**Description:** If *[RELAX\_SMOOTH\_OK]* = 1, the delta Remaining Capacity and Full Charge Capacity is smoothed over this set period of time. It is recommended to use the default setting.

### 13.14.12 Condition Flag

| Class       | Subclass       | Name            | Start Address | Type | Min | Max | Default | Unit |
|-------------|----------------|-----------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | Condition Flag | Max Error Limit | 0x483A        | U1   | 0   | 100 | 100     | %    |

**Description:** Max Error Limit Percentage

### 13.14.13 Max Error

#### 13.14.13.1 Time Cycle Equivalent

| Class       | Subclass  | Name                  | Start Address | Type | Min | Max | Default | Unit |
|-------------|-----------|-----------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | Max Error | Time Cycle Equivalent | 0x483E        | U1   | 1   | 255 | 12      | 2 h  |

**Description:** After valid QMax update, each passed time period of **Time Cycle Equivalent** will increment of *MaxError()* by **Cycle Delta**.

### 13.14.13.2 Cycle Delta

| Class       | Subclass  | Name        | Start Address | Type | Min | Max | Default | Unit  |
|-------------|-----------|-------------|---------------|------|-----|-----|---------|-------|
| Gas Gauging | Max Error | Cycle Delta | 0x483F        | U1   | 0   | 255 | 5       | 0.01% |

**Description:** Each increment of *CycleCount()* after valid QMax update will increment of *MaxError()* by **Cycle Delta**. Setting this parameter to 0 disables the *MaxError()* increment by time or cycle increment.

### 13.14.14 SoH

| Class       | Subclass | Name          | Start Address | Type | Min | Max | Default | Unit       |
|-------------|----------|---------------|---------------|------|-----|-----|---------|------------|
| Gas Gauging | SoH      | SoH Load Rate | 0x47A3        | U1   | 1   | 255 | 50      | 0.1 h rate |

**Description:** Current rate used in SoH simulation specified in hour-rate (that is, current = C/**SoH Load Rate**)

### 13.14.15 TURBO Cfg

#### 13.14.15.1 Min Turbo Power

| Class       | Subclass  | Name            | Start Address | Type | Min    | Max   | Default | Unit |
|-------------|-----------|-----------------|---------------|------|--------|-------|---------|------|
| Gas Gauging | Turbo Cfg | Min Turbo Power | 0x44C0        | I2   | -32768 | 32767 | -1000   | cW   |

**Description:** This is the minimal Turbo Power for the TURBO BOOST mode used by the system toward the end of discharge. This value is used when **[TDELATV] = 1** to calculate the **Delta Voltage**. Using the lowest turbo level instead of the regular learned **Delta Voltage** for IT simulation can avoid unnecessary SOC jumps when the system is switching from higher to lower turbo mode levels, reducing its power approaching the end of discharge.

#### 13.14.15.2 Max Current C Rate

| Class       | Subclass  | Name               | Start Address | Type | Min  | Max | Default | Unit |
|-------------|-----------|--------------------|---------------|------|------|-----|---------|------|
| Gas Gauging | Turbo Cfg | Max Current C Rate | 0x44C2        | I1   | -127 | 00  | -4      | C    |

**Description:** This value specifies the maximal discharge current. If *TURBO\_CURRENT()* > **Max Current C Rate**, the *TURBO\_CURRENT()* will be capped to this setting and the *TURBO\_POWER()* will be adjusted accordingly.

#### 13.14.15.3 High Frequency Resistance

| Class       | Subclass  | Name                      | Start Address | Type | Min | Max   | Default | Unit |
|-------------|-----------|---------------------------|---------------|------|-----|-------|---------|------|
| Gas Gauging | Turbo Cfg | High Frequency Resistance | 0x44C3        | I2   | 0   | 32767 | 20      | mΩ   |

**Description:** This is the high frequency resistance related the specific cell chemistry and pack configuration.

### 13.14.15.4 Reserve Energy %

| Class       | Subclass  | Name             | Start Address | Type | Min | Max | Default | Unit |
|-------------|-----------|------------------|---------------|------|-----|-----|---------|------|
| Gas Gauging | Turbo Cfg | Reserve Energy % | 0x44C9        | I1   | 0   | 100 | 2       | %    |

**Description:** Remaining energy at present average discharge rate (as defined in **Load Select**) until the maximal peak power will reach the value reported by **MAX\_POWER()**.

## 13.15 RA Table

### 13.15.1 R\_a0

| Class    | Subclass | Name            | Start Address | Type | Min    | Max    | Default | Unit               |
|----------|----------|-----------------|---------------|------|--------|--------|---------|--------------------|
| RA Table | R_a0     | Cell 0 R_A Flag | 0x4100        | H2   | 0x0000 | 0xFFFF | 0xFF55  | 2 <sup>-10</sup> Ω |

**Description:**

This value indicates the validity of the cell impedance table for Cell1. It is recommended not to change this value manually.

| High Byte |                                           | Low Byte |                                                    |
|-----------|-------------------------------------------|----------|----------------------------------------------------|
| 0x00      | Cell impedance and QMax updated           | 0x00     | Table not used and QMax updated                    |
| 0x05      | RELAX mode and QMax update in progress    | 0x55     | Table being used                                   |
| 0x55      | DISCHARGE mode and cell impedance updated | 0xFF     | Table never used, no QMax or cell impedance update |
| 0xFF      | Cell impedance never updated              |          |                                                    |

The gauge stores and updates the impedance profile for Cell1 as shown in the following table.

| Class    | Subclass | Name          | Start Address | Type | Min | Max   | Default | Unit               | Description                        |
|----------|----------|---------------|---------------|------|-----|-------|---------|--------------------|------------------------------------|
| RA Table | R_a0     | Cell 0 R_A 0  | 0x4102        | I2   | 0   | 32767 | 38      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 0  |
| RA Table | R_a0     | Cell 0 R_A 1  | 0x4104        | I2   | 0   | 32767 | 41      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 1  |
| RA Table | R_a0     | Cell 0 R_A 2  | 0x4106        | I2   | 0   | 32767 | 43      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 2  |
| RA Table | R_a0     | Cell 0 R_A 3  | 0x4108        | I2   | 0   | 32767 | 44      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 3  |
| RA Table | R_a0     | Cell 0 R_A 4  | 0x410A        | I2   | 0   | 32767 | 42      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 4  |
| RA Table | R_a0     | Cell 0 R_A 5  | 0x410C        | I2   | 0   | 32767 | 42      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 5  |
| RA Table | R_a0     | Cell 0 R_A 6  | 0x410E        | I2   | 0   | 32767 | 45      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 6  |
| RA Table | R_a0     | Cell 0 R_A 7  | 0x4110        | I2   | 0   | 32767 | 48      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 7  |
| RA Table | R_a0     | Cell 0 R_A 8  | 0x4112        | I2   | 0   | 32767 | 49      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 8  |
| RA Table | R_a0     | Cell 0 R_A 9  | 0x4114        | I2   | 0   | 32767 | 52      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 9  |
| RA Table | R_a0     | Cell 0 R_A 10 | 0x4116        | I2   | 0   | 32767 | 56      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 10 |
| RA Table | R_a0     | Cell 0 R_A 11 | 0x4118        | I2   | 0   | 32767 | 64      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 11 |
| RA Table | R_a0     | Cell 0 R_A 12 | 0x411A        | I2   | 0   | 32767 | 74      | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 12 |
| RA Table | R_a0     | Cell 0 R_A 13 | 0x411C        | I2   | 0   | 32767 | 128     | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 13 |
| RA Table | R_a0     | Cell 0 R_A 14 | 0x411E        | I2   | 0   | 32767 | 378     | 2 <sup>-10</sup> Ω | Cell 0 resistance at grid point 14 |

### 13.15.2 R\_a1

| Class    | Subclass | Name            | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|-----------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a1     | Cell 1 R_A Flag | 0x4140        | H2   | 0x0000 | 0xFFFF | 0xFF55  | —    |

**Description:**

This value indicates the validity of the cell impedance table for Cell2. It is recommended not to change this value manually.

| High Byte |                                           | Low Byte |                                                    |
|-----------|-------------------------------------------|----------|----------------------------------------------------|
| 0x00      | Cell impedance and QMax updated           | 0x00     | Table not used and QMax updated                    |
| 0x05      | RELAX mode and QMax update in progress    | 0x55     | Table being used                                   |
| 0x55      | DISCHARGE mode and cell impedance updated | 0xFF     | Table never used, no QMax or cell impedance update |
| 0xFF      | Cell impedance never updated              |          |                                                    |

The gauge stores and updates the impedance profile for Cell2, as shown in the following table.

| Class    | Subclass | Name          | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|---------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a1     | Cell 1 R_A 0  | 0x4142        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 0  |
| RA Table | R_a1     | Cell 1 R_A 1  | 0x4144        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 1  |
| RA Table | R_a1     | Cell 1 R_A 2  | 0x4146        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 2  |
| RA Table | R_a1     | Cell 1 R_A 3  | 0x4148        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 3  |
| RA Table | R_a1     | Cell 1 R_A 4  | 0x414A        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 4  |
| RA Table | R_a1     | Cell 1 R_A 5  | 0x414C        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 5  |
| RA Table | R_a1     | Cell 1 R_A 6  | 0x414E        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 6  |
| RA Table | R_a1     | Cell 1 R_A 7  | 0x4150        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 7  |
| RA Table | R_a1     | Cell 1 R_A 8  | 0x4152        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 8  |
| RA Table | R_a1     | Cell 1 R_A 9  | 0x4154        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 9  |
| RA Table | R_a1     | Cell 1 R_A 10 | 0x4156        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 10 |
| RA Table | R_a1     | Cell 1 R_A 11 | 0x4158        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 11 |
| RA Table | R_a1     | Cell 1 R_A 12 | 0x415A        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 12 |
| RA Table | R_a1     | Cell 1 R_A 13 | 0x415C        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 13 |
| RA Table | R_a1     | Cell 1 R_A 14 | 0x415E        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 14 |

### 13.15.3 R\_a2

| Class    | Subclass | Name            | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|-----------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a2     | Cell 2 R_A Flag | 0x4180        | H2   | 0x0000 | 0xFFFF | 0xFF55  | —    |

**Description:**

This value indicates the validity of the cell impedance table for Cell3. It is recommended not to change this value manually.

| High Byte |                                           | Low Byte |                                                    |
|-----------|-------------------------------------------|----------|----------------------------------------------------|
| 0x00      | Cell impedance and QMax updated           | 0x00     | Table not used and QMax updated                    |
| 0x05      | RELAX mode and QMax update in progress    | 0x55     | Table being used                                   |
| 0x55      | DISCHARGE mode and cell impedance updated | 0xFF     | Table never used, no QMax or cell impedance update |
| 0xFF      | Cell impedance never updated              |          |                                                    |



The gauge stores and updates the impedance profile for Cell3 as shown in the following table.

| Class    | Subclass | Name          | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|---------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a2     | Cell 2 R_A 0  | 0x4182        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 0  |
| RA Table | R_a2     | Cell 2 R_A 1  | 0x4184        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 1  |
| RA Table | R_a2     | Cell 2 R_A 2  | 0x4186        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 2  |
| RA Table | R_a2     | Cell 2 R_A 3  | 0x4188        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 3  |
| RA Table | R_a2     | Cell 2 R_A 4  | 0x418A        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 4  |
| RA Table | R_a2     | Cell 2 R_A 5  | 0x418C        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 5  |
| RA Table | R_a2     | Cell 2 R_A 6  | 0x418E        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 6  |
| RA Table | R_a2     | Cell 2 R_A 7  | 0x4190        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 7  |
| RA Table | R_a2     | Cell 2 R_A 8  | 0x4192        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 8  |
| RA Table | R_a2     | Cell 2 R_A 9  | 0x4194        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 9  |
| RA Table | R_a2     | Cell 2 R_A 10 | 0x4196        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 10 |
| RA Table | R_a2     | Cell 2 R_A 11 | 0x4198        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 11 |
| RA Table | R_a2     | Cell 2 R_A 12 | 0x419A        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 12 |
| RA Table | R_a2     | Cell 2 R_A 13 | 0x419C        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 13 |
| RA Table | R_a2     | Cell 2 R_A 14 | 0x419E        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 14 |

### 13.15.4 R\_a3

| Class    | Subclass | Name            | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|-----------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a3     | Cell 3 R_A Flag | 0x41C0        | H2   | 0x0000 | 0xFFFF | 0xFF55  | —    |

#### Description:

This value indicates the validity of the cell impedance table for Cell4. It is recommended not to change this value manually.

| High Byte | Low Byte                                  |
|-----------|-------------------------------------------|
| 0x00      | Cell impedance and QMax updated           |
| 0x05      | RELAX mode and QMax update in progress    |
| 0x55      | DISCHARGE mode and cell impedance updated |
| 0xFF      | Cell impedance never updated              |

The gauge stores and updates the impedance profile for Cell4 as shown in the following table.

| Class    | Subclass | Name         | Start Address | Type | Min    | Max   | Default | Unit               | Description                       |
|----------|----------|--------------|---------------|------|--------|-------|---------|--------------------|-----------------------------------|
| RA Table | R_a3     | Cell 3 R_A 0 | 0x41C2        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 0 |
| RA Table | R_a3     | Cell 3 R_A 1 | 0x41C4        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 1 |
| RA Table | R_a3     | Cell 3 R_A 2 | 0x41C6        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 2 |
| RA Table | R_a3     | Cell 3 R_A 3 | 0x41C8        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 3 |
| RA Table | R_a3     | Cell 3 R_A 4 | 0x41CA        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 4 |
| RA Table | R_a3     | Cell 3 R_A 5 | 0x41CC        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 5 |
| RA Table | R_a3     | Cell 3 R_A 6 | 0x41CE        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 6 |
| RA Table | R_a3     | Cell 3 R_A 7 | 0x41D0        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 7 |
| RA Table | R_a3     | Cell 3 R_A 8 | 0x41D2        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 8 |
| RA Table | R_a3     | Cell 3 R_A 9 | 0x41D4        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 9 |

| Class    | Subclass | Name          | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|---------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a3     | Cell 3 R_A 10 | 0x41D6        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 10 |
| RA Table | R_a3     | Cell 3 R_A 11 | 0x41D8        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 11 |
| RA Table | R_a3     | Cell 3 R_A 12 | 0x41DA        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 12 |
| RA Table | R_a3     | Cell 3 R_A 13 | 0x41DC        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 13 |
| RA Table | R_a3     | Cell 3 R_A 14 | 0x41DE        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 14 |

### 13.15.5 R\_a0x

| Class    | Subclass | Name             | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|------------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a0x    | xCell 0 R_A Flag | 0x4200        | H2   | 0x0000 | 0xFFFF | 0xFFFF  | —    |

#### Description:

This value indicates the validity of the cell impedance table for Cell1. It is recommended not to change this value manually.

| High Byte | Low Byte                                  |
|-----------|-------------------------------------------|
| 0x00      | Cell impedance and QMax updated           |
| 0x05      | RELAX mode and QMax update in progress    |
| 0x55      | DISCHARGE mode and cell impedance updated |
| 0xFF      | Cell impedance never updated              |

The gauge stores and updates the impedance profile for Cell1 as shown in the following table.

| Class    | Subclass | Name           | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|----------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a0x    | xCell 0 R_A 0  | 0x4202        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 0  |
| RA Table | R_a0x    | xCell 0 R_A 1  | 0x4204        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 1  |
| RA Table | R_a0x    | xCell 0 R_A 2  | 0x4206        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 2  |
| RA Table | R_a0x    | xCell 0 R_A 3  | 0x4208        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 3  |
| RA Table | R_a0x    | xCell 0 R_A 4  | 0x420A        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 4  |
| RA Table | R_a0x    | xCell 0 R_A 5  | 0x420C        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 5  |
| RA Table | R_a0x    | xCell 0 R_A 6  | 0x420E        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 6  |
| RA Table | R_a0x    | xCell 0 R_A 7  | 0x4210        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 7  |
| RA Table | R_a0x    | xCell 0 R_A 8  | 0x4212        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 8  |
| RA Table | R_a0x    | xCell 0 R_A 9  | 0x4214        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 9  |
| RA Table | R_a0x    | xCell 0 R_A 10 | 0x4216        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 10 |
| RA Table | R_a0x    | xCell 0 R_A 11 | 0x4218        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 11 |
| RA Table | R_a0x    | xCell 0 R_A 12 | 0x421A        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 12 |
| RA Table | R_a0x    | xCell 0 R_A 13 | 0x421C        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 13 |
| RA Table | R_a0x    | xCell 0 R_A 14 | 0x421E        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 0 resistance at grid point 14 |

### 13.15.6 R\_a1x

| Class    | Subclass | Name             | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|------------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a1x    | xCell 1 R_A Flag | 0x4240        | H2   | 0x0000 | 0xFFFF | 0xFFFF  | —    |

**Description:**

This value indicates the validity of the cell impedance table for Cell2. It is recommended not to change this value manually.

| High Byte |                                           | Low Byte |                                                    |
|-----------|-------------------------------------------|----------|----------------------------------------------------|
| 0x00      | Cell impedance and QMax updated           | 0x00     | Table not used and QMax updated                    |
| 0x05      | RELAX mode and QMax update in progress    | 0x55     | Table being used                                   |
| 0x55      | DISCHARGE mode and cell impedance updated | 0xFF     | Table never used, no QMax or cell impedance update |
| 0xFF      | Cell impedance never updated              |          |                                                    |

The gauge stores and updates the impedance profile for Cell2 as shown in the following table.

| Class    | Subclass | Name           | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|----------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a1x    | xCell 1 R_A 0  | 0x4242        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 0  |
| RA Table | R_a1x    | xCell 1 R_A 1  | 0x4244        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 1  |
| RA Table | R_a1x    | xCell 1 R_A 2  | 0x4246        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 2  |
| RA Table | R_a1x    | xCell 1 R_A 3  | 0x4248        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 3  |
| RA Table | R_a1x    | xCell 1 R_A 4  | 0x424A        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 4  |
| RA Table | R_a1x    | xCell 1 R_A 5  | 0x424C        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 5  |
| RA Table | R_a1x    | xCell 1 R_A 6  | 0x424E        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 6  |
| RA Table | R_a1x    | xCell 1 R_A 7  | 0x4250        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 7  |
| RA Table | R_a1x    | xCell 1 R_A 8  | 0x4252        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 8  |
| RA Table | R_a1x    | xCell 1 R_A 9  | 0x4254        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 9  |
| RA Table | R_a1x    | xCell 1 R_A 10 | 0x4256        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 10 |
| RA Table | R_a1x    | xCell 1 R_A 11 | 0x4258        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 11 |
| RA Table | R_a1x    | xCell 1 R_A 12 | 0x425A        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 12 |
| RA Table | R_a1x    | xCell 1 R_A 13 | 0x425C        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 13 |
| RA Table | R_a1x    | xCell 1 R_A 14 | 0x425E        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 1 resistance at grid point 14 |

### 13.15.7 R\_a2x

| Class    | Subclass | Name             | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|------------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a2x    | xCell 2 R_A Flag | 0x4280        | H2   | 0x0000 | 0xFFFF | 0xFFFF  | —    |

**Description:**

This value indicates the validity of the cell impedance table for Cell3. It is recommended not to change this value manually.

| High Byte |                                        | Low Byte |                                 |
|-----------|----------------------------------------|----------|---------------------------------|
| 0x00      | Cell impedance and QMax updated        | 0x00     | Table not used and QMax updated |
| 0x05      | RELAX mode and QMax update in progress | 0x55     | Table being used                |

| High Byte |                                           | Low Byte |                                                    |
|-----------|-------------------------------------------|----------|----------------------------------------------------|
| 0x55      | DISCHARGE mode and cell impedance updated | 0xFF     | Table never used, no QMax or cell impedance update |
| 0xFF      | Cell impedance never updated              |          |                                                    |

The gauge stores and updates the impedance profile for Cell3 as shown in the following table.

| Class    | Subclass | Name           | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|----------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a2x    | xCell 2 R_A 0  | 0x4282        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 0  |
| RA Table | R_a2x    | xCell 2 R_A 1  | 0x4284        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 1  |
| RA Table | R_a2x    | xCell 2 R_A 2  | 0x4286        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 2  |
| RA Table | R_a2x    | xCell 2 R_A 3  | 0x4288        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 3  |
| RA Table | R_a2x    | xCell 2 R_A 4  | 0x428A        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 4  |
| RA Table | R_a2x    | xCell 2 R_A 5  | 0x428C        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 5  |
| RA Table | R_a2x    | xCell 2 R_A 6  | 0x428E        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 6  |
| RA Table | R_a2x    | xCell 2 R_A 7  | 0x4290        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 7  |
| RA Table | R_a2x    | xCell 2 R_A 8  | 0x4292        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 8  |
| RA Table | R_a2x    | xCell 2 R_A 9  | 0x4294        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 9  |
| RA Table | R_a2x    | xCell 2 R_A 10 | 0x4296        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 10 |
| RA Table | R_a2x    | xCell 2 R_A 11 | 0x4298        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 11 |
| RA Table | R_a2x    | xCell 2 R_A 12 | 0x429A        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 12 |
| RA Table | R_a2x    | xCell 2 R_A 13 | 0x429C        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 13 |
| RA Table | R_a2x    | xCell 2 R_A 14 | 0x429E        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 2 resistance at grid point 14 |

### 13.15.8 R\_a3x

| Class    | Subclass | Name             | Start Address | Type | Min    | Max    | Default | Unit |
|----------|----------|------------------|---------------|------|--------|--------|---------|------|
| RA Table | R_a3x    | xCell 3 R_A Flag | 0x42C0        | H2   | 0x0000 | 0xFFFF | 0xFFFF  | —    |

#### Description:

This value indicates the validity of the cell impedance table for Cell4. It is recommended not to change this value manually.

| High Byte |                                           | Low Byte |                                                    |
|-----------|-------------------------------------------|----------|----------------------------------------------------|
| 0x00      | Cell impedance and QMax updated           | 0x00     | Table not used and QMax updated                    |
| 0x05      | RELAX mode and QMax update in progress    | 0x55     | Table being used                                   |
| 0x55      | DISCHARGE mode and cell impedance updated | 0xFF     | Table never used, no QMax or cell impedance update |
| 0xFF      | Cell impedance never updated              |          |                                                    |

The gauge stores and updates the impedance profile for Cell4 as shown in the following table.

| Class    | Subclass | Name          | Start Address | Type | Min    | Max   | Default | Unit               | Description                       |
|----------|----------|---------------|---------------|------|--------|-------|---------|--------------------|-----------------------------------|
| RA Table | R_a3x    | xCell 3 R_A 0 | 0x42C2        | I2   | –32768 | 32768 | 38      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 0 |
| RA Table | R_a3x    | xCell 3 R_A 1 | 0x42C4        | I2   | –32768 | 32768 | 41      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 1 |
| RA Table | R_a3x    | xCell 3 R_A 2 | 0x42C6        | I2   | –32768 | 32768 | 43      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 2 |

| Class    | Subclass | Name           | Start Address | Type | Min    | Max   | Default | Unit               | Description                        |
|----------|----------|----------------|---------------|------|--------|-------|---------|--------------------|------------------------------------|
| RA Table | R_a3x    | xCell 3 R_A 3  | 0x42C8        | I2   | –32768 | 32768 | 44      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 3  |
| RA Table | R_a3x    | xCell 3 R_A 4  | 0x42CA        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 4  |
| RA Table | R_a3x    | xCell 3 R_A 5  | 0x42CC        | I2   | –32768 | 32768 | 42      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 5  |
| RA Table | R_a3x    | xCell 3 R_A 6  | 0x42CE        | I2   | –32768 | 32768 | 45      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 6  |
| RA Table | R_a3x    | xCell 3 R_A 7  | 0x42D0        | I2   | –32768 | 32768 | 48      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 7  |
| RA Table | R_a3x    | xCell 3 R_A 8  | 0x42D2        | I2   | –32768 | 32768 | 49      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 8  |
| RA Table | R_a3x    | xCell 3 R_A 9  | 0x42D4        | I2   | –32768 | 32768 | 52      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 9  |
| RA Table | R_a3x    | xCell 3 R_A 10 | 0x42D6        | I2   | –32768 | 32768 | 56      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 10 |
| RA Table | R_a3x    | xCell 3 R_A 11 | 0x42D8        | I2   | –32768 | 32768 | 64      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 11 |
| RA Table | R_a3x    | xCell 3 R_A 12 | 0x42DA        | I2   | –32768 | 32768 | 74      | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 12 |
| RA Table | R_a3x    | xCell 3 R_A 13 | 0x42DC        | I2   | –32768 | 32768 | 128     | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 13 |
| RA Table | R_a3x    | xCell 3 R_A 14 | 0x42DE        | I2   | –32768 | 32768 | 378     | 2 <sup>–10</sup> Ω | Cell 3 resistance at grid point 14 |

## 13.16 SBS Configuration

### 13.16.1 Data

#### 13.16.1.1 Remaining Capacity Alarm

##### 13.16.1.1.1 Remaining Ah Capacity Alarm

| Class             | Subclass | Name                        | Start Address | Type | Min | Max | Default | Unit |
|-------------------|----------|-----------------------------|---------------|------|-----|-----|---------|------|
| SBS Configuration | Data     | Remaining Ah Capacity Alarm | 0x47FC        | U2   | 0   | 700 | 300     | mAh  |

**Description:** *RemainingCapacityAlarm()* value in mAh

##### 13.16.1.1.2 Remaining Wh Capacity Alarm

| Class             | Subclass | Name                        | Start Address | Type | Min | Max  | Default | Unit |
|-------------------|----------|-----------------------------|---------------|------|-----|------|---------|------|
| SBS Configuration | Data     | Remaining Wh Capacity Alarm | 0x47FE        | U2   | 0   | 1000 | 432     | cWh  |

**Description:** *RemainingCapacityAlarm()* value in 10 mWh

##### 13.16.1.2 RemainingTimeAlarm

| Class             | Subclass | Name                 | Start Address | Type | Min | Max | Default | Unit |
|-------------------|----------|----------------------|---------------|------|-----|-----|---------|------|
| SBS Configuration | Data     | Remaining Time Alarm | 0x4800        | U2   | 0   | 30  | 10      | min  |

**Description:** *RemainingTimeAlarm()* value

##### 13.16.1.3 Initial Battery Mode

| Class             | Subclass | Name                 | Start Address | Type | Min    | Max    | Default | Unit |
|-------------------|----------|----------------------|---------------|------|--------|--------|---------|------|
| SBS Configuration | Data     | Initial Battery Mode | 0x4802        | H2   | 0x0000 | 0xFFFF | 0x0081  | —    |

|      |      |      |      |      |      |     |     |
|------|------|------|------|------|------|-----|-----|
| 15   | 14   | 13   | 12   | 11   | 10   | 9   | 8   |
| CAPM | CHGM | AM   | RSVD | RSVD | RSVD | PB  | CC  |
| 7    | 6    | 5    | 4    | 3    | 2    | 1   | 0   |
| CF   | RSVD | RSVD | RSVD | RSVD | RSVD | PBS | ICC |

**CAPM (Bit 15):** Capacity\_Mode (R/W)

- 1 = Report in 10 mW or 10 mWh
- 0 = Report in mA or mAh (default)

**CHGM (Bit 14):** Charger\_Mode (R/W)

- 1 = Disable *ChargingVoltage()* and *ChargingCurrent()* broadcasts to host and smart battery charger (default)
- 0 = Enable *ChargingVoltage()* and *ChargingCurrent()* broadcasts to host and smart battery charger

**AM (Bit 13):** ALARM Mode (R/W)

- 1 = Disable AlarmWarning broadcasts to host and smart battery charger
- 0 = Enable AlarmWarning broadcasts to host and smart battery charger (default)

**RSVD (Bits 12–10):** Reserved. Do not use.

**PB (Bit 9):** Primary\_Battery (R/W)

- 1 = Battery operating in its primary role
- 0 = Battery operating in its secondary role (default)

**CC (Bit 8):** Charge\_Controller\_Enabled (R/W)

- 1 = Internal charge control enabled
- 0 = Internal charge control disabled (default)

**CF (Bit 7):** Condition\_Flag (R)

- 1 = Conditioning cycle requested
- 0 = Battery OK

**RSVD (Bits 6–2):** Reserved. Do not use.

**PBS (Bit 1):** Primary\_Battery\_Support (R)

- 1 = Primary or secondary battery support
- 0 = Function not supported (default)

**ICC (Bit 0):** Internal\_Charge\_Controller (R)

- 1 = Function supported
- 0 = Function not supported (default)

### 13.16.1.4 Specification Information

| Class             | Subclass | Name                      | Start Address | Type | Min    | Max    | Default | Unit |
|-------------------|----------|---------------------------|---------------|------|--------|--------|---------|------|
| SBS Configuration | Data     | Specification Information | 0x4804        | H2   | 0x0000 | 0xFFFF | 0x0031  | —    |

|         |         |         |         |          |          |          |          |
|---------|---------|---------|---------|----------|----------|----------|----------|
| 15      | 14      | 13      | 12      | 11       | 10       | 9        | 8        |
| IPScale | IPScale | IPScale | IPScale | VScale   | VScale   | VScale   | VScale   |
| 7       | 6       | 5       | 4       | 3        | 2        | 1        | 0        |
| Version | Version | Version | Version | Revision | Revision | Revision | Revision |

*SpecificationInformation()* values

**IPScale (Bits 15–12):** IP Scale Factor

0,0,0,0 = Reported currents and capacities scaled by 10E0 except *ChargingVoltage()* and *ChargingCurrent()*

0,0,0,1 = Reported currents and capacities scaled by 10E1 except *ChargingVoltage()* and *ChargingCurrent()*

0,0,1,0 = Reported currents and capacities scaled by 10E2 except *ChargingVoltage()* and *ChargingCurrent()*

0,0,1,1 = Reported currents and capacities scaled by 10E3 except *ChargingVoltage()* and *ChargingCurrent()*

**VScale (Bits 11–8):** Voltage Scale Factor

0,0,0,0 = Reported voltages scaled by 10E0

0,0,0,1 = Reported voltages scaled by 10E1

0,0,1,0 = Reported voltages scaled by 10E2

0,0,1,1 = Reported voltages scaled by 10E3

**Version (Bits 7–4):** Version

0,0,0,1 = Version 1.0

0,0,1,1 = Version 1.1

0,0,1,1 = Version 1.1 with optional PEC support

**Revision (Bits 3–0):** Revision

0,0,0,1 = Version 1.0 and 1.1 (default)

### 13.16.1.5 Manufacturer Date

| Class             | Subclass | Name              | Start Address | Type | Min | Max   | Default  | Unit |
|-------------------|----------|-------------------|---------------|------|-----|-------|----------|------|
| SBS Configuration | Data     | Manufacturer Date | 0x4067        | U2   | 0   | 65535 | 01/01/80 | —    |

**Description:** *ManufacturerDate()* value in the following format: Day + Month\*32 + (Year–1980) \* 512

### 13.16.1.6 Serial Number

| Class             | Subclass | Name          | Start Address | Type | Min    | Max    | Default | Unit |
|-------------------|----------|---------------|---------------|------|--------|--------|---------|------|
| SBS Configuration | Data     | Serial Number | 0x4069        | H2   | 0x0000 | 0xFFFF | 0x0001  | —    |

**Description:** *SerialNumber()* value

### 13.16.1.7 Manufacturer Name

| Class             | Subclass | Name              | Start Address | Type  | Min | Max | Default           | Unit  |
|-------------------|----------|-------------------|---------------|-------|-----|-----|-------------------|-------|
| SBS Configuration | Data     | Manufacturer Name | 0x406B        | S20+1 | —   | —   | Texas Instruments | ASCII |

**Description:** *ManufacturerName()* value

### 13.16.1.8 Device Name

| Class             | Subclass | Name        | Start Address | Type  | Min | Max | Default | Unit  |
|-------------------|----------|-------------|---------------|-------|-----|-----|---------|-------|
| SBS Configuration | Data     | Device Name | 0x4080        | S20+1 | —   | —   | bq40z50 | ASCII |

**Description:** *DeviceName()* value

### 13.16.1.9 Device Chemistry

| Class             | Subclass | Name             | Start Address | Type | Min | Max | Default | Unit  |
|-------------------|----------|------------------|---------------|------|-----|-----|---------|-------|
| SBS Configuration | Data     | Device Chemistry | 0x4095        | S4+1 | —   | —   | LION    | ASCII |

**Description:** *DeviceChemistry()* value





## AFE Threshold and Delay Settings

### A.1 Overload in Discharge Protection (AOLD)

**Table A-1. Overload in Discharge Protection Threshold  
(Settings:AFE:AFE Protection Control [RSNS] = 0)<sup>(1)</sup>**

| OLD Threshold ([RSNS] = 0) |           |         |           |
|----------------------------|-----------|---------|-----------|
| Setting                    | Threshold | Setting | Threshold |
| 0x00                       | –8.30 mV  | 0x08    | –30.54 mV |
| 0x01                       | –11.08 mV | 0x09    | –33.32 mV |
| 0x02                       | –13.86 mV | 0x0A    | –36.10 mV |
| 0x03                       | –16.64 mV | 0x0B    | –38.88 mV |
| 0x04                       | –19.42 mV | 0x0C    | –41.66 mV |
| 0x05                       | –22.20 mV | 0x0D    | –44.44 mV |
| 0x06                       | –24.98 mV | 0x0E    | –47.22 mV |
| 0x07                       | –27.76 mV | 0x0F    | –50.00 mV |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:OLD Threshold[3:0]** sets the voltage threshold.

**Table A-2. Overload in Discharge Protection Threshold  
(Settings:AFE:AFE Protection Control [RSNS] = 1)<sup>(1)</sup>**

| OLD Threshold ([RSNS] = 1) |           |         |            |
|----------------------------|-----------|---------|------------|
| Setting                    | Threshold | Setting | Threshold  |
| 0x00                       | –16.60 mV | 0x08    | –61.08 mV  |
| 0x01                       | –22.16 mV | 0x09    | –66.64 mV  |
| 0x02                       | –27.72 mV | 0x0A    | –72.20 mV  |
| 0x03                       | –33.28 mV | 0x0B    | –77.76 mV  |
| 0x04                       | –38.84 mV | 0x0C    | –83.32 mV  |
| 0x05                       | –44.40 mV | 0x0D    | –88.88 mV  |
| 0x06                       | –49.96 mV | 0x0E    | –94.44 mV  |
| 0x07                       | –55.52 mV | 0x0F    | –100.00 mV |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:OLD Threshold[3:0]** sets the voltage threshold.

**Table A-3. Overload in Discharge Protection Delay<sup>(1)</sup>**

| Setting | Time | Setting | Time  | Setting | Time  | Setting | Time  |
|---------|------|---------|-------|---------|-------|---------|-------|
| 0x00    | 1 ms | 0x04    | 9 ms  | 0x08    | 17 ms | 0x0C    | 25 ms |
| 0x01    | 3 ms | 0x05    | 11 ms | 0x09    | 19 ms | 0x0D    | 27 ms |
| 0x02    | 5 ms | 0x06    | 13 ms | 0x0A    | 21 ms | 0x0E    | 29 ms |
| 0x03    | 7 ms | 0x07    | 15 ms | 0x0B    | 23 ms | 0x0F    | 31 ms |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:OLD Threshold[7:4]** sets the delay time.

### A.2 Short Circuit in Charge (ASCC)

**Table A-4. Short Circuit in Charge Threshold**  
(Settings:AFE:AFE Protection Control [RSNS] = 0)<sup>(1)</sup>

| Setting | Threshold | Setting | Threshold |
|---------|-----------|---------|-----------|
| 0x00    | 22.2 mV   | 0x04    | 66.65 mV  |
| 0x01    | 33.3 mV   | 0x05    | 77.75 mV  |
| 0x02    | 44.4 mV   | 0x06    | 88.85 mV  |
| 0x03    | 55.5 mV   | 0x07    | 100 mV    |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCC Threshold[2:0]** sets the voltage threshold.

**Table A-5. Short Circuit in Charge Threshold**  
(Settings:AFE:AFE Protection Control [RSNS] = 1)<sup>(1)</sup>

| Setting | Threshold | Setting | Threshold |
|---------|-----------|---------|-----------|
| 0x00    | 44.4 mV   | 0x04    | 133.3 mV  |
| 0x01    | 66.6 mV   | 0x05    | 155.5 mV  |
| 0x02    | 88.8 mV   | 0x06    | 177.7 mV  |
| 0x03    | 111.1 mV  | 0x07    | 200 mV    |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCC Threshold[2:0]** sets the voltage threshold.

**Table A-6. Short Circuit in Charge Delay<sup>(1)</sup>**

| Setting | Time   | Setting | Time   | Setting | Time   | Setting | Time   |
|---------|--------|---------|--------|---------|--------|---------|--------|
| 0x00    | 0 μs   | 0x04    | 244 μs | 0x08    | 488 μs | 0x0C    | 732 μs |
| 0x01    | 61 μs  | 0x05    | 305 μs | 0x09    | 549 μs | 0x0D    | 793 μs |
| 0x02    | 122 μs | 0x06    | 366 μs | 0x0A    | 610 μs | 0x0E    | 854 μs |
| 0x03    | 183 μs | 0x07    | 427 μs | 0x0B    | 671 μs | 0x0F    | 915 μs |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCC Threshold[7:4]** sets the delay time.

### A.3 Short Circuit in Discharge (ASCD1 and ASCD2)

**Table A-7. Short Circuit in Discharge Threshold**  
(Settings:AFE:AFE Protection Control [RSNS] = 0)<sup>(1)</sup>

| Setting | Threshold | Setting | Threshold |
|---------|-----------|---------|-----------|
| 0x00    | –22.2 mV  | 0x04    | –66.65 mV |
| 0x01    | –33.3 mV  | 0x05    | –77.75 mV |
| 0x02    | –44.4 mV  | 0x06    | –88.85 mV |
| 0x03    | –55.5 mV  | 0x07    | –100 mV   |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCD1 Threshold[2:0]** and **Protection:AFE Thresholds:SCD2 Threshold[2:0]** sets the voltage thresholds.

**Table A-8. Short Circuit in Discharge Threshold**  
(Settings:AFE:AFE Protection Control [RSNS] = 1)<sup>(1)</sup>

| Setting | Threshold | Setting | Threshold |
|---------|-----------|---------|-----------|
| 0x00    | –44.4 mV  | 0x04    | –133.3 mV |
| 0x01    | –66.6 mV  | 0x05    | –155.5 mV |
| 0x02    | –88.8 mV  | 0x06    | –177.7 mV |
| 0x03    | –111.1 mV | 0x07    | –200 mV   |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCD1 Threshold[2:0]** and **Protection:AFE Thresholds:SCD2 Threshold[2:0]** sets the voltage thresholds.

**Table A-9. Short Circuit in Discharge 1 Delay**  
(Settings:AFE:AFE Protection Control [SCDDx2] = 0)<sup>(1)</sup>

| Setting | Time   | Setting | Time   | Setting | Time   | Setting | Time   |
|---------|--------|---------|--------|---------|--------|---------|--------|
| 0x00    | 0 μs   | 0x04    | 244 μs | 0x08    | 488 μs | 0x0C    | 732 μs |
| 0x01    | 61 μs  | 0x05    | 305 μs | 0x09    | 549 μs | 0x0D    | 793 μs |
| 0x02    | 122 μs | 0x06    | 366 μs | 0x0A    | 610 μs | 0x0E    | 854 μs |
| 0x03    | 183 μs | 0x07    | 427 μs | 0x0B    | 671 μs | 0x0F    | 915 μs |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCD1Threshold[7:4]** sets the delay time.

**Table A-10. Short Circuit in Discharge 1 Delay**  
(Settings:AFE:AFE Protection Control [SCDDx2] = 1)<sup>(1)</sup>

| Setting | Time   | Setting | Time   | Setting | Time    | Setting | Time    |
|---------|--------|---------|--------|---------|---------|---------|---------|
| 0x00    | 0 μs   | 0x04    | 488 μs | 0x08    | 976 μs  | 0x0C    | 1464 μs |
| 0x01    | 122 μs | 0x05    | 610 μs | 0x09    | 1098 μs | 0x0D    | 1586 μs |
| 0x02    | 244 μs | 0x06    | 732 μs | 0x0A    | 1220 μs | 0x0E    | 1708 μs |
| 0x03    | 366 μs | 0x07    | 854 μs | 0x0B    | 1342 μs | 0x0F    | 1830 μs |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCD1 Threshold[7:4]** sets the delay time.

**Table A-11. Short Circuit in Discharge 2 Delay**  
(Settings:AFE:AFE Protection Control [SCDDx2] = 0)<sup>(1)</sup>

| Setting | Time  | Setting | Time   | Setting | Time   | Setting | Time   |
|---------|-------|---------|--------|---------|--------|---------|--------|
| 0x00    | 0 μs  | 0x04    | 122 μs | 0x08    | 244 μs | 0x0C    | 366 μs |
| 0x01    | 31 μs | 0x05    | 153 μs | 0x09    | 275 μs | 0x0D    | 396 μs |
| 0x02    | 61 μs | 0x06    | 183 μs | 0x0A    | 305 μs | 0x0E    | 427 μs |
| 0x03    | 92 μs | 0x07    | 214 μs | 0x0B    | 335 μs | 0x0F    | 458 μs |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCD2 Threshold[7:4]** sets the delay time.

**Table A-12. Short Circuit in Discharge 2 Delay**  
(Settings:AFE:AFE Protection Control [SCDDx2] = 1)<sup>(1)</sup>

| Setting | Time   | Setting | Time   | Setting | Time   | Setting | Time   |
|---------|--------|---------|--------|---------|--------|---------|--------|
| 0x00    | 0 μs   | 0x04    | 244 μs | 0x08    | 488 μs | 0x0C    | 732 μs |
| 0x01    | 62 μs  | 0x05    | 306 μs | 0x09    | 550 μs | 0x0D    | 792 μs |
| 0x02    | 122 μs | 0x06    | 366 μs | 0x0A    | 610 μs | 0x0E    | 854 μs |
| 0x03    | 184 μs | 0x07    | 428 μs | 0x0B    | 670 μs | 0x0F    | 916 μs |

<sup>(1)</sup> Data flash setting **Protection:AFE Thresholds:SCD2 Threshold[7:4]** sets the delay time.



## ***Sample Filter Settings***

**Table B-1. Sample V/I/P Filter Settings and Associated Low-Pass Filter Time Constants<sup>(1)</sup>**

| Average V/I/P Filter | Effective Low-Pass Time Constant |
|----------------------|----------------------------------|
| 10                   | 0.25 seconds                     |
| 50                   | 0.5 seconds                      |
| 145                  | 1 second                         |
| 200                  | 3 seconds                        |

<sup>(1)</sup> Data flash setting **Calibration:Filter:Average V/I/P** sets this threshold.